

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LOS ANGELES REGION**

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ORDER R4-2022-0174

NPDES NUMBER CA0054216

**WASTE DISCHARGE REQUIREMENTS
FOR THE SANTA CLARITA VALLEY SANITATION DISTRICT
VALENCIA WATER RECLAMATION PLANT**

The following entity is subject to waste discharge requirements (WDRs) set forth in this Order:

Table 1. Discharger Information

Discharger:	Santa Clarita Valley Sanitation District (SCVSD, Discharger, or Permittee)
Name of Facility:	Valencia Water Reclamation Plant (Valencia WRP or Facility)
Facility Address:	28185 The Old Road Santa Clarita, CA 91355 Los Angeles County

Table 2. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North)	Discharge Point Longitude (West)	Receiving Water
001	Tertiary treated effluent	34.430546°	-118.592546°	Santa Clara River
002	Tertiary treated effluent	34.430075°	-118.592208°	Santa Clara River

Table 3. Administrative Information

This Order was adopted on:	May 12, 2022
This Order shall become effective on:	July 1, 2022
This Order shall expire on:	June 30, 2027
The Discharger shall file a Report of Waste Discharge (ROWD) as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for the Order reissuance of a NPDES permit no later than:	180 days prior to the Order expiration date

The United States Environmental Protection Agency (U.S. EPA)
and the California Regional Water Quality Control Board have
classified this discharge as follows:

Major

I, Renee Purdy, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Los Angeles Region, on the date indicated above.

Renee Purdy, Executive Officer

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1. FACILITY INFORMATION

Information describing the Valencia Water Reclamation Plant (Valencia WRP or Facility) is summarized in Table 1 and in sections 1 and 2 of the Fact Sheet (Attachment F). Section 1 of the Fact Sheet also includes information regarding the Facility's permit application.

2. FINDINGS

The California Regional Water Quality Control Board, Los Angeles Region (Los Angeles Water Board), finds:

- 2.1. Legal Authorities.** This Order serves as waste discharge requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the California Water Code (Water Code) (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as a National Pollutant Discharge Elimination System (NPDES) permit authorizing the Discharger to discharge into waters of the United States at the discharge location described in Table 2 subject to the WDRs in this Order.
- 2.2. Background and Rationale for Requirements.** The Los Angeles Water Board developed the requirements in this Order based on information submitted as part of the application, and monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G and H are also incorporated into this Order.
- 2.3. Provisions and Requirements Implementing State Law.** The provisions and requirements in subsection 4.2, 4.3, and 5.2 are included to implement state law only. These provisions and requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.
- 2.4. Notification of Interested Parties.** The Los Angeles Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.
- 2.5. Consideration of Public Comment.** The Los Angeles Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED that Order Number R4-2015-0071 is rescinded upon the effective date of this order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Permittee shall comply with the requirements in this Order. This action in no way prevents the Los Angeles Water Board from taking enforcement action for past violations of the previous Order.

3. DISCHARGE PROHIBITIONS

- 3.1. Discharge of treated wastewater at a location different from that described in this Order is prohibited.
- 3.2. The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited, except as allowed in section 6.3.5.d of this Order and Standard Provision 1.7 of Attachment D, Standard Provisions.
- 3.3. The monthly average effluent dry weather discharge flow rate from the facility shall not exceed the 21.6 million gallons per day (MGD) design capacity.
- 3.4. The Permittee shall not cause degradation of any water body, except as consistent with State Water Board Resolution Number 68-16.
- 3.5. The treatment or disposal of wastes from the facility shall not cause pollution or nuisance as defined in section 13050, subdivisions (l) and (m), of the Water Code.
- 3.6. The discharge of any substances in concentrations toxic to human, animal, plant, or aquatic life is prohibited.
- 3.7. The discharge of any radiological, chemical, or biological warfare agent or high-level radiological waste is prohibited.
- 3.8. The discharge of trash to surface waters of the State or the deposition of trash where it may be discharged into surface waters of the State is prohibited.

4. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

4.1. Effluent Limitations – Discharge Points 001 and 002 (Santa Clara River)

4.1.1. Final Effluent Limitations – Discharge Points 001 and 002 (Santa Clara River)

- a. The Permittee shall maintain compliance with the following effluent limitations at Discharge Points 001 and 002 into the Santa Clara River, with compliance measured at Monitoring Locations EFF-001 and EFF-002 as described in the Monitoring and Reporting Program (MRP), Attachment E:

Table 4a. Effluent Limitations – Santa Clara River

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily	Notes
Biochemical Oxygen Demand (BOD ₅ 20°C)	mg/L	20	30	45	--
BOD ₅ 20°C	lbs/day	3,600	5,400	8,100	a
Total Suspended Solids (TSS)	mg/L	15	40	45	--
TSS	lbs/day	2,700	7,200	8,100	a
Removal Efficiency for BOD ₅ 20°C and TSS	%	≥85	--	--	--
Temperature	°F	--	--	80	b
Total Coliform	MPN or CFU/100 mL	23	2.2	240	c

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily	Notes
Oil and Grease	mg/L	10	--	15	--
Oil and Grease	lbs/day	1,800	--	2,700	a
Settleable Solids	ml/L	0.1	--	0.3	--
Total Residual Chlorine	mg/L	--	--	0.1	--
Total Residual Chlorine	lbs/day	--	--	18	a
Methylene Blue Activated Substances (MBAS)	mg/L	0.5	--	--	--
MBAS	lbs/day	90	--	--	a
Boron	mg/L	1.5	--	--	--
Boron	lbs/day	270	--	--	a
Total dissolved solids (TDS)	mg/L	1,000	--	--	--
TDS	lbs/day	180,000	--	--	a
Chloride	mg/L	100	--	--	d
Chloride	mg/L	C VAL, 3 mo.av.	--	230	e
Sulfate	mg/L	400	--	--	--
Sulfate	lbs/day	72,000	--	--	a
Ammonia Nitrogen	mg/L	1.75	--	5.2	f
Nitrate + Nitrite (as N)	mg/L	6.8	--	--	f
Nitrite (as N)	mg/L	0.9	--	--	f
Nitrate (as N)	mg/L	6.8	--	--	f
Selenium	µg/L	4.5	--	6.8	--
Selenium	lbs/day	0.81	--	1.2	a
Cyanide	µg/L	4.4	--	8.1	--
Cyanide	lbs/day	0.79	--	1.5	a
2,3,7,8-TCDD (Dioxin)	pg/L	0.014	--	0.028	--
2,3,7,8-TCDD (Dioxin)	lbs/day	2.5×10^{-9}	--	5×10^{-9}	a
Iron	µg/L	300	--	--	--
Iron	lbs/day	54	--	--	a
Total trihalomethanes (TTHMs)	µg/L	80	--	--	g
TTHMs	lbs/day	14	--	--	a
Chronic Toxicity <i>Ceriodaphnia dubia</i> Survival and reproduction endpoint	Pass or Fail, (Test of Significant Toxicity, (TST)), Percent Effect	Pass	--	Pass (TST) or Percent Effect <50% (survival endpoint)	h & i

Footnotes for Table 4a

- a. The mass-based effluent limitations are based on the plant design flow rate of 21.6 MGD and are calculated as follows: Flow (mgd) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations shall be the only applicable effluent limitations.
- b. An interim effluent limitation for temperature is included in section 4.1.2 of this Order for the duration of the compliance schedule.
- c. The wastes discharged to water courses shall be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if: (1) the median number of total coliform bacteria at some point in the treatment process does not exceed a 7-day median of 2.2 Most Probable Number (MPN) or Colony Forming Units (CFU) per 100 milliliters utilizing the bacteriological results of the last seven (7) days for which an analysis has been completed, (2) the number of total coliform bacteria does not exceed 23 MPN or CFU per 100 milliliters in more than one sample within any 30-day period, and (3) no sample shall exceed 240 MPN or CFU of total coliform bacteria per 100 milliliters. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
- d. The 100 mg/L 3-month rolling average is based on the water quality objective in the Basin Plan because the Permittee was not able to meet the July 1, 2019 deadline, specified in the Implementation section of the *Upper Santa Clara River Chloride TMDL*, Resolution No. 2008-012, for completing the ultra violet (UV) disinfection and microfiltration and reverse osmosis (MF/RO) capital improvement projects. On January 1, 2023, this limitation will be superseded by the limitation described below in Footnote e, upon the expiration of Time Schedule Order No. R4-2019-0055-A02 and the completion of the UV and MF/RO construction at the Valencia WRP.
- e. The following three-month rolling average effluent limitation is derived from the waste load allocation for chloride, as set forth in Resolution No. R14-010, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate an Averaging Period for Chloride Water Quality Objectives in Reaches 4B, 5 and 6; Incorporate New Site Specific Objectives for Chloride in Reaches 5 and 6; and Revise the Total Maximum Daily Load for Chloride in the Upper Santa Clara River*, adopted by the Los Angeles Water Board on October 9, 2014 and approved by the State Water Resources Control Board, the Office of Administrative Law (OAL) and USEPA on, December 16, 2014, March 18, 2015, and April 28, 2015, respectively:

$$C_{VAL,3mo.av} = \frac{1}{3} * \sum_{mi=1}^3 \left[\frac{Q_{SAU,mi}(100 - C_{SAU,mi})}{Q_{VAL,mi}} + 100 \right]$$

Where: $Q_{VAL,mi}$ = Valencia WRP monthly effluent flow in MGD;

$C_{SAU,mi}$ = Saugus WRP monthly effluent chloride level in mg/L;

$C_{VAL,mi}$ = Valencia WRP monthly effluent chloride level in mg/L;

$Q_{SAU,mi}$ and $Q_{VAL,mi}$ shall not exceed the design flow during dry-weather periods.

On January 1, 2023, this limitation will supersede the limitation described above in Footnote d, upon the expiration of Time Schedule Order No. R4-2019-0055-A02 and the completion of the UV and MF/RO construction at the Valencia WRP.

- f. This limitation is derived from the final waste load allocation, as set forth in Resolution No. 03-011, *Amendment to the Water Quality Control Plan for the Los Angeles Region to include a TMDL for*

Nitrogen Compounds in the Santa Clara River, adopted by the Los Angeles Water Board on August 7, 2003.

- g. The Total trihalomethanes limitation is the sum of the following trihalomethane compounds: bromodichloromethane, bromoform, chloroform, and dibromochloromethane.
- h. Instead of an average monthly result, the monthly median result is compared to the Median Monthly Effluent Limitation (MMEL) and shall be reported as "Pass" or "Fail." The maximum daily result is compared to Maximum Daily Effluent Limitation (MDEL) and shall be reported as "Pass" or "Fail" and "% Effect." If the chronic aquatic toxicity routine monitoring test conducted in the beginning of the month results in a "Fail" at the instream waste concentration (IWC), then the Valencia WRP shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the "Fail" at the IWC. If the first chronic MMEL compliance test results in a "Fail" at the IWC, then the second MMEL compliance test is not necessary because the "Fail" results from the first two tests would constitute a violation of the chronic toxicity MMEL.
- i. A numeric WQBEL is established because effluent data showed that there was reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The Chronic Toxicity final effluent limitation is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objectives. These final effluent limitations are established using current USEPA guidance in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June/2010) and [EPA Regions 8, 9, and 10 Toxicity Training Tool](#) (January 2010), <https://www.epa.gov/sites/production/files/documents/ToxTrainingTool10Jan2010.pdf>.

End of Footnotes for Table 4a

- b. The effluent values for pH shall be maintained within the limits of 6.5 standard units and 8.5 standard units.
- c. For the protection of the water contact recreation beneficial use, the wastes discharged to water courses shall have received adequate treatment, so that the turbidity of the treated wastewater does not exceed any of the following: (a) an average of 2 Nephelometric turbidity units (NTU) within a 24-hour period, (b) 5 NTU more than 5 percent of the time (72 minutes) within a 24-hour period, and (c) 10 NTU at any time.

Table 4b. Effluent Limitations – Santa Clara River

Parameters	Units	Average Monthly	Notes
Combined Radium-226 and Radium-228	pCi/L	5	Note a
Gross Alpha particle activity (excluding radon and uranium)	pCi/L	15	Note a
Uranium	pCi/L	20	Note a
Gross Beta/photon emitters	millirem/ year	4	Note a
Strontium-90	pCi/L	8	Note a
Tritium	pCi/L	20,000	Note a

Footnotes for Table 4b

- a. The radioactivity final effluent limitations are derived from 22 CCR § 64442 and § 64443. The incorporation by reference is prospective including future changes to the incorporated provisions as changes take effect.

End of Footnotes for Table 4b

4.1.2. Interim Effluent Limitations – Discharge Points 001 & 002 (Santa Clara River)

This Order includes a new, more stringent effluent limitation for temperature based on a new interpretation of the water quality objective for temperature contained in the Basin Plan. Consistent with section 1.e. of the State Water Board's Resolution 2008-0025 - *Policy for Compliance Schedules in National Pollutant Discharge Elimination System Permits* (Compliance Schedule Policy), the Discharger shall have up to ten years to implement actions specified in Table 6 to comply with a more stringent permit limitation. The interim effluent limitation in Table 5 shall apply from the effective date of Order Number R4-2022-0174 until the expiration date of the compliance schedule.

Table 5. Interim Effluent Limitation

Constituent	Units	Daily Maximum	Note
Temperature	°F	86	a

Footnotes for Table 5

- a. The temperature of wastes discharge shall not exceed 86°F except as a result of external ambient temperature. This interim limitation is based on the final effluent limitation for temperature in Order Number R4-2015-0071.

End of Footnotes for Table 5

4.2. Land Discharge Specifications – Not Applicable

4.3. Recycling Specifications

The Permittee shall continue to investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or capture and treatment of dry-weather urban runoff and stormwater on a permissive basis for beneficial reuse. The Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.

5. RECEIVING WATER LIMITATIONS

5.1. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. The discharge shall not cause the following in the receiving water:

- 5.1.1. The natural receiving water temperature of all regional waters shall not be altered unless it can be demonstrated to the satisfaction of the Los Angeles Water Board that such alteration in temperature does not adversely affect beneficial uses. Additionally, for waters designated with a warm freshwater habitat (WARM) beneficial use, water temperature shall not be altered by more than 5°F above the natural temperature. At no time shall these WARM-designated waters be raised above 80°F as a result of waste discharge except during the term of the compliance schedule set forth in Table 6, section 6.3.7, when the following interim receiving

water limitation is in effect: at no time shall these WARM-designated waters be raised above 86°F as a result of waste discharge except as a result of external ambient temperature.

- 5.1.2. The pH of inland surface waters shall not be depressed below 6.5 or raised above 8.5 as a result of wastes discharged. Ambient pH levels shall not be changed more than 0.5 units from natural conditions as a result of wastes discharged. Natural conditions shall be determined on a case-by-case basis.
- 5.1.3. The dissolved oxygen in the receiving water shall not be depressed below 5 mg/L as a result of the wastes discharged.
- 5.1.4. Total residual chlorine shall not persist in the receiving water at any concentration that causes impairment of beneficial uses as a result of the wastes discharged.
- 5.1.5. Waters shall be free of changes in turbidity that cause nuisance or adversely affect beneficial uses. Increases in natural turbidity attributable to controllable water quality factors shall not exceed the following limits, as a result of wastes discharged:
 - a. Where natural turbidity is between 0 and 50 NTU, increases shall not exceed 20%.
 - b. Where natural turbidity is greater than 50 NTU, increases shall not exceed 10%.
- 5.1.6. The wastes discharged shall not produce concentrations of substances in the receiving water that are toxic to or cause detrimental physiological responses in human, animal, or aquatic life.
- 5.1.7. The wastes discharged shall not cause concentrations of substances to occur at levels that are harmful to human health in waters which are existing or potential sources of drinking water.
- 5.1.8. The concentrations of toxic pollutants in the water column, sediments, or biota shall not adversely affect beneficial uses as a result of the wastes discharged.
- 5.1.9. The wastes discharged shall not contain substances that result in increases in BOD, which adversely affect the beneficial uses of the receiving waters.
- 5.1.10. Waters discharged shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses.
- 5.1.11. The dissolved sulfide concentration of waters in and near sediments shall not be significantly increased above that present under natural conditions as a result of waters discharged.
- 5.1.12. The wastes discharged shall not cause the receiving waters to contain any substance in concentrations that adversely affect any designated beneficial use.
- 5.1.13. The wastes discharged shall not alter the natural taste, odor, or color of fish, shellfish, or other surface water resources used for human consumption.
- 5.1.14. The wastes discharged shall not result in problems due to breeding of mosquitoes, gnats, black flies, midges, or other pests.

- 5.1.15. The wastes discharged shall not result in visible floating particulates, foams, or oil and grease in the receiving waters that cause nuisance or adversely affect beneficial uses.
- 5.1.16. The wastes discharged shall not alter the color of the receiving waters; create a visual contrast with the natural appearance of the water; or cause aesthetically undesirable discoloration of the receiving waters.
- 5.1.17. The wastes discharged shall not contain any individual pesticide or combination of pesticides in concentrations that adversely affect beneficial uses of the receiving waters. There shall be no increase in pesticide concentrations found in bottom sediments or aquatic life as a result of the wastes discharged.
- 5.1.18. There shall be no chronic toxicity in ambient waters as a result of wastes discharged.
- 5.1.19. The wastes discharged shall not cause the ammonia water quality objective in the Basin Plan to be exceeded in the receiving waters. Compliance with the ammonia water quality objectives shall be determined by comparing the receiving water ammonia concentration to the ammonia water quality objective in the Basin Plan. The ammonia water quality objective can also be calculated using the pH and temperature of the receiving water at the time of collection of the ammonia sample.

5.2. Groundwater Limitations – Not Applicable

6. PROVISIONS

6.1. Standard Provisions

- 6.1.1. The Permittee shall comply with all Standard Provisions included in Attachment D.
- 6.1.2. **Los Angeles Water Board Standard Provisions.** The Permittee shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
 - a. Neither the treatment nor the discharge of pollutants shall create a pollution, contamination, or nuisance as defined by section 13050 of the Water Code.
 - b. Odors, vectors, and other nuisances of sewage or sludge origin beyond the limits of the treatment plant site or the sewage collection system due to improper operation of facilities, as determined by the Los Angeles Water Board, are prohibited.
 - c. All facilities used for collection, transport, treatment, or disposal of wastes shall be adequately protected against damage resulting from overflow, washout, or inundation from a storm or flood having a recurrence interval of once in 100 years.
 - d. Collection, treatment, and disposal systems shall be operated in a manner that precludes or impedes public contact with wastewater.
 - e. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer of the Los Angeles Water Board.
 - f. The provisions of this Order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.

- g. Nothing in this Order shall be construed to preclude the institution of any legal action or relieve the Permittee from any responsibilities, liabilities or penalties established pursuant to any applicable state law or regulation under authority preserved by section 311 of the CWA, related to oil and hazardous substances liability.
- h. The Discharger must comply with the lawful requirements of municipalities, counties, drainage districts, and other local agencies regarding discharges of stormwater to storm drain systems or other water courses under their jurisdiction, including applicable requirements in municipal stormwater management programs developed to comply with NPDES permits issued by the Los Angeles Water Board to local agencies.
- i. Discharge of wastes to any point other than specifically described in this Order is prohibited and constitutes a violation thereof.
- j. The Permittee shall comply with all applicable effluent limitations, national standards of performance, toxic effluent standards, and all federal regulations established pursuant to sections 301, 302, 303(d), 304, 306, 307, 316, 403, and 405 of the federal CWA and amendments thereto.
- k. These requirements do not exempt the operator of the waste disposal facility from compliance with any other laws, regulations, or ordinances which may be applicable; they do not legalize this waste disposal facility; and they leave unaffected any further restraints on the disposal of wastes at this site which may be contained in other statutes or required by other agencies.
- l. The Discharger shall make efforts to reduce Facility infrastructure vulnerability to current and future impacts resulting from climate change, including but not limited to extreme wet weather events, flooding, storm surges, and projected sea level rise when the facility is located near the ocean or discharges to the ocean.
- m. Oil or oily material, chemicals, refuse, or other polluting materials shall not be stored or deposited in areas where they may be picked up by rainfall and carried off of the property and/or discharged to surface waters. Any such spill of such materials shall be contained and removed immediately.
- n. A copy of these waste discharge specifications shall be maintained at the discharge Facility so as to be available at all times to operating personnel.
- o. If there is any storage of hazardous or toxic materials or hydrocarbons at this Facility and if the Facility is not manned at all times, a 24-hour emergency response telephone number shall be prominently posted where it can easily be read from the outside.
- p. The Permittee shall file with the Los Angeles Water Board a report of waste discharge at least 120 days before making any proposed change in the character, location or volume of the discharge.
- q. In the event of any change in name, ownership, or control of these waste disposal facilities, the Permittee shall notify the Los Angeles Water Board of such change and shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Los Angeles Water Board, 30 days prior to taking effect.

- r. The discharge of any waste resulting from the combustion of toxic or hazardous wastes to any waste stream that ultimately discharges to waters of the United States is prohibited, unless specifically authorized elsewhere in this Order.
- s. The Permittee shall notify the Executive Officer in writing no later than 6 months prior to planned discharge of any chemical, other than the products previously reported to the Executive Officer, which may be toxic to aquatic life. Such notification shall include:
 - i. Name and general composition of the chemical,
 - ii. Frequency of use,
 - iii. Quantities to be used,
 - iv. Proposed discharge concentrations, and
 - v. USEPA registration number, if applicable.
- t. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this Facility, may subject the Permittee to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Permittee to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- u. Water Code section 13385(h)(i) requires the Los Angeles Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each serious violation. Pursuant to Water Code section 13385(h)(2), a “serious violation” is defined as any waste discharge that violates the effluent limitations contained in the applicable waste discharge requirements for a Group II pollutant by 20 percent or more, or for a Group I pollutant by 40 percent or more. Appendix A of 40 CFR section 123.45 specifies the Group I and II pollutants. Pursuant to Water Code section 13385.1(a)(1), a “serious violation” is also defined as “a failure to file a discharge monitoring report required pursuant to section 13383 for each complete period of 30 days following the deadline for submitting the report, if the report is designed to ensure compliance with limitations contained in waste discharge requirements that contain effluent limitations.”
- v. Water Code section 13385(i) requires the Los Angeles Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each violation whenever a person violates a waste discharge requirement effluent limitation in any period of six consecutive months, except that the requirement to assess the mandatory minimum penalty shall not be applicable to the first three non-serious violations within that time period.
- w. Pursuant to Water Code section 13385.1(d), for the purposes of section 13385.1 and subdivisions (h), (i), and (j) of section 13385, “effluent limitation” means a numeric restriction or a numerically expressed narrative restriction, on the quantity, discharge rate, concentration, or toxicity units of a pollutant or pollutants that may be discharged from an authorized location. An effluent limitation may be final or interim and may be expressed as a prohibition. An effluent limitation, for these purposes,

does not include a receiving water limitation, a compliance schedule, or a best management practice.

- x. Water Code section 13387(e) provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this order, including monitoring reports or reports of compliance or noncompliance, or who knowingly falsifies, tampers with, or renders inaccurate any monitoring device or method required to be maintained in this order shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000), imprisonment pursuant to subdivision (h) of section 1170 of the Penal Code for 16, 20, or 24 months, or by both that fine and imprisonment. For a subsequent conviction, such a person shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000) per day of violation, by imprisonment pursuant to subdivision (h) of section 1170 of the Penal Code for two, three, or four years, or by both that fine and imprisonment.
- y. In the event the Permittee does not comply or will be unable to comply for any reason, with any prohibition, effluent limitation, or receiving water limitation of this Order, the Permittee shall notify the Manager of the Watershed Regulatory section at the Los Angeles Water Board by telephone (213) 576-6616 or by fax at (213) 576-6660 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing to the Los Angeles Water Board within five days, unless the Los Angeles Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and prevent recurrence including, where applicable, a schedule of implementation. The written notification shall also be submitted via email with reference to CI-4993 to losangeles@waterboards.ca.gov. Other noncompliance requires written notification as above at the time of the normal monitoring report.

6.2. Monitoring and Reporting Program (MRP) Requirements

The Permittee shall comply with the MRP, and future revisions thereto, in Attachment E.

6.3. Special Provisions

6.3.1. Reopener Provisions

- a. This Order may be modified, revoked and reissued, or terminated for cause, including, but not limited to:
 - i. Violation of any term or condition contained in this Order.
 - ii. Obtaining this Order by misrepresentation, or by failure to disclose fully all relevant facts; or
 - iii. A change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge.

The filing of a request by the Permittee for an Order modification, revocation, and issuance or termination, or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.

- b. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity testing, monitoring of internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.
- c. This Order may be modified, in accordance with the provisions set forth in title 40 of the Code of Federal Regulations (40 CFR) parts 122 and 124 to include requirements for the implementation of a watershed protection management approach.
- d. The Board may modify, or revoke and reissue this Order if present or future investigations demonstrate that the discharge(s) governed by this Order have or will have a reasonable potential to cause or contribute to adverse impacts on water quality or beneficial uses of the receiving waters.
- e. This Order may also be modified, revoked, and reissued or terminated in accordance with the provisions of 40 CFR sections 122.44, 122.62 to 122.64, 125.62, and 125.64. Causes for taking such actions include, but are not limited to, failure to comply with any condition of this Order, endangerment to human health or the environment resulting from the permitted activity, or acquisition of newly obtained information which would have justified the application of different conditions if known at the time of Order adoption. The filing of a request by the Permittee for an Order modification, revocation and issuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.
- f. This Order may be modified, in accordance with the provisions set forth in 40 CFR parts 122 to 124, to include new minimum levels (MLs).
- g. If an applicable toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is promulgated under section 307(a) of the CWA for a toxic pollutant and that standard or prohibition is more stringent than any limitation on the pollutant in this Order, the Los Angeles Water Board may institute proceedings under these regulations to modify or revoke and reissue the Orders to conform to the toxic effluent standard or prohibition.
- h. If more stringent applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments, thereto, the Los Angeles Water Board will revise and modify this Order in accordance with such standards.
- i. This Order may be reopened and modified, to add or revise effluent limitations as a result of future additions or amendments to a statewide water quality control plan or the Los Angeles Region's Basin Plan, or the adoption or revision of a TMDL.
- j. This Order may be reopened and modified to revise any and all of the chronic toxicity testing provisions and effluent limitations, to the extent necessary, to incorporate applicable elements contained in the State Water Board adopted Toxicity Provisions following USEPA-approval of such Toxicity Provisions to be consistent with the State Water Board precedential decisions, new policies, a new state-wide plan, new laws, or new regulations.

- k. This Order will be reopened and modified to the extent necessary, to be consistent with new policies, new state-wide plans, new laws, or new regulations.

6.3.2. Special Studies, Technical Reports and Additional Monitoring Requirements

a. Toxicity Reduction Requirements

The Permittee shall prepare and submit a copy of the Permittee's initial investigation Toxicity Reduction Evaluation (TRE) workplan in accordance with Monitoring and Reporting Program section 5.6.

b. Treatment Plant Capacity

The Permittee shall submit a written report to the Executive Officer of the Los Angeles Water Board within 90 days after the "30-day (monthly) average" daily dry-weather flow equals or exceeds 75 percent of the design capacity of waste treatment and/or disposal facilities. The Permittee's senior administrative officer shall sign a letter, which transmits that report and certify that the Permittee's policy-making body is adequately informed of the report's contents. The report shall include the following:

- i. The average daily flow for the month, the date on which the peak flow occurred, the rate of that peak flow, and the total flow for the day.
- ii. The best estimate of when the monthly average daily dry-weather flow rate will equal or exceed the design capacity of the facilities.
- iii. A schedule for studies, design, and other steps needed to provide additional capacity for waste treatment and/or disposal facilities before the waste flow rate equals the capacity of present units.

If the facility has reached 75 percent of capacity by the effective date of this Order but has not previously submitted such report, such a report shall be filed within 90 days of the issuance of this Order.

6.3.3. Best Management Practices and Pollution Prevention

a. Stormwater Pollution Prevention Plan (SWPPP) – (Not Applicable)

b. Spill Clean-up Contingency Plan (SCCP)

Within 90 days of the effective date of this Order, the Permittee is required to update and submit a SCCP, which describes the activities and protocols to address clean-up of spills, overflows, and bypasses of untreated or partially treated wastewater from the Permittee's collection system or treatment facilities that reach water bodies, including dry channels and beach sands. At a minimum, the plan shall include sections on spill clean-up and containment measures, public notification, and monitoring. The Permittee shall review and amend the plan as appropriate after each spill from the Facility or in the service area of the Facility. The Permittee shall include a discussion in the annual summary report of any modifications to the Plan and the application of the Plan to all spills during the year.

c. Pollutant Minimization Program (PMP)

Reporting protocols in MRP section 10.2.4 describe sample results that are to be reported as Detected but Not Quantified (DNQ) or Not Detected (ND). Definitions for

a reported Minimum Level (ML) and Method Detection Limit (MDL) are provided in Attachment A. These reporting protocols and definitions are used in determining the need to conduct a PMP as follows:

The Permittee shall develop and conduct a PMP as further described below when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL; sample results from analytical methods more sensitive than those methods required by this Order; presence of whole effluent toxicity; health advisories for fish consumption; or, results of benthic or aquatic organism tissue sampling) that a pollutant is present in the effluent above an effluent limitation and either of the following is true:

- i. The concentration of the pollutant is reported as DNQ and the effluent limitation is less than the reported ML; or,
- ii. The concentration of the pollutant is reported as ND and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in the MRP.

The goal of the PMP shall be to reduce all potential sources of a pollutant through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Los Angeles Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan (PPP), if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Los Angeles Water Board:

- i. An annual review and semi-annual monitoring of potential sources of the reportable pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling.
- ii. Quarterly monitoring for the reportable pollutant(s) in the influent to the wastewater treatment system.
- iii. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable pollutant(s) in the effluent at or below the effluent limitation.
- iv. Implementation of appropriate cost-effective control measures for the reportable pollutant(s), consistent with the control strategy.

An annual status report that shall be sent to the Los Angeles Water Board including:

- (a) All PMP monitoring results for the previous year.
- (b) A list of potential sources of the reportable pollutant(s).
- (c) A summary of all actions undertaken pursuant to the control strategy.
- (d) A description of actions to be taken in the following year.

6.3.4. Construction, Operation and Maintenance Specifications

- a. Certified Wastewater Treatment Plant Operator:** Wastewater treatment facilities subject to this Order shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to California Code of Regulations (CCR), title 23, division 3, chapter 26 (Water Code sections 13625 – 13633).
- b. Climate Change Effects Vulnerability Assessment and Mitigation Plan:** The Permittee shall consider the impacts of climate change as they affect the operation of the treatment facility due to flooding, wildfires, or other climate-related changes. The Permittee shall develop a Climate Change Effects Vulnerability Assessment and Mitigation Plan (Climate Change Plan) to assess and manage climate change-related effects that may impact the wastewater treatment facility's operation, water supplies, its collection system, and water quality, including any projected changes to the influent water temperature and pollutant concentrations, and beneficial uses. For facilities that discharge to the ocean including desalination plants, the Climate Change Plan shall also include the impacts from sea level rise. The Climate Change Plan is due 12 months after effective date of this Order.
- c. Alternate Power Source:** The Permittee shall maintain in good working order a sufficient alternate power source for operating the wastewater treatment and disposal facilities. All equipment shall be located and secured to minimize failure due to moisture, liquid spray, flooding, wildfires, and other physical phenomena. The alternate power source shall be designed to allow inspection and maintenance and shall provide for periodic testing. If such alternate power source is not in existence, the Discharger shall halt, reduce, or otherwise control all discharges upon the reduction, loss, or failure of the primary source of power. The Permittee shall provide standby or emergency power facilities and/or storage capacity or other means so that in the event of plant upset or outage due to power failure or other cause, discharge of raw or inadequately treated sewage does not occur.
- d. Routine Maintenance and Operational Testing for Emergency Infrastructure/Equipment:** The Permittee shall perform monthly maintenance and operational testing for all emergency infrastructure and equipment at the facility, including but not limited to any bypass gate/weir in the headworks, alarm systems, backup pumps, standby power generators, and other critical emergency pump station components. The Permittee shall update the Operation and Maintenance Plan to include monthly maintenance and operational testing of emergency infrastructure and equipment, and shall keep the records of all operational testing for emergency systems, repairs, and modifications.

6.3.5. Special Provisions for Municipal Facilities (POTWs Only)

- a. Biosolids Disposal Requirements (Not Applicable) – Refer to Attachment I
- b. Pretreatment Requirements
 - i. The Permittee has developed and implemented an approved Pretreatment Program that was submitted to the Los Angeles Water Board. This Order requires implementation of the approved Pretreatment Program. Any violation of the Pretreatment Program will be considered a violation of this Order.

- ii. In 1972, the County Sanitation Districts of Los Angeles County's (Sanitation Districts) Board of Directors adopted the Wastewater Ordinance. The purpose of this Ordinance is to establish controls on users of the Sanitation Districts sewerage system in order to protect the environment and public health, and to provide for the maximum beneficial use of the Sanitation Districts' facilities. This Wastewater Ordinance, as amended July 1, 1998, superseded all previous regulations and policies of the Sanitation Districts governing items covered in this Ordinance. Specifically, the provisions of this Ordinance superseded the Sanitation Districts' "Policy Governing Use of District Trunk Sewers" dated December 6, 1961 and amended the Sanitation Districts' "An Ordinance Regulating Sewer Construction, Sewer Use and Industrial Wastewater Discharges," dated April 1, 1972, and as amended July 1, 1975, July 1, 1980, July 1, 1983, and November 1, 1989.
- iii. Any change to the program shall be reported to the Los Angeles Water Board in writing and shall not become effective until approved by the Executive Officer in accordance with procedures established in 40 CFR section 403.18.
- iv. Applications for renewal or modification of this Order must contain information about industrial discharges to the POTW pursuant to 40 CFR section 122.21(j)(6). Pursuant to 40 CFR section 122.42(b) and provision 7.1 of Attachment D, Standard Provisions, of this Order, the Permittee shall provide adequate notice of any new introduction of pollutants or substantial change in the volume or character of pollutants from industrial discharges which were not included in the permit application. Pursuant to 40 CFR section 122.44(j)(1), the Permittee shall annually identify and report, in terms of character and volume of pollutants, any Significant Industrial Users discharging to the POTW subject to Pretreatment Standards under section 307(b) of the CWA and 40 CFR part 403.
- v. The Permittee shall evaluate whether its pretreatment local limits are adequate to meet the requirements of this Order and shall submit a written technical report as required under section 2.1 of Attachment H. The Valencia WRP is part of the Santa Clarita Valley Sanitation District (SCVSD), consisting of the Saugus WRP, the Valencia WRP, and the collection system. In the reevaluation of the local limits, the Permittee shall consider the effluent limitations contained in this Order, the contributions from the upstream WRPs in the SCVSD, and other relevant factors due to the interconnection of the Districts' WRPs within the SCVSD. The Permittee shall submit to the Los Angeles Water Board revised local limits, as necessary, for Los Angeles Water Board approval within 180 days of the effective date of Order No. R4-2022-0174. In addition, the Permittee shall consider collection system overflow protection from such constituents as large debris, oil and grease, etc.
- vi. The Permittee shall comply with Attachment H – Pretreatment Reporting Requirements.

c. Collection System Requirements

The Permittee's collection system is part of the system that is subject to this Order. As such, the Permittee must properly operate and maintain its collection system (40 CFR section 122.41(e)). The Permittee must report any non-compliance (40 CFR section 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 CFR section 122.41(d)). On October 20, 2006, the

Permittee submitted a Notice of Intent to enroll under the Statewide General Waste Discharge Requirements for Sanitary Sewer System, Water Quality Order Number 2006-0003.

d. Filter Bypass

Conditions pertaining to bypass are contained in Attachment D, section 1 Standard Provisions – Permit Compliance, subsection 7. The bypass or overflow of untreated or partially treated wastewater to waters of the State is prohibited, except as allowed under conditions stated in 40 CFR section 122.41(m) and (n). During periods of elevated, wet weather flows, a portion of the secondary treated wastewater is diverted around the tertiary filters as a necessary means to avoid loss of life, personal injury or severe property damage. There are no feasible alternatives to this diversion. These anticipated discharges are approved under the bypass conditions when all storage has been utilized and the resulting combined discharge of fully treated (tertiary) and partially treated (secondary) wastewater complies with the effluent and receiving water limitations in this Order. The ROWD constitutes notice to these anticipated bypasses.

6.3.6. Spill Reporting Requirements

a. Initial Notification

Although State and Los Angeles Water Board staff do not have duties as first responders, this requirement is an appropriate mechanism to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and beneficial uses. For certain spills, overflows and bypasses, the Permittee shall make notifications as required below:

- i. In accordance with the requirements of Health and Safety Code section 5411.5, the Permittee shall provide notification to the local health officer or the director of environmental health with jurisdiction over the affected water body of any unauthorized release of sewage or other waste that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but no later than two hours after becoming aware of the release.
- ii. In accordance with the requirements of Water Code section 13271, the Permittee shall provide notification to the California Office of Emergency Services (OES) of the release of reportable amounts of hazardous substances or sewage that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but not later than two hours after becoming aware of the release. The CCR, title 23, section 2250, defines a reportable amount of sewage as being 1,000 gallons. The phone number for reporting these releases to the OES is (800) 852-7550. In addition, the Permittee shall notify other interested persons of any such sewage spill by maintaining an email list of those interested persons that have requested such notification.
- iii. The Permittee shall notify the Los Angeles Water Board of any unauthorized release of sewage from its POTW that causes, or probably will cause, a discharge to a water of the state as soon as possible, but not later than two hours after becoming aware of the release. This initial notification does not need to be made if the Permittee has notified OES and the local health officer or the director of

environmental health with jurisdiction over the affected waterbody. The phone number for reporting these releases of sewage to the Los Angeles Water Board is (213) 576-6657. The phone numbers for after hours and weekend reporting of releases of sewage to the Los Angeles Water Board are (213) 305-2284 and (213) 305-2253.

At a minimum, the following information shall be provided to the Los Angeles Water Board:

- The location, date, and time of the release.
- The water body that received or will receive the discharge.
- An estimate of the amount of sewage or other waste released and the amount that reached a surface water at the time of notification.
- If ongoing, the estimated flow rate of the release at the time of the notification.
- The name, organization, phone number and email address of the reporting representative.

b. Monitoring

For spills, overflows and bypasses reported under section 6.3.6.a, the Permittee shall monitor as required below:

To define the geographical extent of the spill's impact, the Permittee shall obtain grab samples for all spills, overflows or bypasses of any volume that reach any waters of the state (including surface and ground waters). If a grab sample cannot be obtained due to accessibility or safety concerns that cannot be addressed with the appropriate personal protective equipment or following proper sampling procedures, the sample shall be obtained as soon as it becomes safe to do so. The Permittee shall analyze the samples for total coliform, *E. coli* (if total coliform tests positive), *Enterococcus* (if spill reaches the marine waters, where the salinity is greater than 1 part per thousand more than 5 percent of time), and relevant pollutants of concern, upstream and downstream of the point of entry of the spill (if feasible, accessible, and safe). Daily monitoring shall be conducted from the time the spill is known until the results of two consecutive sets of bacteriological monitoring indicate the return to the background level or the County Department of Public Health authorizes cessation of monitoring.

c. Reporting

The initial notification required under section 6.3.6.a shall be followed by:

- i. As soon as possible, but not later than twenty-four hours after becoming aware of an unauthorized discharge of sewage or other waste from its wastewater treatment plant to a water of the state, the Permittee shall [submit a statement to the Los Angeles Water Board by email](mailto:augustine.anijelo@waterboards.ca.gov) at augustine.anijelo@waterboards.ca.gov. If the discharge is 1,000 gallons or more, this statement shall certify that OES has been notified of the discharge in accordance with Water Code section 13271. The statement shall also certify that the local health officer or director of environmental health with jurisdiction over the affected water bodies has been notified of the

discharge in accordance with Health and Safety Code section 5411.5. The statement shall also include at a minimum the following information:

- Agency, NPDES Number, Order Number, and MRP CI Number, if applicable.
 - The location, date, and time of the discharge.
 - The water body that received the discharge.
 - A description of the level of treatment of the sewage or other waste discharged.
 - An initial estimate of the amount of sewage or other waste released and the amount that reached a surface water.
 - The OES control number and the date and time that notification of the incident was provided to OES.
 - The name of the local health officer or director of environmental health representative notified (if contacted directly); the date and time of notification; and the method of notification (e.g., phone, fax, email).
- ii. A written preliminary report five business days after disclosure of the incident is required. Submission to the Los Angeles Water Board of the California Integrated Water Quality System (CIWQS) Sanitary Sewer Overflow (SSO) event number shall satisfy this requirement. Within 30 days after submitting the preliminary report, the Permittee shall submit the final written report to this Los Angeles Water Board. (A copy of the final written report, for a given incident, already submitted pursuant to a statewide General WDRs for Sanitary Sewer Systems (SSS WDR), may be submitted to the Los Angeles Water Board to satisfy this requirement.) The written report shall document the information required in paragraph d below, monitoring results and any other information required in provisions of the Standard Provisions document including corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences. The Executive Officer for just cause can grant an extension for submittal of the final written report.
- iii. The Permittee shall include a certification in the annual summary report (due according to the schedule in the MRP) that states that the sewer system emergency equipment, including alarm systems, backup pumps, standby power generators, and other critical emergency pump station components were maintained and tested in accordance with the Permittee's preventive maintenance plan. Any deviations from or modifications to the plan shall be discussed.

d. Records

The Permittee shall develop and maintain a record of all spills, overflows or bypasses of raw or partially treated sewage from its collection system or treatment plant. This record shall be made available to the Los Angeles Water Board upon request and a spill summary shall be included in the annual summary report. The records shall contain:

- i. The date and time of each spill, overflow, or bypass.
- ii. The location of each spill, overflow, or bypass.

- iii. The estimated volume of each spill, overflow, and bypass including gross volume, amount recovered and amount not recovered, monitoring results as required by section 6.3.6.b.
- iv. The cause of each spill, overflow, or bypass.
- v. Whether each spill, overflow, or bypass entered a receiving water and, if so, the name of the water body and whether it entered via storm drains or other man-made conveyances.
- vi. Any mitigation measures implemented.
- vii. Any corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences.
- viii. The mandatory information included in SSO online reporting for finalizing and certifying the SSO report for each spill, overflow, or bypass under the SSO WDR.

e. Activities Coordination

Although not required by this Order, Los Angeles Water Board expects the POTW's owners/operators will coordinate their compliance activities for consistency and efficiency with other entities that have responsibilities to implement: (1) this NPDES permit, including the Pretreatment Program, (ii) a Municipal Separate Storm Sewer Systems (MS4) NPDES permit that may contain spill prevention, sewer maintenance, reporting requirements and (iii) the State Water Board Number WQ 2006-0003 DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems (SSS WDRs) or subsequent updates. The Los Angeles Water Board also expects that POTW's owners/operators to consider coordination with other agencies regarding the potential for the permissive integration of the MS4 with the wastewater collection system.

f. Consistency with SSS WDRs

The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311, 1342). The Permittee must comply with State Water Board Order Number 2006-0003-DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, (SSS WDRs) as amended by State Water Board Order Number WQ 2013-0058-exec and any subsequent order updating these requirements. These statewide WDRs require public agencies that own or operate sanitary sewer systems with greater than one mile of sewer lines to enroll for coverage and comply with requirements, to develop and implement sewer system management plans, and report all Sanitary Sewer Overflows (SSOs) to the State Water Board's online SSOs database. The Permittee enrolled in the SSS WDRs in 2006, and the collection systems of the Permittee are covered under the SSS WDRs. The Permittee must properly operate and maintain its collection system (40 CFR section 122.41 (e)), report any non-compliance (40 CFR section 122.41(1)(6) and (7)), and mitigate any discharge from the collection system in violation of this NPDES permit (40 CFR section 122.41(d)).

The requirements contained in this Order in sections 6.3.3.b (SCCP Plan section), 6.3.4 (Construction, Operation and Maintenance Specifications section), and 6.3.6

(Spill Reporting Requirements section) are intended to be consistent with the requirements of the SSS WDRs. The Los Angeles Water Board recognizes that there may be some overlap between these NPDES permit provisions and SSS WDRs requirements, related to the collection systems. The requirements of the SSS WDRs are considered the minimum thresholds (see finding 11 of State Water Board Order Number 2006-0003-DWQ). To encourage efficiency, the Los Angeles Water Board will accept the documentation prepared by the Permittees under the SSS WDRs for compliance purposes as satisfying the requirements in sections 6.3.3.b, 6.3.4, and 6.3.6 provided the more stringent provisions contained in this NPDES permit are also addressed. Pursuant to SSS WDRs, section D, provision 2(iii) and (iv), the provisions of this NPDES permit supersede the SSS WDRs, for all purposes, including enforcement, to the extent the requirements may be deemed duplicative.

6.3.7. Compliance Schedule

- a. The compliance schedule and the interim limit in section 4.1.2 of this Order are authorized under section 1.e. of the State Water Board's Resolution 2008-0025 - Policy for Compliance Schedules in National Pollutant Discharge Elimination System Permits (Compliance Schedule Policy).
- b. The Discharger shall notify the Los Angeles Water Board in a written compliance report, no later than 14 days following each interim milestone date, of its compliance or noncompliance with the interim requirements.
- c. In order to monitor compliance with the interim and final effluent limitations for temperature, the Discharger shall monitor the influent and effluent for temperature at the frequencies required in Tables E-2 and E-3. Each result shall be reported in the monthly report to track progress in achieving compliance with the final effluent limitations.
- d. The Permittee may be subject to enforcement action for failure to complete the tasks by the given milestone dates, as specified in Table 6.

Table 6. Compliance Schedule & Milestone Dates

Task	Completion Date
Submit and Begin Implementation of Pollution Prevention Plan (PPP) for Source Control	August 1, 2022
Convene Technical Advisory Committee and Stakeholder Committee	October 1, 2022
Release the Request for Proposal to Retain Consultant to Evaluate Temperature Impacts in the Watershed and Management Options	February 28, 2023
Finalize the Technical Workplan	February 28, 2024
Prepare a Technical Workplan Progress Report	February 28, 2025
Complete Implementation of Technical Workplan	February 28, 2026

Task	Completion Date
Notify Los Angeles Water Board of Selected Preferred Project and Identify Regulatory Approval Process and Present Results of Technical Workplan at Next Scheduled Board Meeting	April 1, 2026
Complete Preliminary Design	April 1, 2027
Complete Environmental Review	October 1, 2027
Complete Design of Preferred Project	July 1, 2028
Issue Notice to Proceed for Project Work	July 1, 2029
Complete Preferred Project	July 1, 2030

6.3.8. Implementation Provisions for Upper Santa Clara River Chloride TMDL

The Discharger shall comply with the following Implementation Tasks associated with Resolution No. R4-2014-010, Revision of the TMDL for Chloride in the Upper Santa Clara River:

Table 7. Remaining Tasks Under Upper Santa Clara Chloride TMDL

Task No.	Task	Completion Date
1	Alternate Water Supply a) Should (1) the in-river concentration at Blue Cut, the Reach 4B boundary exceed the water quality objective of 100 mg/L, measured for the purposes of this TMDL as a 3-month rolling average, (2) each agricultural diverter provide records of the diversion dates and amounts to the Regional Board and Santa Clarita Valley County Sanitation Districts of Los Angeles County (SCVSD) for at least 2 years after the effective date of the TMDL and (3) each agricultural diverter provides photographic evidence that diverted water is applied to avocado, strawberry or other chloride sensitive crop and evidence of a water right to divert, then the SCVSD will be responsible for providing an alternative water supply, negotiating the delivery of alternative water by a third party, or providing fiscal remediation to be quantified in negotiations between the SCVSD and the agricultural diverter at the direction of the Regional Board until such time as the in-river chloride concentrations do not exceed the conditional SSO. b) Should the instream concentration exceed 230 mg/L more than two times in the three-year period, the discharger identified by the Regional Board Executive Officer shall be required to submit, within ninety days of a request by the Regional Board Executive Officer, a work plan for an accelerated schedule to reduce chloride discharges.	(05/04/2005) (Does not apply upon completion of Task 4)

Task No.	Task	Completion Date
2	Trend monitoring: The SCVSD will submit to the Regional Board and upon approval implement a revised monitoring plan to conduct chloride trend monitoring to ensure that water quality objectives and waste load allocations are being met, downstream surface water quality is not degraded, and groundwater underlying Reach 5 upstream of the Valencia WRP outfall 001 and Reach 6 is not degraded due to implementation of compliance measures by SCVSD. The monitoring plan shall include a plan to collect water samples and analyze them for chloride in surface water for Reaches 4B, 5, and 6 at a minimum of once per month. The monitoring plan shall also include a plan for chloride trend monitoring in the alluvial groundwater basins underlying Reaches 5 and 6 and in the Saugus Aquifer at a minimum of twice a year. At a minimum, the monitoring plan should include a network of three groundwater wells with multiple screens to evaluate impacts to groundwater. The plan will include a monitoring schedule that extends beyond the final implementation deadline of this TMDL to support continual evaluation of impacts of compliance measures to surface water and groundwater quality. This TMDL shall be reconsidered if chloride trend monitoring indicates degradation of groundwater or surface water due to implementation of compliance measures	09/20/2015
3	Begin monitoring per approved SCVSD revised monitoring plan completed in Task 2.	Six months after Executive Officer approval of Task 2 revised monitoring plan for SCVSD
4	Implementation of Compliance Measures by SCVSD c) Microfiltration/Reverse Osmosis (MF/RO) and Brine Minimization Facilities i. Complete design of MF/RO and brine minimization facilities ii. Award contract for MF/RO and brine minimization facilities iii. Start onsite construction of MF/RO and brine minimization facilities iv. Start-up of MF/RO and brine minimization facilities	 4/12/2017 7/10/2017 3/10/2018 7/1/2019
5	Progress reports will be submitted by the SCVSD to the Regional Board on a semiannual basis for Task 4 and an annual basis for Task 3. Progress reports shall include supporting documentation that tasks were completed by the deadline	Semiannually for Task 4 Annually for Task 3 (Eighteen months after Executive Officer approval of

Task No.	Task	Completion Date
		Task 2 monitoring plan for SCVSD, and annually thereafter)

7. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section 4 of this Order will be determined as specified below:

7.1. General

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Permittee shall be out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

7.2. Multiple Sample Data

When determining compliance with a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses and the data set contains one or more reported determinations of DNQ or ND. In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure:

- 7.2.1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
- 7.2.2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

7.3. Average Monthly Effluent Limitation (AMEL)

If the average (or when applicable, the median determined by section 7.2 above for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation for the purpose of calculating mandatory minimum penalties, though the Permittee may be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month) in cases where discretionary administrative civil liabilities are appropriate. If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Permittee may be considered out of compliance for that calendar month. The Permittee will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which

no sample (daily discharge) is taken, no compliance determination can be made for that calendar month with respect to the AMEL.

If the analytical result of a single sample, monitored monthly, quarterly, semiannually, or annually, does not exceed the AMEL for a given parameter, the Permittee will have demonstrated compliance with the AMEL for each day of that month for that parameter.

If the analytical result of any single sample, monitored monthly, quarterly, semiannually, or annually, exceeds the AMEL for any parameter, the Permittee may collect up to four additional samples within the same calendar month. All analytical results shall be reported in the monitoring report for that month. The concentration of pollutant (an arithmetic mean or a median) in these samples estimated from the "Multiple Sample Data Reduction" section above, will be used for compliance determination.

In the event of noncompliance with an AMEL, the sampling frequency for that parameter shall be increased to weekly and shall continue at this level until compliance with the AMEL has been demonstrated.

7.4. Average Weekly Effluent Limitation (AWEL)

If the average of daily discharges over a calendar week exceeds the AWEL for a given parameter, an alleged violation will be flagged and the Permittee will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. The average of daily discharges over the calendar week that exceeds the AWEL for a parameter will be considered out of compliance for that week only. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Permittee will be considered out of compliance for that calendar week. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week with respect to the AWEL.

A calendar week will begin on Sunday and end on Saturday. Partial calendar weeks at the end of calendar month will be carried forward to the next month in order to calculate and report a consecutive seven-day average value on Saturday.

7.5. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge on a calendar day exceeds the MDEL for a given parameter, an alleged violation will be flagged, and the Permittee will be considered out of compliance for that day for that parameter. If no sample (daily discharge) is taken over a calendar day, no compliance determination can be made for that day with respect to effluent violation determination, but compliance determination can be made for that day with respect to reporting violation determination.

7.6. Instantaneous Minimum Effluent Limitation

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, a potential violation will be flagged, and the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

7.7. Instantaneous Maximum Effluent Limitation

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, a potential violation will be flagged, and the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

7.8. Six-month Median Effluent Limitation

If the median of daily discharges over any 180-day period exceeds the six-month median effluent limitation for a given parameter, a potential violation will be flagged, and the Permittee will be considered out of compliance for each day of that 180-day period for that parameter. The next assessment of compliance will occur after the next sample is taken. If only a single sample is taken during a given 180-day period and the analytical result for that sample exceeds the six-month median, the Permittee will be considered out of compliance for the 180-day period. For any 180-period during which no sample is taken, no compliance determination can be made for the six-month median effluent limitation.

7.9. Median Monthly Effluent Limitation (MMEL)

If the median of daily discharges over a calendar month exceeds the MMEL for a given parameter, an alleged violation will be flagged and the Permittee will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). However, an alleged violation of the MMEL will be considered one violation for the purpose of assessing State mandatory minimum penalties. If no sample (daily discharge) is taken over a calendar month, no compliance determination can be made for that month with respect to effluent violation determination, but compliance determination can be made for that month with respect to reporting violation determination.

7.10. Chronic Toxicity

The discharge is subject to determination of "Pass" or "Fail" and "Percent Effect" from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t-test approach described in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010), Appendix A, Figure A-1, Table A-1, and Appendix B, Table B-1. The null hypothesis (H_0) for the TST statistical approach is: Mean ambient water response $\leq 0.75 \times$ Mean control response. A test result that rejects this null hypothesis is reported as "Pass." A test result that does not reject this null hypothesis is reported as "Fail." The relative "Percent Effect" at the discharge IWC is defined and reported as: $((\text{Mean control response} - \text{Mean discharge IWC response}) \div \text{Mean control response}) \times 100$. This is a t-test, a statistical analysis comparing two sets of replicate observations - in the case of Whole Effluent Toxicity (WET), only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is "Pass" or "Fail")). The Welch's t-test employed by the TST statistical approach is an adaptation of Student's t-test and is used with two samples having unequal variances.

The MDEL for chronic toxicity is exceeded and a violation will be flagged when a chronic toxicity test, analyzed at the IWC for the reproduction endpoint using the TST statistical approach, results in “Fail” and the “Percent Effect” of the survival endpoint is $\geq 50\%$.

The MMEL for chronic toxicity is exceeded and a violation will be flagged when the median of no more than three independent chronic toxicity tests, initiated in a calendar month and analyzed using the TST statistical approach result in “Fail” for any endpoint.

If a chronic aquatic toxicity routine monitoring test results in a “Fail” at the IWC, the Permittee may complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the “Fail” at the IWC. If the first chronic MMEL compliance test results in a “Fail” at the IWC, then the second MMEL compliance test is not necessary because the “Fail” results from the first two tests would constitute a violation of the chronic toxicity MMEL.

The chronic toxicity MDEL and MMEL are set at the IWC for the discharge (100% effluent) and expressed in units of the TST statistical approach (“Pass” or “Fail”, “Percent Effect”). All NPDES effluent compliance monitoring for the chronic toxicity MDEL and MMEL shall be reported using only the 100% effluent concentration and negative control, expressed in units of the TST, using the *Ceriodaphnia dubia*, which was determined to be the most sensitive species for the Valencia WRP discharge. The TST hypothesis (H_0) (see above) is statistically analyzed using the IWC and a negative control. Effluent toxicity tests shall be run using a multi-concentration test design when required by Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms (USEPA 2002, EPA-821-R-02-013). However, if the USEPA approves the Alternative Test Procedure, the Discharger may use a two-concentration test design. The Los Angeles Water Board’s review of reported toxicity test results will not include review of concentration-response patterns as appropriate (see Fact Sheet discussion at 4.3.6). As described in the bioassay laboratory audit correspondence from the State Water Resources Control Board dated August 7, 2014, and from the USEPA dated December 24, 2013, the Percent Minimum Significant Difference (PMSD) criteria only apply to compliance reporting for the No Observable Effect Concentration (NOEC) and the sublethal statistical endpoints of the NOEC, and therefore are not used to interpret results using the TST statistical approach. Standard Operating Procedures used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent (and receiving water) toxicity test measurement results from the TST statistical approach must be submitted to the Los Angeles Water Board (40 CFR section 122.41(h)). The Los Angeles Water Board will make a final determination as to whether a toxicity test result is valid, and may consult with the Permittee, the USEPA, the State Water Board’s Quality Assurance Officer, or the State Water Board’s Environmental Laboratory Accreditation Program (ELAP) as needed. The Board may consider the results of any TIE/TRE studies in an enforcement action.

7.11. Percent Removal

The average monthly percent removal is the removal efficiency expressed in percentage across a treatment plant for a given pollutant parameter, as determined from the 30-day average values of pollutant concentrations (C in mg/L) of influent and effluent samples collected at about the same time using the following equation:

$$\text{Percent Removal (\%)} = [1 - (C_{\text{Effluent}}/C_{\text{Influent}})] \times 100\%$$

When preferred, the Permittee may substitute mass loadings and mass emissions for the concentrations.

7.12. Mass and Concentration Limitations

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding mass emission rate determined from that sample concentration shall also be reported as ND or DNQ.

7.13. Compliance with Single Constituent Effluent Limitations

Permittees may be considered out of compliance with the effluent limitation if the concentration of the pollutant (see section 7.2 “Multiple Sample Data Reduction” above) in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

7.14. Compliance with Effluent Limitations Expressed as a Sum of Several Constituents

Permittees are out of compliance with an effluent limitation which applies to the sum of a group of chemicals (e.g., PCBs) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

7.15. Compliance with 2,3,7,8-TCDD and its Equivalents

Compliance with the dioxin effluent limitation shall be determined based on 2,3,7,8-TCDD alone. However, TCDD equivalents shall be monitored and calculated using the following formula, where the MLs, and toxicity equivalency factors (TEFs) are as provided in the table below. The Permittee shall report all measured values of individual congeners, including data qualifiers. When calculating TCDD equivalents, the Permittee shall set congener concentrations below the minimum levels to zero. USEPA method 1613 may be used to analyze dioxin and furan congeners.

$$\text{Dioxin Concentration} = \sum_{i=1}^{17} (TEQi) = \sum_{i=1}^{17} (Ci)(TEFi)$$

where:

Ci = individual concentration of a dioxin or furan congener

TEFi = individual TEF for a congener

MLs and TEFs

Congeners	MLs (pg/L)	TEFs
2,3,7,8-TetraCDD	10	1.0
1,2,3,7,8-PentaCDD	50	1.0

Congeners	MLs (pg/L)	TEFs
1,2,3,4,7,8-HexaCDD	50	0.1
1,2,3,6,7,8-HexaCDD	50	0.1
1,2,3,7,8,9-HexaCDD	50	0.1
1,2,3,4,6,7,8-HeptaCDD	50	0.01
OctaCDD	100	0.0001
2,3,7,8-TetraCDF	10	0.1
1,2,3,7,8-PentaCDF	50	0.05
2,3,4,7,8-PentaCDF	50	0.5
1,2,3,4,7,8-HexaCDF	50	0.1
1,2,3,6,7,8-HexaCDF	50	0.1
1,2,3,7,8,9-HexaCDF	50	0.1
2,3,4,6,7,8-HexaCDF	50	0.1
1,2,3,4,6,7,8-HeptaCDFs	50	0.01
1,2,3,4,7,8,9-HeptaCDFs	50	0.01
OctaCDF	100	0.0001

7.16. Compliance with Gross Beta/photon Emitters

The monthly average effluent limitation for gross beta/photon is equal to 4 millirem/year with a screening level of 50 picoCuries per liter (pCi/L). Due to naturally occurring Potassium-40, the results of the Potassium-40 may be subtracted from the total gross beta activity to determine if the screening level is exceeded. The Potassium-40 beta particle activity must be calculated by multiplying elemental potassium concentration (in mg/L) by a factor of 0.82 to determine activity from Potassium-40. The Potassium-40 must be analyzed from the same or equivalent sample used for the gross beta analysis.

If the gross beta particle activity minus the naturally occurring Potassium-40 is less than or equal to 50 pCi/L, the facility is in compliance and the value shall be reported as <4 millirem/year. If the gross beta particle activity minus the naturally occurring Potassium-40 beta particle activity exceeds the screening level, the Discharger must have the samples further analyzed for the *individual* nuclides. The calculation for the sum of the fractions is presented below.

The maximum contaminant level (MCL) for gross beta/photon emitters is equal 4 millirem per year. A millirem is a dose of energy to the body or any internal organ. USEPA regulates 179 man-made nuclides, and each of them has a concentration of radiation measured in pCi/L, which produces the 4 millirem dose. These concentrations are listed on table, *Derived Concentrations of (pCi/L) of Beta and Photon Emitters in Drinking Water*, which shall be used to determine compliance.

Derived Concentrations (pCi/l) of Beta and Photon Emitters in Drinking Water

Yielding a Dose of 4 mrem/yr to the Total Body or to any Critical Organ as defined in NBS Handbook 69

Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l
H-3	20,000	Ni-65	300	Nb-95	300	Sb-124	60	Nd-147	200	Os-191	600
Be-7	6,000	Cu-64	900	Nb-97	3,000	Sb-125	300	Nd-149	900	Os-191m	9,000
C-14	2,000	Zn-65	300	Mo-99	600	Te-125m	600	Pm-147	600	Os-193	200
F-18	2,000	Zn-69	6,000	Tc-96	300	Te-127	900	Pm-149	100	Ir-190	600
Na-22	400	Zn-69m	200	Tc-96m	30,000	Te-127m	200	Sm-151	1,000	Ir-192	100
Na-24	600	Ga-72	100	Tc-97	6,000	Te-129	2,000	Sm-153	200	Ir-194	90
Si-31	3,000	Ge-71	6,000	Tc-97m	1,000	Te-129m	90	Eu-152	200	Pt-191	300
P-32	30	As-73	1,000	Tc-99	900	Te-131m	200	Eu-154	60	Pt-193	3,000
S-35 inorg	500	As-74	100	Tc-99m	20,000	Te-132	90	Eu-155	600	Pt-193m	3,000
Cl-36	700	As-76	60	Ru-97	1,000	I-126	3	Gd-153	600	Pt-197	300
Cl-38	1,000	As-77	200	Ru-103	200	I-129	1	Gd-159	200	Pt-197m	3,000
K-42	900	Se-75	900	Ru-105	200	I-131	3	Tb-160	100	Au-196	600
Ca-45	10	Br-82	100	Ru-106	30	I-132	90	Dy-165	1,000	Au-198	100
Ca-47	80	Rb-86	600	Rh-103m	30,000	I-133	10	Dy-166	100	Au-199	600
Sc-46	100	Rb-87	300	Rh-105	300	I-134	100	Ho-166	90	Hg-197	900
Sc-47	300	Sr-85m	20,000	Pd-103	900	I-135	30	Er-169	300	Hg-197m	600
Sc-48	80	Sr-85	900	Pd-109	300	Cs-131	20,000	Er-171	300	Hg-203	60
V-48	90	Sr-89	20	Ag-105	300	Cs-134	80	Tm-170	100	Tl-200	1,000
Cr-51	6,000	Sr-90	8	Ag-110m	90	Cs-134m	20,000	Tm-171	1,000	Tl-201	900
Mn-52	90	Sr-91	200	Ag-111	100	Cs-135	900	Yb-175	300	Tl-202	300
Mn-54	300	Sr-92	200	Cd-109	600	Cs-136	800	Lu-177	300	Tl-204	300
Mn-56	300	Y-90	60	Cd-115	90	Cs-137	200	Hf-181	200	Pb-203	1,000
Fe-55	2,000	Y-91	90	Cd-115m	90	Ba-131	600	Ta-182	100	Bi-206	100
Fe-59	200	Y-91m	9,000	In-113m	3,000	Ba-140	90	W-181	1,000	Bi-207	200
Co-57	1,000	Y-92	200	In-114m	60	La-140	60	W-185	300	Pa-230	600
Co-58	300	Y-93	90	In-115	300	Ce-141	300	W-187	200	Pa-233	300
Co-58m	9000	Zr-93	2,000	In-115m	1,000	Ce-143	100	Re-186	300	Np-239	300
Co-60	100	Zr-95	200	Sn-113	300	Ce-144	30	Re-187	9,000	Pu-241	300
Ni-59	300	Zr-97	60	Sn-125	60	Pr-142	90	Re-188	200	Bk-249	2,000
Ni-63	50	Nb-93m	1,000	Sb-122	90	Pr-143	100	Os-185	200		

The sum of the fraction method is used because each photon emitter targets a different organ of the body, which results in a different magnitude of risk. The sum of the beta and photon emitters shall not exceed 4 millirem/year (40 CFR section 141.66(d)(2)).

Each nuclide has a different concentration that produces 4 millirem dose because different radionuclides have different energy levels. Some nuclides need to be in a higher concentration to give the same 4 millirem dose.

The laboratory shall measure the nuclide concentration in the water and compare this result to the concentration allowed for that particular nuclide (see table below). The comparison results in a fraction. This is shown in calculation below:

Fraction of the maximum

$$4 \text{ millirem/year exposure limit} = \frac{\text{pCi/L found in sample (from laboratory results)}}{\text{pCi/L equivalent from 4 millirem of exposure (from conversion table)}}$$

Each fraction must then be converted to a dose equivalent of 4 millirem/year by multiplying the fraction by 4. The results for each emitter must be summed to determine compliance.

A sample calculation is presented in the table below:

---	X	Y	X/Y	4(X/Y)
Emitter	Lab Analysis (pCi/L)	Conversion from table (pCi/4millirem)	Calculate Fraction	Calculate Total (millirem)
Cs-134	5,023	20,000	0.25115	1.0

---	X	Y	X/Y	4(X/Y)
Emitter	Lab Analysis (pCi/L)	Conversion from table (pCi/4millirem)	Calculate Fraction	Calculate Total (millirem)
Cs-137	30	200	0.150	0.6
Sr-90	4	8	0.5	2.0
I-131	2	3	0.7	2.8
Sum of the Fractions	---	---	1.60115	6.4

In the example above, the system would be considered in violation of the gross beta/photon effluent limitation because the "sum-of-the-fractions" is 6.4 millirem, which means that the sum of the annual dose equivalent to the total body, or to any internal organ, exceeds 4 millirem/year.

7.17. Mass Emission Rate

The mass emission rate shall be obtained from the following calculation for any calendar day:

$$\text{Mass emission rate (lb/day)} = \frac{8.34}{N} \sum_{i=1}^N Q_i C_i$$

$$\text{Mass emission rate (kg/day)} = \frac{3.79}{N} \sum_{i=1}^N Q_i C_i$$

in which 'N' is the number of samples analyzed in any calendar day. 'Q_i' and 'C_i' are the flow rate (MGD) and the constituent concentration (mg/L), respectively, which are associated with each of the 'N' grab samples, which may be taken in any calendar day. If a composite sample is taken, 'C_i' is the concentration measured in the composite sample and 'Q_i' is the average flow rate occurring during the period over which samples are composited.

The daily concentration of all constituents shall be determined from the flow-weighted average of the same constituents in the combined waste streams as follows:

$$\text{Daily concentration} = \frac{1}{Q_t} \sum_{i=1}^N Q_i C_i$$

in which 'N' is the number of component waste streams. 'Q_i' and 'C_i' are the flow rate (MGD) and the constituent concentration (mg/L), respectively, which are associated with each of the 'N' waste streams. 'Q_t' is the total flow rate of the combined waste streams.

7.18. Bacterial Standards and Analysis

7.18.1. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$$

where n is the number of days samples were collected during the period and C is the concentration of bacteria (MPN/100 mL or CFU/100 mL) found on each day of sampling.

7.18.2. For bacterial analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total and fecal coliform, at a minimum, and 1 to 1000 per 100 ml for enterococcus). The detection methods used for each analysis shall be reported with the results of the analyses.

7.18.3. Detection methods used for total coliform shall be those presented in Table 1A of 40 CFR part 136, unless alternate methods have been approved by USEPA pursuant to 40 CFR part 136, or improved methods have been determined by the Executive Officer and/or USEPA.

7.18.4. Detection methods used for enterococcus shall be those presented in Table 1A of 40 CFR part 136 or in the USEPA publication EPA 600/4-85/076, Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure or any improved method determined by the Executive Officer and/or USEPA to be appropriate.

7.19. Single Operational Upset (SOU)

A SOU that leads to simultaneous violations of more than one pollutant parameter shall be treated as a single violation and limits the Permittee's liability in accordance with the following conditions:

7.19.1. A SOU is broadly defined as a single unusual event that temporarily disrupts the usually satisfactory operation of a system in such a way that it results in violation of multiple pollutant parameters.

7.19.2. A Permittee may assert SOU to limit liability only for those violations which the Permittee submitted notice of the upset as required in Provision 5.5.2(b) of Attachment D – Standard Provisions.

7.19.3. For purpose outside of Water Code section 13385 subdivisions (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Permittees to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with USEPA Memorandum "Issuance of Guidance Interpreting Single Operational Upset" (September 27, 1989).

7.19.4. For purpose of Water Code section 13385 (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Permittees to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with Water Code section 13385 (f)(2).

ATTACHMENT A. DEFINITIONS

Arithmetic Mean (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean = $\mu = \sum x/n$ Where: $\sum x$ is the sum of the measured ambient water concentrations, and n is the number of samples.

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Bioaccumulative

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Biosolids

Biosolids refer to sewage sludge that has been treated and tested and shown to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agricultural, silvicultural, horticultural, and land reclamation activities as specified under 40 CFR part 503.

Carcinogenic

Carcinogenic pollutants are substances that are known to cause cancer in living organisms.

Coefficient of Variation (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Detected, but Not Quantified (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

Dilution Credit

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Effluent Concentration Allowance (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document for Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Enclosed Bays

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

Estimated Chemical Concentration

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Estuaries

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

Inland Surface Waters

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

Maximum Daily Flow

The maximum daily flow means the maximum instantaneous flow of the day.

Median Monthly Effluent Limitation (MMEL)

For the purposes of chronic aquatic toxicity, MMEL is an effluent limitation based on a maximum of three independent toxicity tests, analyzed using the TST.

Median

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL)

MDL is defined as the minimum measured concentration of a substance that can be reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR part 136, Attachment B, revised as of July 3, 1999.

Minimum Level (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Mixing Zone

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

Not Detected (ND)

Sample results which are less than the laboratory's MDL.

PCBs (polychlorinated biphenyls) as Aroclors

The sum of chlorinated biphenyls whose analytical characteristics resemble those of Aroclor-1016, Aroclor-1221, Aroclor-1232, Aroclor-1242, Aroclor-1248, Aroclor-1254, and Aroclor-1260.

PCBs as Congeners

The sum of the following 41 individually quantified PCB congeners or mixtures of isomers of a single congeners in a co-elution: PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206.

Persistent Pollutants

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

Pollutant Minimization Program (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Los Angeles Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

Pollution Prevention

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Los Angeles Water Board.

Reporting Level (RL)

The RL is the ML (and its associated analytical method) chosen by the Permittee for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Los Angeles Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Source of Drinking Water

Any water designated as municipal or domestic supply (MUN) in a Los Angeles Water Board Basin Plan.

Standard Deviation (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

x is the observed value;

μ is the arithmetic mean of the observed values; and

n is the number of samples.

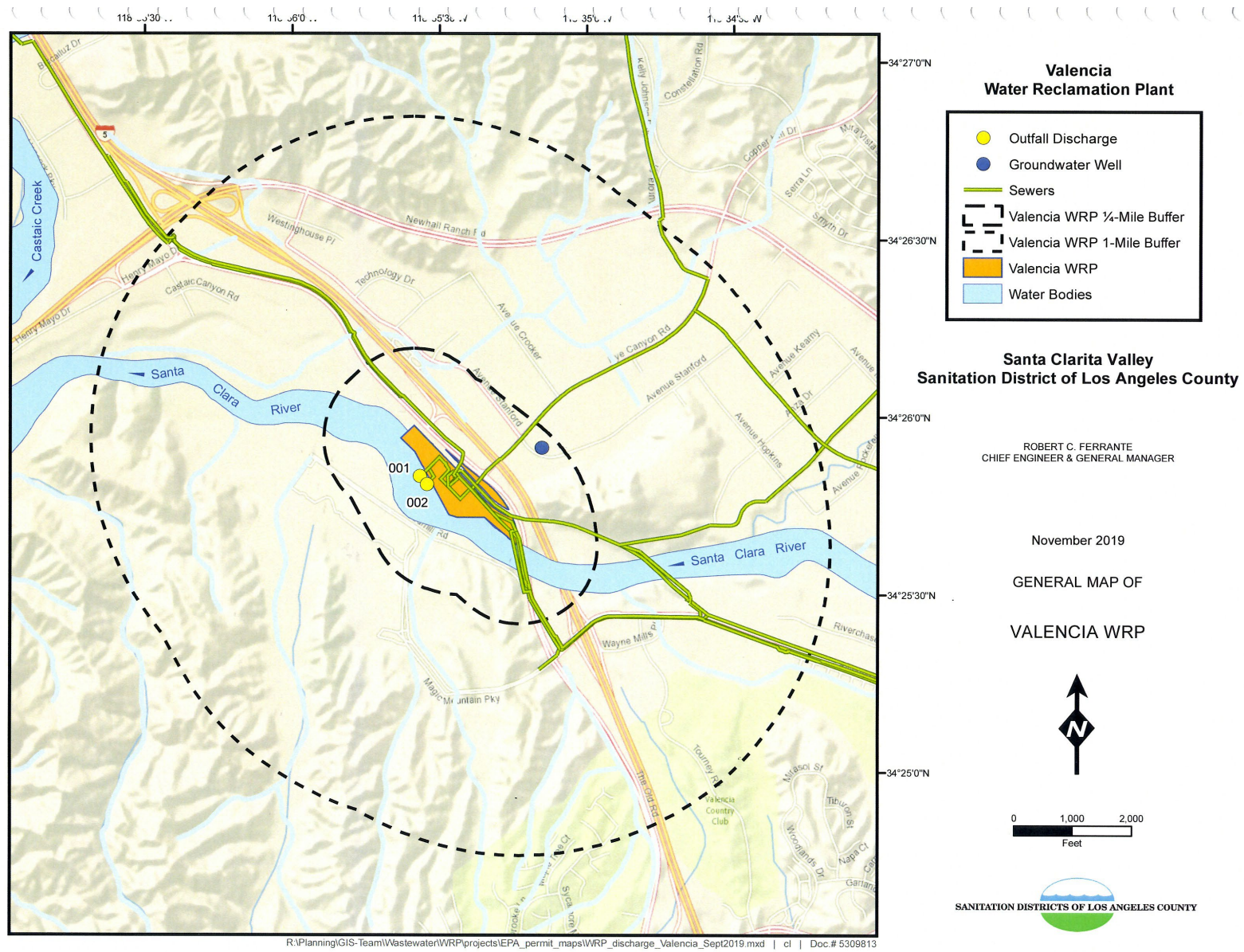
Statistical Threshold Value (STV)

The STV for the bacteria water quality objectives is a set value that approximates the 90th percentile of the water quality distribution of a bacterial population.

Toxicity Reduction Evaluation (TRE)

TRE is a study conducted in a stepwise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

ATTACHMENT B. MAPS



ATTACHMENT C. FLOW SCHEMATIC

DIAGRAM C1a

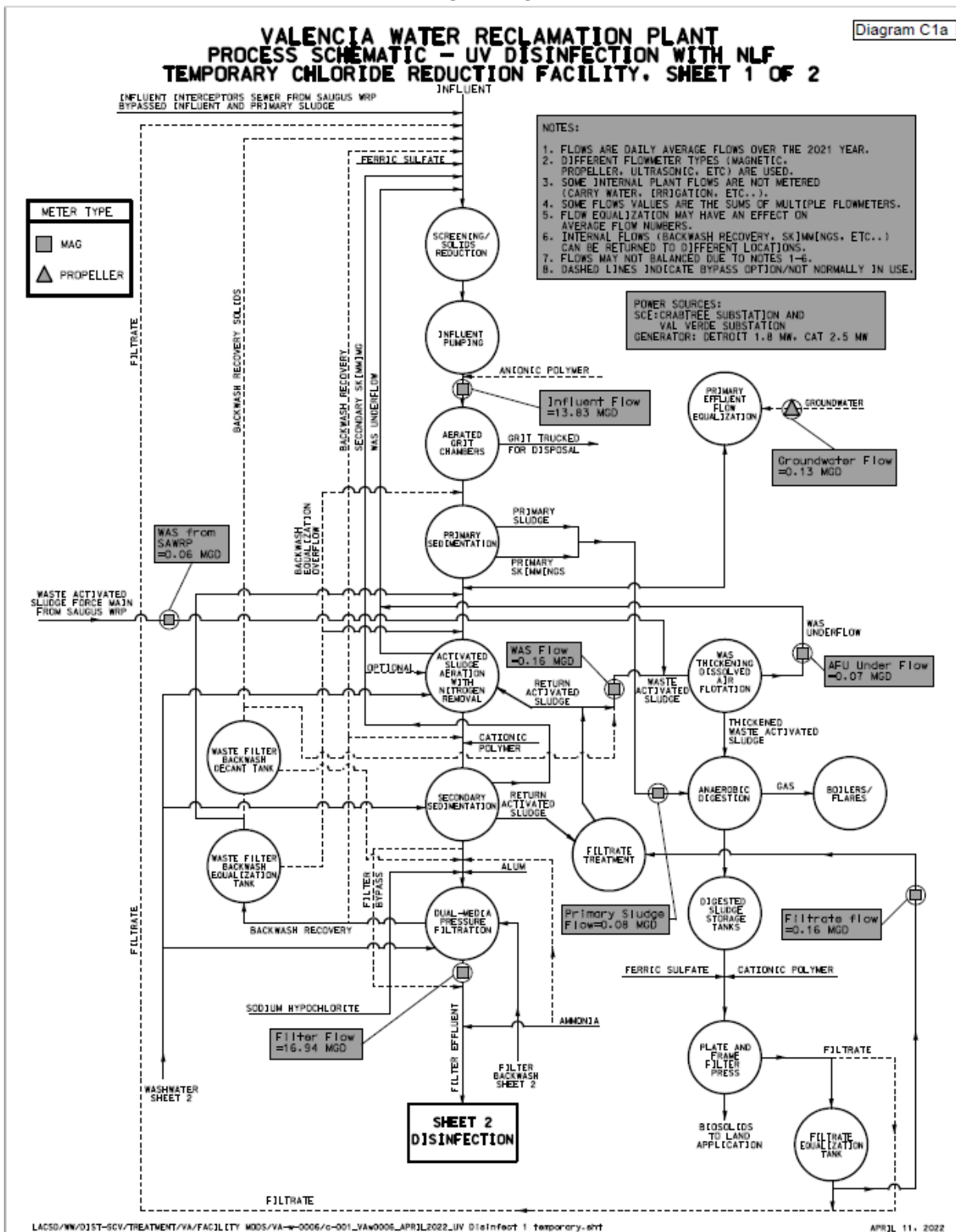
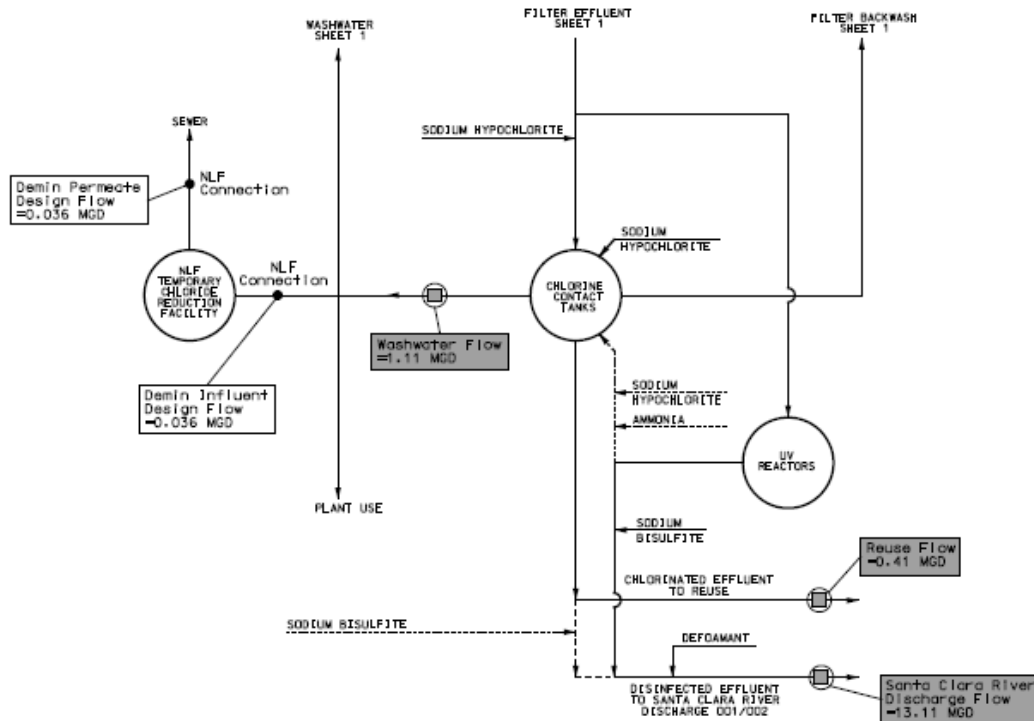


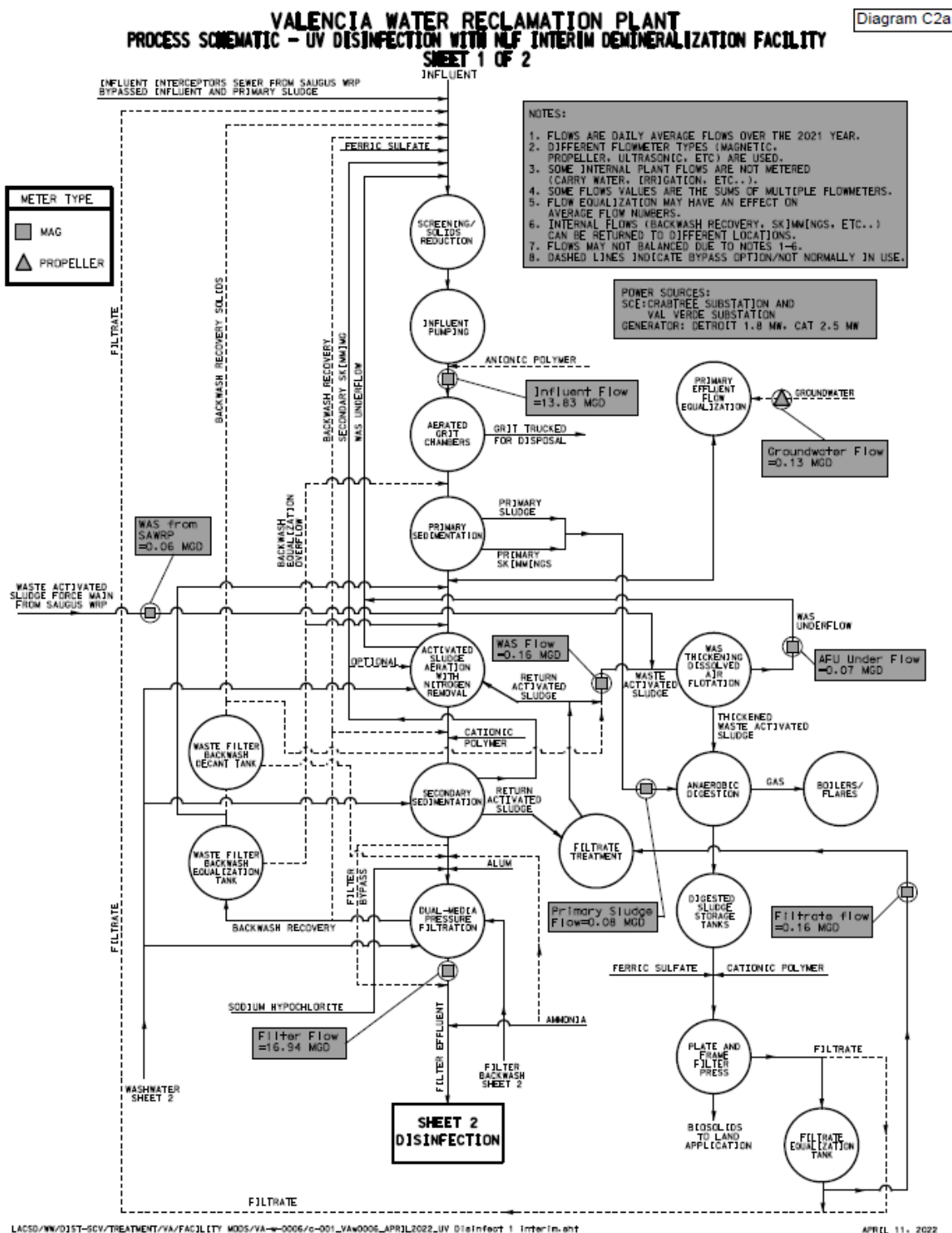
DIAGRAM C1b

VALENCIA WATER RECLAMATION PLANT
PROCESS SCHEMATIC – UV DISINFECTION WITH NLF
TEMPORARY CHLORIDE REDUCTION FACILITY, SHEET 2 OF 2

Diagram C1b



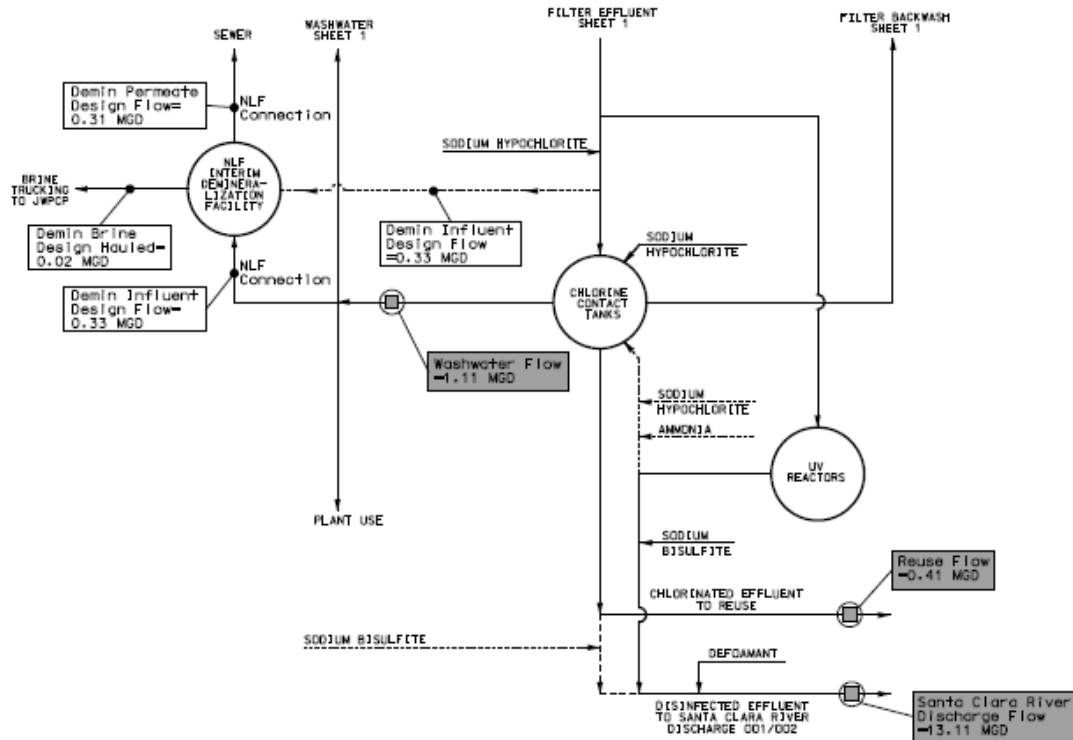
ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C2a



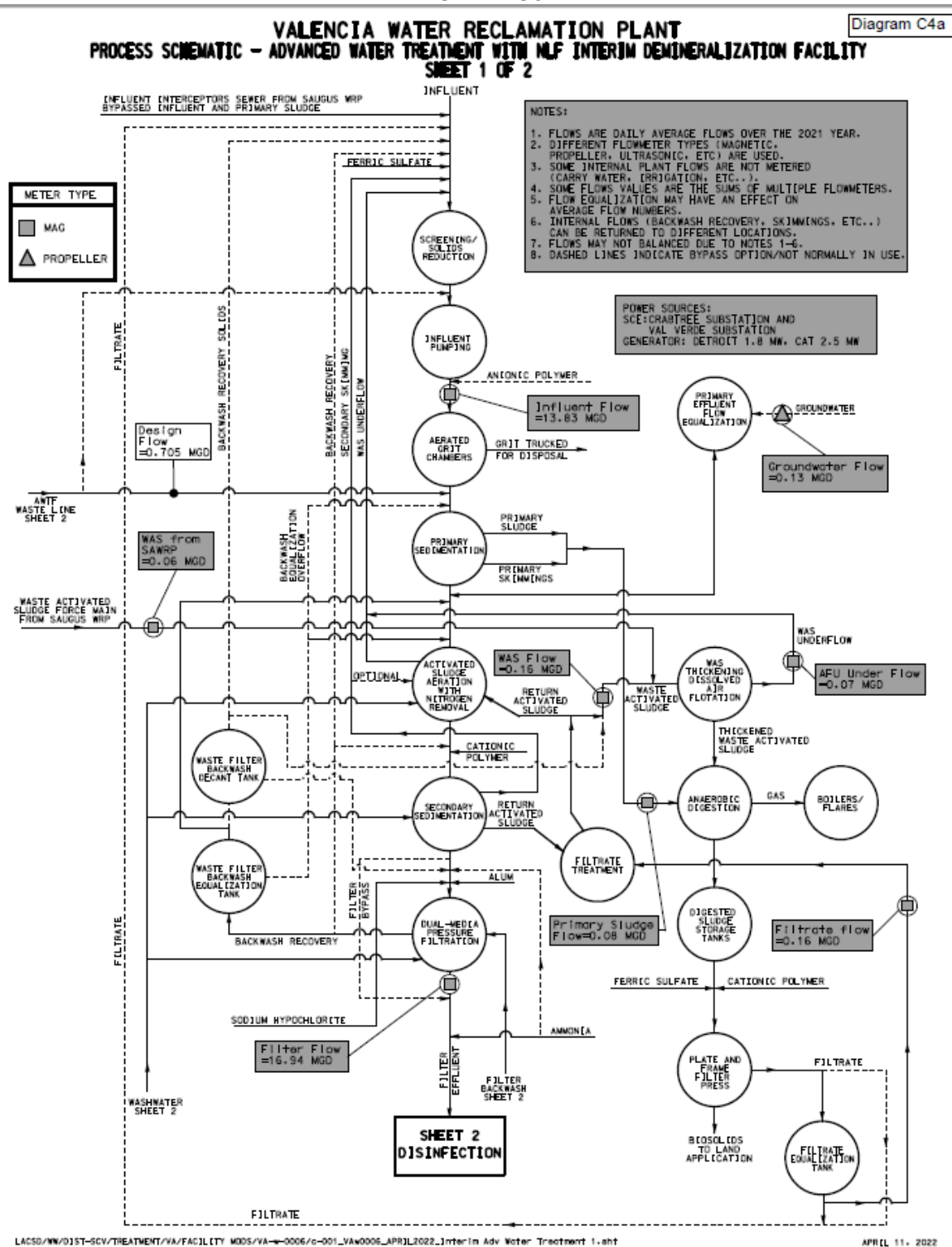
ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C2b

VALENCIA WATER RECLAMATION PLANT
PROCESS SCHEMATIC – UV DISINFECTION WITH NLF INTERIM DEMINERALIZATION FACILITY
SHEET 2 OF 2

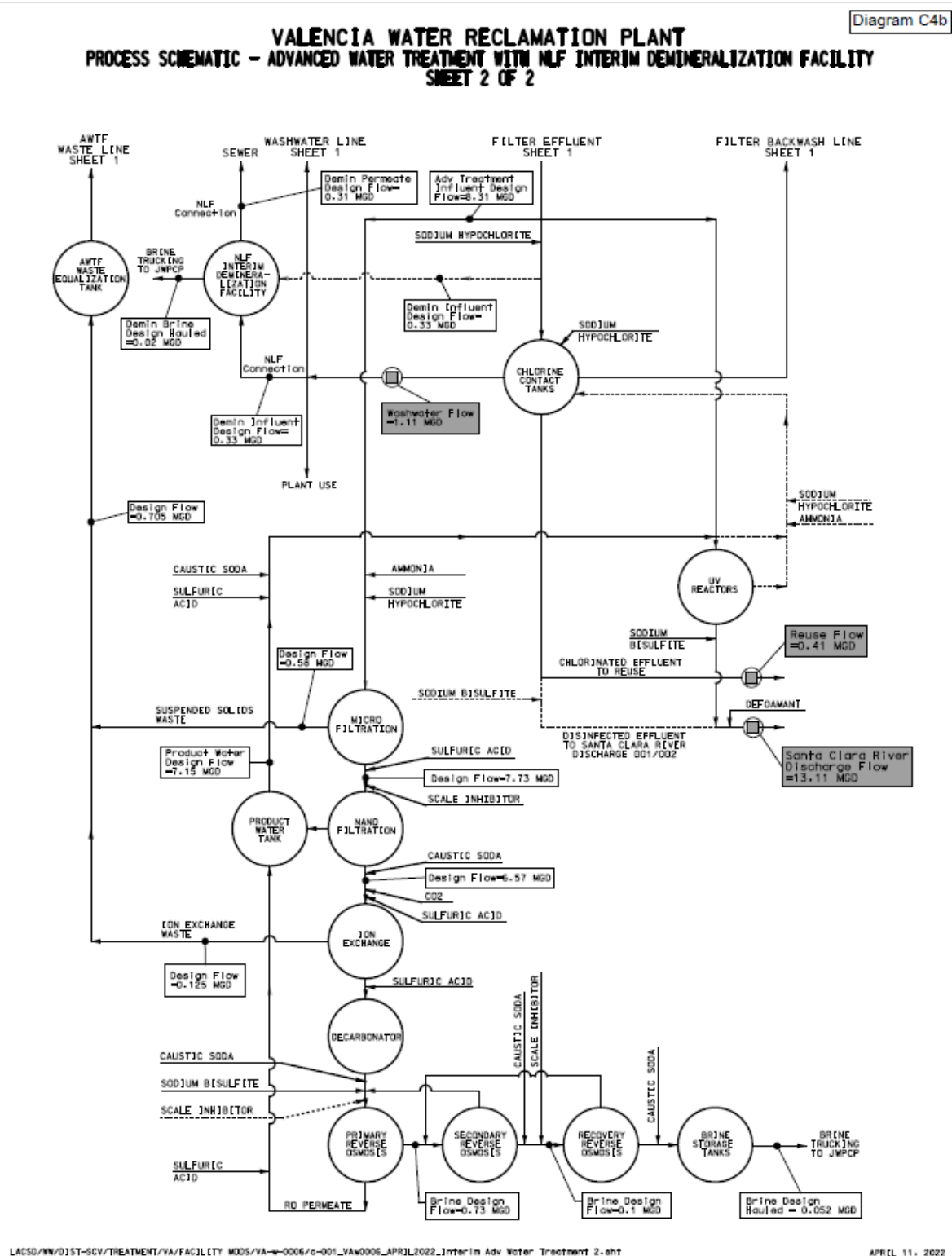
Diagram C2b



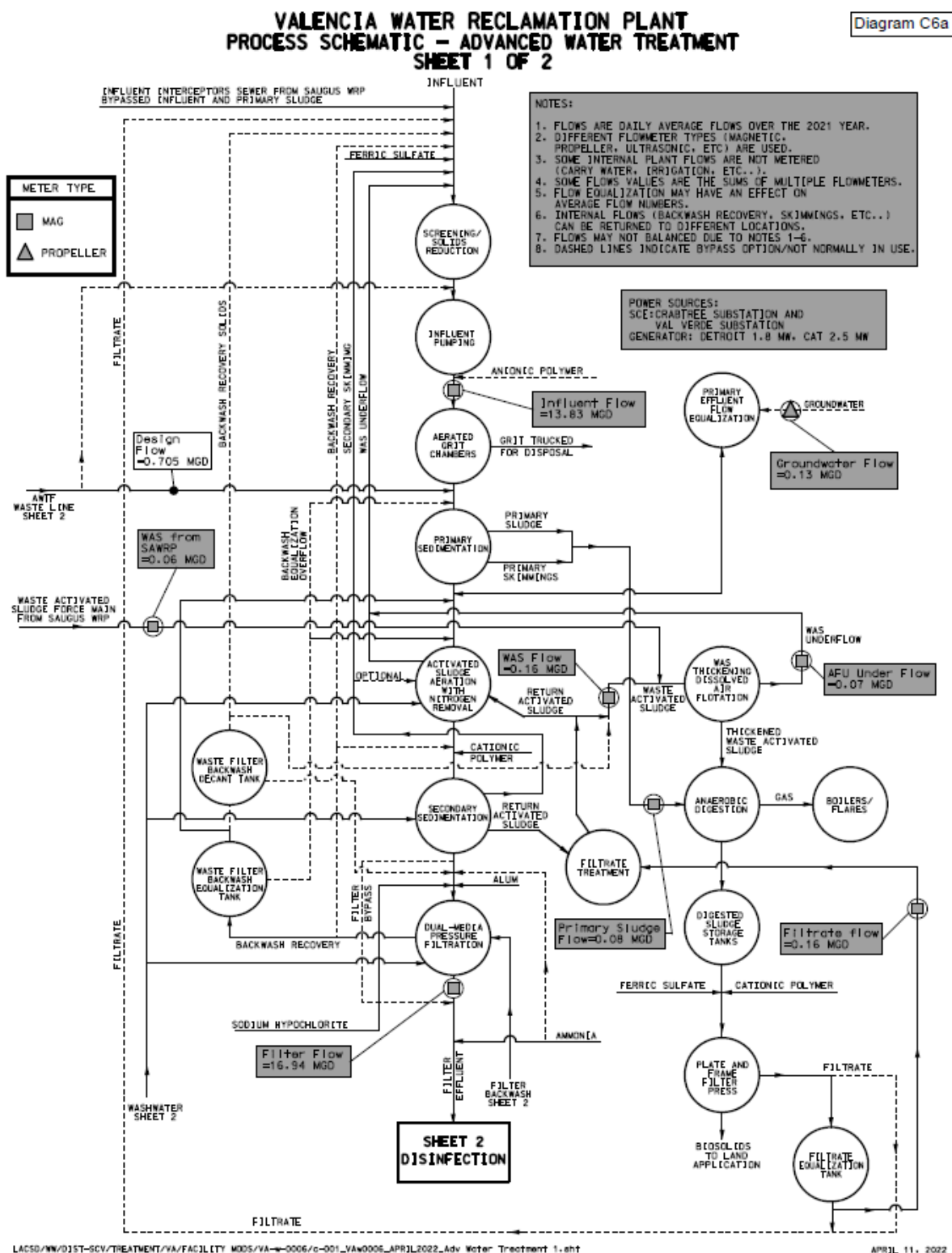
ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C3a



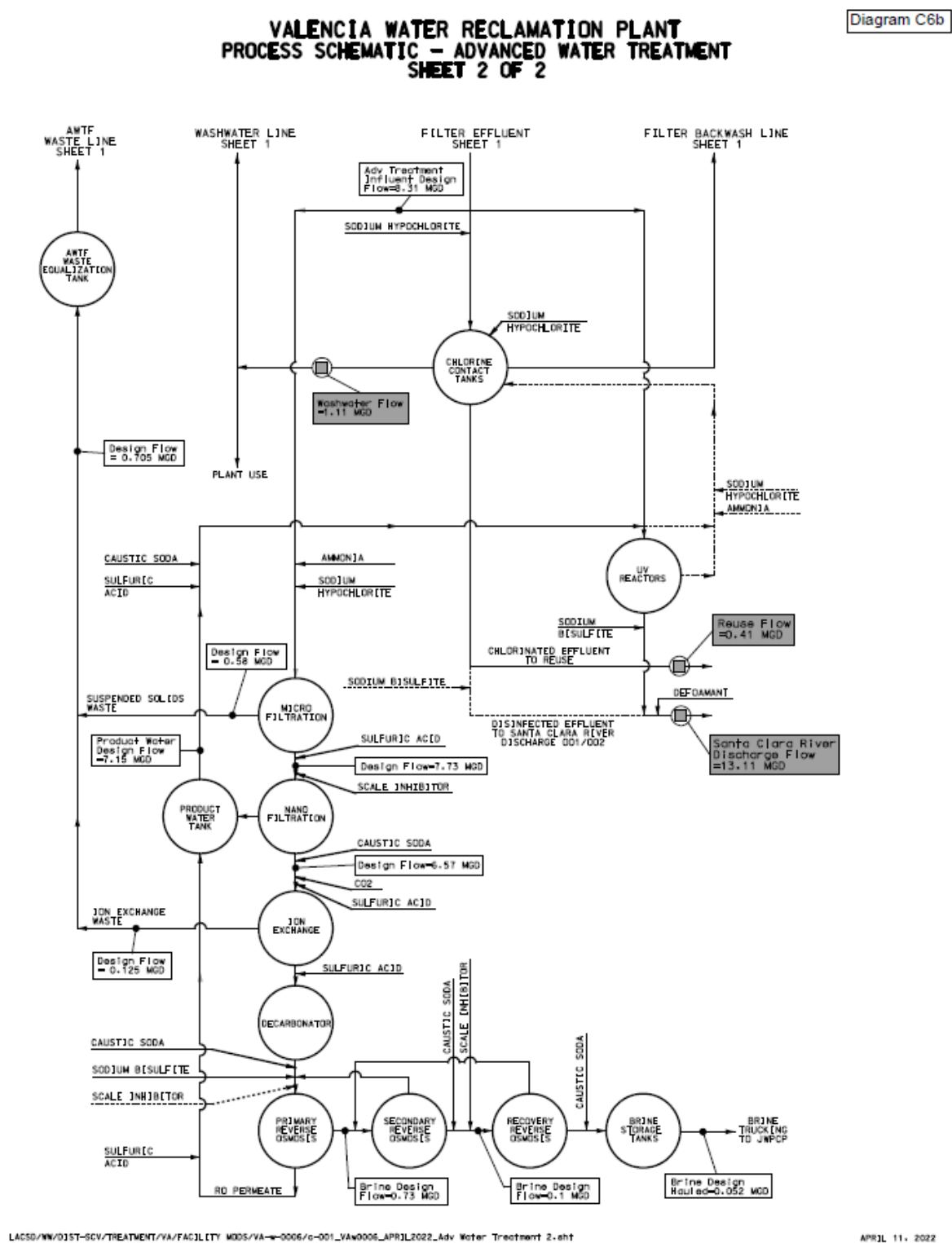
ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C3b



ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C4a



ATTACHMENT C – FLOW SCHEMATIC
DIAGRAM C4b



ATTACHMENT D. STANDARD PROVISIONS

1. STANDARD PROVISIONS – PERMIT COMPLIANCE

1.1. Duty to Comply

- 1.1.1. The Permittee must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA), its regulations, and the California Water Code (Water Code) and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 CFR section 122.41(a); California Water Code (Water Code) sections 13261, 13263, 13264, 13265, 13268, 13000, 13001, 13304, 13350, 13385)
- 1.1.2. The Permittee shall comply with effluent standards or prohibitions established under part 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (title 40 of the Code of Federal Regulations (40 CFR) section 122.41(a)(1))

1.2. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR section 122.41(c).)

1.3. Duty to Mitigate

The Permittee shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR section 122.41(d).)

1.4. Proper Operation and Maintenance

The Permittee shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Permittee to achieve compliance with the conditions of this Order. Proper operation and maintenance also include adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Permittee only when necessary to achieve compliance with the conditions of this Order. (40 CFR section 122.41(e).)

1.5. Property Rights

- 1.5.1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR section 122.41(g).)
- 1.5.2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 CFR section 122.5(c).)

1.6. Inspection and Entry

The Permittee shall allow the Los Angeles Water Board, State Water Board, USEPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. section 1318(a)(B); 40 CFR section 122.41(i); Water Code sections 13267 and 13383):

- 1.6.1. Enter upon the Permittee's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. section 1318(a)(B)(i); 40 CFR section 122.41(i)(1); Water Code sections 13267 and 13383);
- 1.6.2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. section 1318(a)(B)(ii); 40 CFR section 122.41(i)(2); Water Code sections 13267 and 13383);
- 1.6.3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. section 1318(a)(B)(ii); 40 CFR section 122.41(i)(3); Water Code sections 13267 and 13383); and
- 1.6.4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C. section 1318(a)(B); 40 CFR section 122.41(i)(4); Water Code sections 13267 and 13383)

1.7. Bypass

1.7.1. Definitions

- a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR section 122.41(m)(1)(i).)
- b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR section 122.41(m)(1)(ii).)

1.7.2. Bypass not exceeding limitations. The Permittee may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance 1.7.3, 1.7.4, and 1.7.5 below. (40 CFR section 122.41(m)(2).)

1.7.3. Prohibition of bypass. Bypass is prohibited, and the Los Angeles Water Board may take enforcement action against a Permittee for bypass, unless (40 CFR section 122.41(m)(4)(i)):

- a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR section 122.41(m)(4)(i)(A));

- b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR section 122.41(m)(4)(i)(B)); and
 - c. The Permittee submitted notice to the Los Angeles Water Board as required under Standard Provisions – Permit Compliance 1.7.5 below. (40 CFR section 122.41(m)(4)(i)(C).)
- 1.7.4. The Los Angeles Water Board may approve an anticipated bypass, after considering its adverse effects, if the Los Angeles Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance 1.7.3 above. (40 CFR section 122.41(m)(4)(ii).)
- 1.7.5. Notice
- a. Anticipated bypass. If the Permittee knows in advance of the need for a bypass, it shall submit a notice, if possible, at least 10 days before the date of the bypass. As of December 21, 2020, all notices must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10 below. Notices shall comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR section 127. (40 CFR section 122.41(m)(3)(i).)
 - b. Unanticipated bypass. The Permittee shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting 5.5 below (24-hour notice). As of December 21, 2020, all notices must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10 below. Notices shall comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR section 127. (40 CFR section 122.41(m)(3)(ii).)

1.8. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the Permittee. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR section 122.41(n)(1).)

1.8.1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology-based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance 1.8.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR section 122.41(n)(2).)

1.8.2. Conditions necessary for a demonstration of upset. A Permittee who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 CFR section 122.41(n)(3)):

- a. An upset occurred and that the Permittee can identify the cause(s) of the upset (40 CFR section 122.41(n)(3)(i));
- b. The permitted facility was, at the time, being properly operated (40 CFR section 122.41(n)(3)(ii));
- c. The Permittee submitted notice of the upset as required in Standard Provisions – Reporting 5.5.2.b below (24-hour notice) (40 CFR section 122.41(n)(3)(iii)); and
- d. The Permittee complied with any remedial measures required under Standard Provisions – Permit Compliance 1.3 above. (40 CFR section 122.41(n)(3)(iv).)

1.8.3. Burden of proof. In any enforcement proceeding, the Permittee seeking to establish the occurrence of an upset has the burden of proof. (40 CFR section 122.41(n)(4).)

2. STANDARD PROVISIONS – PERMIT ACTION

2.1. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Permittee for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR section 122.41(f).)

2.2. Duty to Reapply

If the Permittee wishes to continue an activity regulated by this Order after the expiration date of this Order, the Permittee must apply for and obtain a new permit. (40 CFR section 122.41(b).)

2.3. Transfers

This Order is not transferable to any person except after notice to the Los Angeles Water Board. The Los Angeles Water Board may require modification or revocation and reissuance of the Order to change the name of the Permittee and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 CFR sections 122.41(l)(3); and 122.61.)

3. STANDARD PROVISIONS – MONITORING

- 3.1. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR section 122.41(j)(1).)
- 3.2. Monitoring must be conducted according to sufficiently sensitive test methods approved under 40 CFR part 136 for the analysis of pollutants or pollutant parameters or as required under 40 CFR chapter 1, subchapter N. For the purposes of this paragraph, a method is sufficiently sensitive when:
 - 3.2.1. The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the

pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or

- 3.2.2. The method has the lowest ML of the analytical methods approved under 40 CFR part 136 when approved by this Los Angeles Water Board and the State Water Board, or required under 40 CFR chapter 1, subchapter N or O for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 CFR part 136, or otherwise required under 40 CFR chapter 1, subchapters N or O, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 CFR section 122.21(e)(3).

4. STANDARD PROVISIONS – RECORDS

- 4.1. Except for records of monitoring information required by this Order related to the Permittee's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 CFR part 503), the Permittee shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Los Angeles Water Board Executive Officer at any time. (40 CFR part 122.41(j)(2).)
- 4.2. Records of monitoring information shall include:
- 4.2.1. The date, exact place, and time of sampling or measurements (40 CFR section 122.41(j)(3)(i));
 - 4.2.2. The individual(s) who performed the sampling or measurements (40 CFR section 122.41(j)(3)(ii));
 - 4.2.3. The date(s) analyses were performed (40 CFR section 122.41(j)(3)(iii));
 - 4.2.4. The individual(s) who performed the analyses (40 CFR section 122.41(j)(3)(iv));
 - 4.2.5. The analytical techniques or methods used (40 CFR section 122.41(j)(3)(v)); and
 - 4.2.6. The results of such analyses. (40 CFR section 122.41(j)(3)(vi).)
- 4.3. Claims of confidentiality for the following information will be denied (40 CFR section 122.7(b)):
- 4.3.1. The name and address of any permit applicant or Permittee (40 CFR section 122.7(b)(1)); and
 - 4.3.2. Permit applications and attachments, permits and effluent data. (40 CFR section 122.7(b)(2).)

5. STANDARD PROVISIONS – REPORTING

5.1. Duty to Provide Information

The Permittee shall furnish to the Los Angeles Water Board, State Water Board, or USEPA within a reasonable time, any information which the Los Angeles Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Permittee shall also furnish to the Los Angeles Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR section 122.41(h); Water Code sections 13267 and 13383.)

5.2. Signatory and Certification Requirements

- 5.2.1. All applications, reports, or information submitted to the Los Angeles Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting 5.2.2, 5.2.3, 5.2.4, and 5.2.5 below. (40 CFR section 122.41(k).)
- 5.2.2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR section 122.22(a)(3).)
- 5.2.3. All reports required by this Order and other information requested by the Los Angeles Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting 5.2.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting 5.2.2 above (40 CFR section 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR section 122.22(b)(2)); and
 - c. The written authorization is submitted to the Los Angeles Water Board and State Water Board. (40 CFR section 122.22(b)(3).)
- 5.2.4. If an authorization under Standard Provisions – Reporting 5.2.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting 5.2.3 above must be submitted to the Los Angeles Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR section 122.22(c).)

- 5.2.5. Any person signing a document under Standard Provisions – Reporting 5.2.2 or 5.2.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 CFR section 122.22(d).)

- 5.2.6. Any person providing the electronic signature for documents described in Standard Provisions – 5.2.1, 5.2.2, or 5.2.3 that are submitted electronically shall meet all relevant requirements of Standard Provisions – Reporting 5.2, and shall ensure that all relevant requirements of 40 CFR section 3 (Cross-Media Electronic Reporting) and 40 CFR section 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 CFR section 122.22(e).)

5.3. Monitoring Reports

- 5.3.1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR section 122.41(l)(4).)
- 5.3.2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Los Angeles Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. All reports and forms must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10 and comply with 40 CFR section 3, 40 CFR section 122.22, and 40 CFR section 127.
- 5.3.3. If the Permittee monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR section 136, or another method required for an industry-specific waste stream under 40 CFR chapter 1, subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Los Angeles Water Board.
- 5.3.4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order.

5.4. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR section 122.41(l)(5).)

5.5. Twenty-Four Hour Reporting

- 5.5.1. The Discharger shall report any noncompliance which may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A report shall also be provided

within five (5) days of the time the Discharger becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (i.e., combined sewer overflow, sanitary sewer overflow, or bypass event), type of overflow structure (e.g., manhole, combined sewer overflow outfall), discharge volume untreated by the treatment works treating domestic sewage, types of human health and environmental impacts of the event, and whether the noncompliance was related to wet weather.

As of December 21, 2020, all reports related to combined sewer overflows, sanitary sewer overflows, or bypass events must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting 5.10. The reports shall comply with 40 CFR section 3, 40 CFR section 122.22, and 40 CFR section 127. The Los Angeles Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 CFR section 122.41(l)(6)(i).)

5.5.2. The following shall be included as information that must be reported within 24 hours:

- a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR section 122.41(l)(6)(ii)(A).)
- b. Any upset that exceeds any effluent limitation in this Order. (40 CFR section 122.41(l)(6)(ii)(B).)

5.5.3. The Los Angeles Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR section 122.41(l)(6)(iii).)

5.6. Planned Changes

The Permittee shall give notice to the Los Angeles Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR section 122.41(l)(1)):

5.6.1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 CFR section 122.41(l)(1)(i)); or

5.6.2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 CFR section 122.41(l)(1)(ii).)

5.6.3. The alteration or addition results in a significant change in the Permittee's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during

the permit application process or not reported pursuant to an approved land application plan. (40 CFR section 122.41(l)(1)(iii).)

5.7. Anticipated Noncompliance

The Permittee shall give advance notice to the Los Angeles Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 CFR section 122.41(l)(2).)

5.8. Other Noncompliance

The Permittee shall report all instances of noncompliance not reported under Standard Provisions – Reporting 5.3, 5.4, and 5.5 above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting 5.5 above. For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports shall contain the information described in Standard Provision – Reporting 5.5 and the applicable required data in appendix A to 40 CFR section 127. The Los Angeles Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section (40 CFR section 122.41(l)(7).)

5.9. Other Information

When the Permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Los Angeles Water Board, State Water Board, or USEPA, the Permittee shall promptly submit such facts or information. (40 CFR section 122.41(l)(8).)

5.10. Initial Recipient for Electronic Reporting Data

The owner, operator, or the duly authorized representative is required to electronically submit NPDES information specified in appendix A to 40 CFR section 127 to the initial recipient defined in 40 CFR section 127.2(b). USEPA will identify and publish the list of initial recipients on its website and in the Federal Register, by state and by NPDES data group [see 40 CFR section 127.2(c)]. U.S. EPA will update and maintain this listing. (40 CFR § 122.41(l)(9).)

6. STANDARD PROVISIONS – ENFORCEMENT

- 6.1. The Los Angeles Water Board is authorized to enforce the terms of this Order under several provisions of the Water Code, including, but not limited to, sections 13268, 13385, 13386, and 13387.
- 6.2. The CWA provides that any person who violates section 301, 302, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any such sections in a permit issued under section 402, or any requirement imposed in a pretreatment program approved under sections 402(a)(3) or 402(b)(8) of the CWA, is subject to a civil penalty not to exceed \$25,000 per day for each violation. The CWA provides that any person who *negligently* violates sections 301, 302, 306, 307, 308, 318, or 405 of the CWA, or any condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA, or any requirement imposed in a pretreatment program approved under section 402(a)(3) or 402(b)(8) of the CWA, is subject to criminal

penalties of \$2,500 to \$25,000 per day of violation, or imprisonment of not more than one year, or both. In the case of a second or subsequent conviction for a negligent violation, a person shall be subject to criminal penalties of not more than \$50,000 per day of violation, or by imprisonment of not more than two years, or both. Any person who *knowingly* violates such conditions or limitations is subject to criminal penalties of \$5,000 to \$50,000 per day of violation, or imprisonment for not more than three years, or both. In the case of a second or subsequent conviction for a knowing violation, a person shall be subject to criminal penalties of not more than \$100,000 per day of violation, or imprisonment of not more than 6 years, or both. Any person who *knowingly* violates section 301, 302, 303, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA, and who knows at that time that he thereby places another person in imminent danger of death or serious bodily injury, shall, upon conviction, be subject to a fine of not more than \$250,000 or imprisonment of not more than 15 years, or both. In the case of a second or subsequent conviction for a knowing endangerment violation, a person shall be subject to a fine of not more than \$500,000 or by imprisonment of not more than 30 years, or both. An organization, as defined in section 309(c)(3)(B)(iii) of the CWA, shall, upon conviction of violating the imminent danger provision, be subject to a fine of not more than \$1,000,000 and can be fined up to \$2,000,000 for second or subsequent convictions (40 CFR section 122.41(a)(2); Water Code section 13385 and 13387).

- 6.3. Any person may be assessed an administrative penalty by the Administrator of USEPA, the Los Angeles Water Board, or State Water Board for violating section 301, 302, 306, 307, 308, 318 or 405 of this CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA. Administrative penalties for Class I violations are not to exceed \$10,000 per violation, with the maximum amount of any Class I penalty assessed not to exceed \$25,000. Penalties for Class II violations are not to exceed \$10,000 per day for each day during which the violation continues, with the maximum amount of any Class II penalty not to exceed \$125,000. (40 CFR section 122.41(a)(3))
- 6.4. The CWA provides that any person who falsifies, tampers with, or knowingly renders inaccurate any monitoring device or method required to be maintained under this Order shall, upon conviction, be punished by a fine of not more than \$10,000, or by imprisonment for not more than two years, or both. If a conviction of a person is for a violation committed after a first conviction of such person under this paragraph, punishment is a fine of not more than \$20,000 per day of violation, or by imprisonment of not more than four years, or both. (40 CFR section 122.41(j)(5)).
- 6.5. The CWA provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or non-compliance shall, upon conviction, be punished by a fine of not more than \$10,000 per violation, or by imprisonment for not more than six months per violation, or by both. (40 CFR section 122.41(k)(2)).

7. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

7.1. Publicly Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Los Angeles Water Board of the following (40 CFR section 122.42(b)):

- 7.1.1. Any new introduction of pollutants into the POTW from an indirect Permittee that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR section 122.42(b)(1)); and
- 7.1.2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 CFR section 122.42(b)(2).)
- 7.1.3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR section 122.42(b)(3).)

ATTACHMENT E. MONITORING AND REPORTING PROGRAM

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP), (CI-4993)

Section 308(a) of the federal Clean Water Act and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of title 40 of the Code of Federal Regulations (40 CFR) requires that all NPDES permits specify monitoring and reporting requirements. California Water Code (Water Code) section 13383 also authorizes the Los Angeles Water Board to establish monitoring, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement federal and California laws and/or regulations.

1. GENERAL MONITORING PROVISIONS

- 1.1. All samples shall be representative of the waste discharge under conditions of peak load. Results of monthly, quarterly, semiannual, and annual analyses shall be reported by the due date specified in Table E-7 of the MRP.
- 1.2. Pollutants, except those analyzed in the field shall be analyzed using the analytical methods described in 40 CFR sections 136.3, 136.4, and 136.5; or where no methods are specified for a given pollutant, by methods approved by this Los Angeles Water Board or the State Water Board.
- 1.3. **Laboratory Certification.** Laboratories analyzing monitoring samples shall be certified by the State Water Resources Control Board, Division of Drinking Water (DDW) Environmental Laboratory Accreditation Program (ELAP) in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control (QA/QC) data with their reports. A copy of the laboratory certification shall be provided in the Annual Report due to the Los Angeles Water Board each time a new certification and/or renewal of the certification is obtained.
- 1.4. Water/wastewater samples must be analyzed within allowable holding time limits as specified in 40 CFR section 136.3. All QA/QC analyses must be run on the same dates that samples are actually analyzed. The Permittee shall retain the QA/QC documentation in its files and make available for inspection and/or submit them when requested by the Los Angeles Water Board. Proper chain of custody procedures must be followed, and a copy of that documentation shall be submitted with the monthly report.
- 1.5. The Permittee shall calibrate and perform maintenance procedures on all monitoring instruments and to ensure accuracy of measurements or shall ensure that both equipment activities will be conducted.
- 1.6. For any analyses performed for which no procedure is specified in the United States Environmental Protection Agency (USEPA) guidelines, or in the MRP, the constituent or parameter analyzed, and the method or procedure used must be specified in the monitoring report.
- 1.7. Each monitoring report must affirm in writing that “with the exception of field tests, all analyses were conducted at a laboratory certified for such analyses under ELAP through the State Water Board DDW: or, were approved by the Executive Officer and in accordance with current USEPA guideline procedures or as specified in this Monitoring and Reporting Program.”

- 1.8. The monitoring report shall specify the analytical method used, the Method Detection Limit (MDL), and the Reporting Level (RL) [the applicable minimum level (ML) or reported Minimum Level (RML)] for each pollutant. The MLs are those published by the State Water Board in the *Policy for the Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California, (State Implementation Policy or SIP)*, February 9, 2005, Appendix 4. The ML represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interference. When all specific analytical steps are followed and after appropriate application of method specific factors, the ML also represents the lowest standard in the calibration curve for that specific analytical technique. When there is deviation from the method analytical procedures, such as dilution or concentration of samples, other factors may be applied to the ML depending on the sample preparation. The resulting value is the reported ML.
- 1.9. The Permittee shall select the analytical method that provides a ML lower than the permit limit established for a given parameter, unless the Permittee can demonstrate that a particular ML is not attainable, in accordance with procedures set forth in 40 CFR part 136, and obtains approval for a higher ML from the Executive Officer, as provided for in section 1.11, below. If the effluent limitation is lower than all the MLs in Appendix 4 of the SIP, the Discharge must select the method with the lowest ML for compliance purposes. The Permittee shall include in the Annual Summary Report a list of the analytical methods employed for each test.
- 1.10. The Permittee shall instruct its laboratories to establish calibration standards so that the ML (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve. In accordance with section 1.11, below, the Permittee's laboratory may employ a calibration standard lower than the ML in Appendix 4 of the SIP.
- 1.11. In accordance with section 2.4.3 of the SIP, the Los Angeles Water Board Executive Officer, in consultation with the State Water Board's Quality Assurance Program Manager, may establish an ML that is not contained in Appendix 4 of the SIP to be included in the Permittee's permit in any of the following situations:
 - 1.11.1. When the pollutant under consideration is not included in Appendix 4 of the SIP;
 - 1.11.2. When the Permittee and the Los Angeles Water Board agree to include in the permit a test method that is more sensitive than those specified in 40 CFR part 136;
 - 1.11.3. When the Permittee agrees to use an ML that is lower than those listed in Appendix 4;
 - 1.11.4. When the Permittee demonstrates that the calibration standard matrix is sufficiently different from that used to establish the ML in Appendix 4 and proposes an appropriate ML for the matrix; or,
 - 1.11.5. When the Permittee uses a method for which quantification practices are not consistent with the definition of the ML. Examples of such methods are USEPA-approved method 1613 for dioxins, and furans, method 1624 for volatile organic substances, and method 1625 for semi-volatile organic substances. In such cases,

the Permittee, the Los Angeles Water Board, and the State Water Board shall agree on a lowest quantifiable limit and that limit will substitute for the ML for reporting and compliance determination purposes.

If there is any conflict between foregoing provisions and the SIP, the provisions stated in the SIP (section 2.4) shall prevail.

- 1.12. If the Permittee samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this MRP using approved analytical methods, the results of those analyses shall be included in the report. These results shall be reflected in the calculation of the average used in demonstrating compliance with limitations set forth in this Order.
- 1.13. The Permittee shall develop and maintain a record of all spills or bypasses of raw or partially treated sewage from its collection system or treatment plant according to the requirements in the WDR section of this Order. This record shall be made available to the Los Angeles Water Board upon request and a spill summary shall be included in the annual summary report.
- 1.14. For all bacteriological analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total coliform, at a minimum, and 1 to 1000 per 100 ml for enterococcus). The detection methods used for each analysis shall be reported with the results of the analyses.
 - 1.14.1. Detection methods used for total coliform shall be those presented in Table 1A of 40 CFR part 136, unless alternate methods have been approved in advance by the USEPA pursuant to 40 CFR part 136.
 - 1.14.2. Detection methods used for E.coli shall be those presented in Table 1A of 40 CFR part 136 or in the USEPA publication EPA 600/4-85/076, *Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure*, or any improved method determined by the Los Angeles Water Board to be appropriate.
- 1.15. The Permittee shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board at the following address:

State Water Resources Control Board
Quality Assurance Program Officer
Office of Information Management and Analysis
1001 I Street, Sacramento, CA 95814.

2. MONITORING LOCATIONS

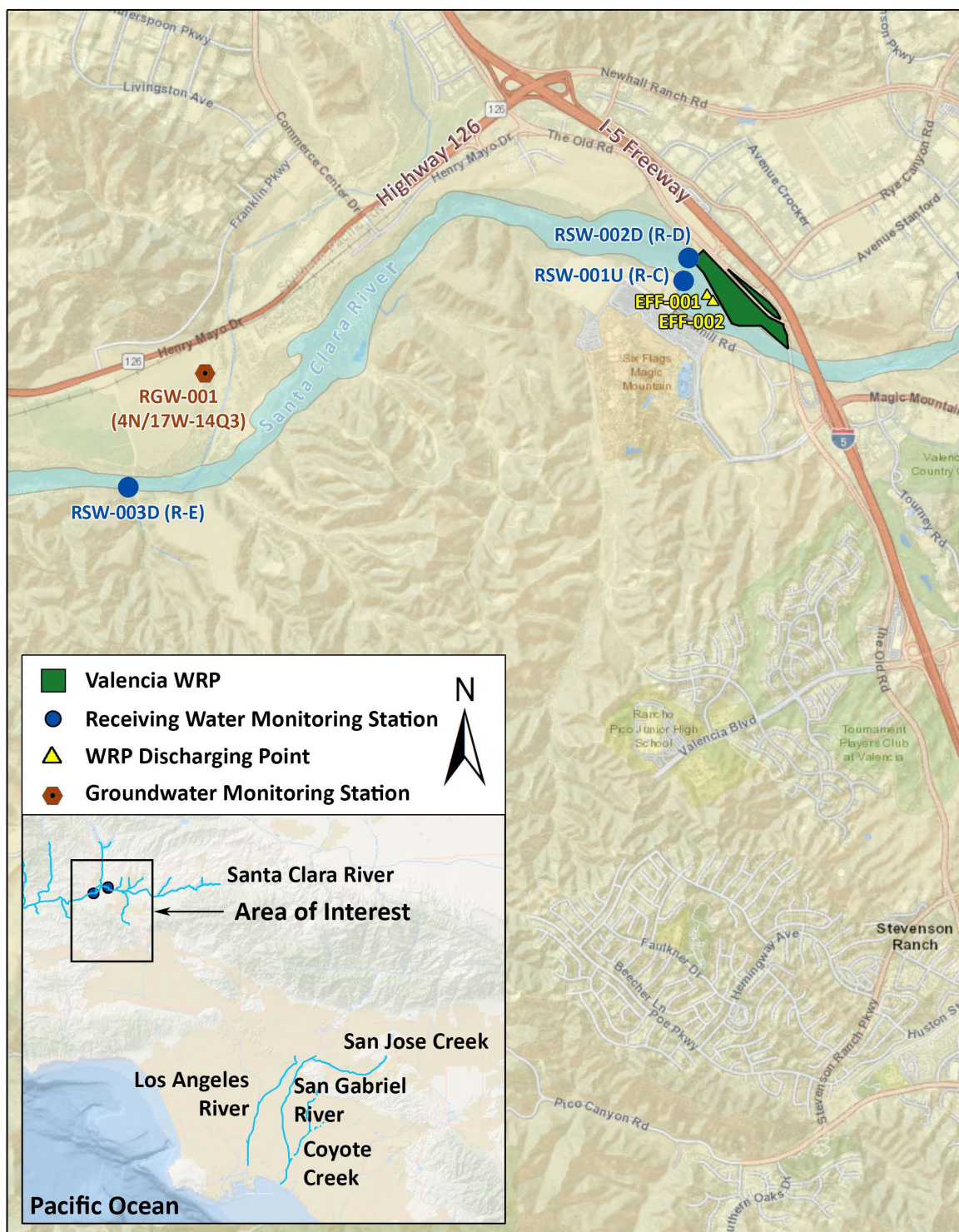
The Permittee shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	Influent Monitoring Station INF-001	Sampling stations shall be established at each point of inflow to the sewage treatment plant and shall be located upstream of any in-plant return flows and where representative samples of the influent can be obtained. Latitude: 34.4303° Longitude: -118.5906°
001	Effluent Monitoring Station EFF-001	The effluent sampling station shall be located downstream of any in-plant return flows and after the final disinfection process, where representative samples of the effluent can be obtained. Under normal conditions, treated effluent is discharged through the following location: Latitude: 34.430546° Longitude: -118.592546°
002	Effluent Monitoring Station EFF-002	The effluent sampling station shall be located downstream of any in-plant return flows and after the final disinfection process, where representative samples of the effluent can be obtained. Under high stream conditions, treated effluent is discharged through the following location: Latitude: 34.430075°, Longitude: -118.592208°
--	Receiving Water Monitoring Station RSW-001U	Santa Clara River, approximately 300 feet upstream of the Valencia WRP Discharge Point 001 (Station R-C). Latitude: 34.4312°, Longitude: -118.5944°
	Receiving Water Monitoring Station RSW-002D	Santa Clara River, approximately 300 feet downstream of the Valencia WRP Discharge Point 001 (Station R-D). Latitude: 34.4326°, Longitude: -118.5940°
	Receiving Water Monitoring Station RSW-003D	Santa Clara River, approximately 1.6 miles upstream of Chiquita canyon Road (Station R-E). Latitude: 34.4186°, Longitude: -118.6357°
--	Receiving Groundwater Monitoring Station RGW-001	Groundwater sampled at Well No. 4N/17W-14Q3

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes. These monitoring locations are indicated in Figure E-1 below.

Figure E-1. Valencia WRP Receiving Water Stations



**Figure 2.1 Valencia WRP
Effluent Discharge Points & Receiving Water Stations**

0 0.25 0.5 1 Miles

Doc#: 4955936

3. INFLUENT MONITORING REQUIREMENTS

Influent monitoring is required to:

- Determine compliance with NPDES permit conditions.
- Assess treatment plant performance.
- Assess effectiveness of the Pretreatment Program.

3.1. Monitoring Location INF-001

The Permittee shall monitor influent to the facility at INF-001 as follows:

Table E-2. Influent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Note
Flow	MGD	recorder	continuous	a
pH	pH unit	grab	weekly	b
Total suspended solids	mg/L	24-hour composite	weekly	b
Biochemical oxygen demand (BOD ₅ 20°C)	mg/L	24-hour composite	weekly	b
Temperature	°F	grab	weekly	b
Chloride	mg/L	24-hour composite	monthly	b
Selenium	µg/L	24-hour composite	monthly	b
Cyanide	µg/L	grab	monthly	b
Iron	µg/L	24-hour composite	monthly	b
Total trihalomethanes	µg/L	calculated	monthly	b
TCDD equivalents	pg/L	grab or 24-hour composite	quarterly	b, c & f
PCBs as aroclors	µg/L	24-hour composite	annually	b and d
PCBs as congeners	pg/L	24-hour composite	annually	b and d
Remaining USEPA priority pollutants excluding asbestos	µg/L	24-hour composite; Grab for VOCs, Chromium VI, and grab or composite for bis(2-ethylhexyl)phthalate	semiannually	b, e, and f

Footnotes for Table E-2

- Total daily flow and instantaneous peak daily flow (24-hr basis) shall be reported. Actual monitored flow shall also be reported (not the maximum flow, i.e., design capacity).
- Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Los Angeles Water Board or State Water Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Appendix 4 of the SIP, the analytical method with the lowest ML must be selected.

- c. The Permittee shall conduct monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the influent. The Permittee shall use the appropriate Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i) concentration analytical result (C_i) and their corresponding Toxicity Equivalence Factor (TEFi), (i.e., $TEQ_i = C_i \times TEF_i$). The Discharger shall report all measured values of individual congeners, including data qualifiers. TCDD equivalents concentration shall be reported as the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

- d. PCBs as aroclors shall be analyzed using USEPA method 608.3. PCBs as congeners shall be analyzed using method 1668c. USEPA recommends that until the USEPA proposed method 1668c is incorporated into 40 CFR 136, permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608.3 for monitoring data, reported as aroclor results, that will be used for determining compliance with WQBELs (if applicable) and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes.
- e. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.
- f. The 40 CFR part 136 method for phthalate esters including bis (2-ethylhexyl) phthalate and TCDD equivalents requires samples to be collected in glass sample containers to avoid interference, which can lead to artifacts and/or elevated baselines in gas chromatograms. Sample collection must be done using glass sample containers for all phthalate esters including bis (2-ethylhexyl) phthalate and TCDD equivalents unless analytical methods for these pollutants in 40 CFR part 136 specify that other means of sample collection are approved. Grab sample type is recommended, but an automatic sampler (composite sample) can be used to collect samples for all phthalate esters including bis (2-ethylhexyl) phthalate, and TCDD equivalents as long as the sample bottles are glassware.

End of Footnotes for Table E-2

4. EFFLUENT MONITORING REQUIREMENTS

Effluent monitoring is required to:

- Determine compliance with National Pollutant Discharge Elimination System (NPDES) permit conditions and water quality standards.
- Assess plant performance and identify operational problems.
- Provide information on wastewater characteristics and flows for use in interpreting water quality and biological data.
- Determine reasonable potential analysis for toxic pollutants.
- Determine waste load allocation compliance and TMDL effectiveness.

4.1. Monitoring Locations EFF-001 and EFF-002

The Permittee shall monitor the discharge of tertiary-treated effluent at EFF-001 and EFF-002. If more than one analytical test method is listed for a given parameter, the Permittee must select from the listed methods and corresponding ML:

Table E-3. Effluent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Total flow	MGD	recorder	continuous	a
Turbidity	NTU	recorder	continuous	a & b
Total residual chlorine	mg/L	recorder	continuous	c
Total residual chlorine	mg/L	grab	daily	b & c
Total coliform	MPN/100mL or CFU/100ml	grab	weekly	b & d
<i>E. coli</i>	MPN/100mL or CFU/100ml	grab	weekly	b & d
Temperature	°F	grab	weekly	b
pH	pH units	grab	weekly	b
Settleable solids	mL/L	grab	weekly	b
Total suspended solids (TSS)	mg/L	24-hour composite	weekly	b
BOD ₅ 20°C	mg/L	24-hour composite	weekly	b & e
Oil and grease	mg/L	grab	quarterly	b
Dissolved oxygen	mg/L	grab	monthly	b
Total dissolved solids	mg/L	24-hour composite	monthly	b
Sulfate	mg/L	24-hour composite	monthly	b
Chloride	mg/L	24-hour composite	monthly	b
Boron	mg/L	24-hour composite	monthly	b
Ammonia nitrogen	mg/L	24-hour composite	monthly	b
Nitrite nitrogen	mg/L	24-hour composite	monthly	b
Nitrate nitrogen	mg/L	24-hour composite	monthly	b
Organic nitrogen	mg/L	calculated	monthly	b
Total Kjeldahl nitrogen	mg/L	24-hour composite	monthly	b
Total nitrogen	mg/L	calculated	monthly	b
Total phosphorus	mg/L	24-hour composite	quarterly	b
Orthophosphate-P	mg/L	24-hour composite	quarterly	b
Surfactants (MBAS)	mg/L	24-hour composite	quarterly	b
Surfactants (CTAS)	mg/L	24-hour composite	quarterly	b
Total hardness (CaCO ₃)	mg/L	24-hour composite	monthly	b
Chronic toxicity <i>Ceriodaphnia dubia</i> Survival and reproduction endpoints	Pass or Fail, (TST) and Percent Effect	24-hour composite	monthly	b & f

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Radioactivity (Including gross alpha, gross beta, combined radium-226 and radium-228, tritium, strontium-90 and uranium)	pCi/L	24-hour composite	semiannually	g
TCDD equivalents	pg/L	grab or 24-hour composite	quarterly	b, j & m
Antimony	µg/L	24-hour composite	semiannually	b
Arsenic	µg/L	24-hour composite	quarterly	b
Beryllium	µg/L	24-hour composite	semiannually	b
Cadmium	µg/L	24-hour composite	semiannually	b
Chromium III	µg/L	calculated	semiannually	b
Chromium VI	µg/L	grab	semiannually	b
Total Chromium	µg/L	grab	semiannually	b
Copper	µg/L	24-hour composite	quarterly	b
Lead	µg/L	24-hour composite	semiannually	b
Mercury	µg/L	24-hour composite	quarterly	b & h
Nickel	µg/L	24-hour composite	semiannually	b
Selenium	µg/L	24-hour composite	monthly	b
Silver	µg/L	24-hour composite	quarterly	b
Thallium	µg/L	24-hour composite	semiannually	b
Zinc	µg/L	24-hour composite	quarterly	b
Cyanide	µg/L	grab	monthly	b
Iron	µg/L	24-hour composite	monthly	b
Dichlorobromomethane	µg/L	grab	monthly	b
Dibromochloromethane	µg/L	grab	monthly	b
Bromoform	µg/L	grab	monthly	b
Chloroform	µg/L	grab	monthly	b
Total trihalomethanes (TTHMs)	µg/L	grab/ calculated sum	monthly	b
Bis(2-ethylhexyl)phthalate	µg/L	24-hour composite or grab	quarterly	b
Perchlorate	µg/L	grab	semiannually	k
1,4-Dioxane	µg/L	grab	semiannually	k
1,2,3-Trichloropropane	µg/L	grab	semiannually	k
Methyl tert-butyl-ether (MTBE)	µg/L	grab	semiannually	k

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Chlorpyrifos	µg/L	grab	annually	b
Diazinon	µg/L	grab	annually	b
PCBs as aroclors	µg/L	24-hour composite	annually	b & i
PCBs as congeners	pg/L	24-hour composite	annually	b & i
Fluoride	mg/L	24-hour composite	annually	b
Remaining USEPA priority pollutants excluding asbestos and PCBs	µg/L	24-hour composite; grab for VOCs, and grab or composite for bis(2-ethylhexyl) phthalate	semiannually	b, l, and m

Footnotes for Table E-3

- a. Where continuous monitoring of a constituent is required, the following shall be reported:
 - Total waste flow – Total daily and peak daily flow (24-hr basis).
 - Turbidity – Maximum daily value, total amount of time each day the turbidity exceeded 5 NTU, flow proportioned average daily value. Grab sample can be used to determine compliance with the 10 NTU limit. A flow-weighted 24-hour composite sample may be used in place of the recorder to determine the flow-proportioned average daily value.
- b. Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Los Angeles Water Board or State Water Resources Control Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Appendix 4 of the SIP, the analytical method with the lowest ML must be selected.
- c. Total residual chlorine shall be recorded continuously. The recorded data shall be maintained by the Permittee for at least five years. The Permittee shall extract the maximum daily peak, and average daily from the recorded media and shall be made available upon request of the Los Angeles Water Board. The continuous monitoring data are not intended to be used for compliance determination purposes. Compliance will be determined by daily sample collected at EFF-001 (or EFF-002), when the outfall is actively being used to discharge Monday through Friday only, except for holidays. Analytical results of daily grab samples will be used to determine compliance with total residual chlorine effluent limitation. Furthermore, additional monitoring requirements specified in section 4.1, following the end of the footnotes to Table E-3.
- d. *E. coli* testing shall be conducted only if total coliform testing is positive. If the total coliform analysis results in no detection, a result of less than (<) the reporting limit for total coliform will be reported for *E. coli*.
- e. If the result of the weekly BOD₅20°C analysis yields a value greater than the 30-day average limit, the frequency of analysis shall be increased to daily within one week of knowledge of the test result for at least 30 days and until compliance with the 7-day and 30-day average BOD₅20°C limits is demonstrated; after which the frequency shall revert to weekly.
- f. The Permittee shall conduct whole effluent toxicity monitoring using *Ceriodaphnia dubia* as the test species, as outlined in section 5 of this MRP. For the *Ceriodaphnia dubia* reproduction endpoint, the median monthly effluent limitation (MMEL) summary result shall be reported as “Pass” or “Fail” and the maximum daily single result shall be reported as “Pass” or “Fail” and “% Effect.” The *Ceriodaphnia dubia* reproduction endpoint shall be reported as “Pass” or “Fail” and “% Effect”. The

Ceriodaphnia dubia survival endpoint, shall be reported as “% Effect.” If the chronic aquatic toxicity routine monitoring test results in a “Fail” at the instream waste concentration (IWC), then the Valencia WRP shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the “Fail” at the IWC. If the first chronic MMEL compliance test results in a “Fail” at the IWC, then the second MMEL compliance test is not necessary because the “Fail” results from the first two tests constitutes a violation of the chronic toxicity MMEL.

- g. Analyze these radiochemicals by the following USEPA methods: method 900.0 for gross alpha and gross beta, method 903.0 or 903.1 for radium-226, method 904.0 for radium-228, method 906.0 for tritium, method 905.0 for strontium-90, and method 908.0 for uranium. Analysis for combined Radium-226 & 228 shall be conducted only if gross alpha results for the same sample exceed 15 pCi/L or beta greater than 50 pCi/L. If Radium-226 & 228 exceeds the stipulated criteria, analyze for Tritium, Strontium-90 and uranium.
- h. The mercury effluent samples shall be analyzed using EPA method 1631E, per 40 CFR part 136, with a quantification level lower than 0.5 ng/L, shall be used to analyze total mercury. If an alternative method with an equivalent or more sensitive method detection limit is approved in 40 CFR part 136, the Discharger may use that method in lieu of USEPA Method 1631E.
- i. PCBs as aroclors shall be analyzed using USEPA method 608. PCBs as congeners shall be analyzed using method 1668c. USEPA recommends that until the USEPA proposed method 1668c is incorporated into 40 CFR 136, permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608 for monitoring data, reported as aroclor results, that will be used for determining compliance with WQBELs (if applicable) and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes.
- j. In accordance with the SIP, the Permittee shall conduct monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the effluent and in the receiving water. The Permittee shall use the appropriate Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i) concentration analytical result (Ci) and their corresponding Toxicity Equivalence Factor (TEFi), (i.e., $TEQ_i = C_i \times TEF_i$). The Discharger shall report all measured values of individual congeners, including data qualifiers. TCDD equivalents concentration shall be reported as the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

- k. Emerging chemicals include 1,4-dioxane (USEPA 8270M test method), perchlorate (USEPA 314 test method, or USEPA method 331 if a detection limit of less than 6 µg/L is achieved), 1,2,3-trichloropropane (USEPA 504.1, 8260B test method, or USEPA 524.2 in SIM mode), and methyl tert-butyl ether (USEPA 8260B test method or USEPA method 624 if a detection level of less than 5 µg/L is achieved, and if the Permittee received ELAP certification to run USEPA method 624).
- l. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.
- m. The 40 CFR part 136 method for phthalate esters including bis (2-ethylhexyl) phthalate and TCDD equivalents requires samples to be collected in glass sample containers to avoid interference, which can lead to artifacts and/or elevated baselines in gas chromatograms. Sample collection must be done using glass sample containers for all phthalate esters including bis (2-ethylhexyl) phthalate and TCDD equivalents unless analytical methods for these pollutants in 40 CFR part 136 specify that other means of sample collection are approved. Grab sample type is recommended, but an automatic sampler (composite sample) can be used to collect samples for all phthalate esters

including bis (2-ethylhexyl) phthalate and TCDD equivalents as long as the sample bottles are glassware.

End of Footnotes for Table E-3

Total Residual Chlorine Additional Monitoring:

Continuous monitoring of total residual chlorine at EFF-001 and EFF-002 shall serve as an internal trigger for the increased grab sampling at EFF-001 and EFF-002 if either of the following occurs, except as noted in item iii:

- i. Total residual chlorine concentration excursions of up to 0.3 mg/L lasting greater than 15 minutes; or
- ii. Total residual chlorine concentration peaks in excess of 0.3 mg/L lasting greater than 1 minute.
- iii. Additional grab samples need not be taken if it can be demonstrated that a stoichiometrically appropriate amount of dechlorination chemical has been added to effectively dechlorinate the effluent to 0.1 mg/L or less for peaks in excess of 0.3 mg/L lasting more than 1 minute, but not for more than five minutes.

5. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

5.1. Discharge In-stream Waste Concentration (IWC) for Chronic Toxicity

The aquatic chronic toxicity IWC for this discharge is 100 percent effluent.

5.2. Sample Volume and Holding Time

The total sample volume shall be determined by the specific toxicity test method used. Sufficient sample volume shall be collected to perform the required toxicity test. For the receiving water, sufficient sample volume shall also be collected for subsequent TIE studies, if necessary, at each sampling event. All toxicity tests shall be conducted as soon as possible following sample collection. No more than 36 hours shall elapse before the conclusion of sample collection and test initiation.

5.3. Chronic Freshwater Species and Test Methods

If effluent samples are collected from outfalls discharging to receiving waters with salinity <1 ppt, the Permittee shall conduct the chronic toxicity tests on effluent samples at the in-stream waste concentration for the discharge in accordance with species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R02/013, 2002; Table IA, 40 CFR part 136). In no case shall these species be substituted with another test species unless written authorization from the Executive Officer is received.

- 5.3.1. A static renewal toxicity test with the fathead minnow, *Pimephales promelas* (Larval Survival and Growth Test Method 1000.0).
- 5.3.2. A static renewal toxicity test with the daphnid, *Ceriodaphnia dubia* (Survival and Reproduction Test Method 1002.01).
- 5.3.3. A static toxicity test with the green alga, *Selenastrum capricornutum* (also named *Raphidocelis subcapitata*) (Growth Test Method 1003.0).

5.4. Species Sensitivity Screening

The Permittee may begin a species sensitivity screening for chronic aquatic toxicity prior to permit reissuance, but no later than 18 months prior to the expiration date of Order Number R4-2022-0174. For continuous dischargers, species sensitivity screening includes four sets of testing conducted in each quarter of a year. In each of the four sets, the Permittee shall collect a single effluent sample to initiate and concurrently conduct three toxicity tests using the fish, an invertebrate, and the alga species previously referenced. This sample shall also be analyzed for the parameters required on a monthly frequency for the discharge, during that given month. As allowed under the test method for the *Ceriodaphnia dubia* and the *Pimephales promelas*, a second and third sample may be collected for use as test solution renewal water as the seven-day toxicity test progresses. However, that same sample shall be used to renew both the *Ceriodaphnia dubia* and the *Pimephales promelas*. For non-continuous dischargers, a set of testing shall be conducted in each quarter in which there is expected to be at least 15 days of discharge. For non-continuous dischargers that discharge in only one quarter of the year in which there is expected to be at least 15 days of discharge, two sets of testing shall be conducted within the same quarter.

If the result of all three species is “Pass”, then the species that exhibits the highest “Percent Effect” at the discharge IWC during species sensitivity screening shall be considered the most sensitive species in a given set. If only one species fails, then that species shall be considered the most sensitive species in a given set. The species that had more fails in the four sets shall be considered the most sensitive species and shall be used for routine monitoring during the next permit cycle. If two or more species result in “Fail,” then the species that exhibits the highest “Percent Effect” at the discharge IWC during the four sets of species sensitivity screening shall be used for routine monitoring during the next permit cycle.

During the calendar month, toxicity tests used to determine the most sensitive test species shall be reported as effluent compliance monitoring results for the chronic toxicity MDEL and MMEL.

5.5. Quality Assurance and Additional Requirements

5.5.1. Quality assurance measures, instructions, and other recommendations and requirements are found in the test methods manual previously referenced. Additional requirements are specified below. The discharge is subject to determination of “Pass” or “Fail” and “Percent Effect” from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t-test approach described in the *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, 2010), Appendix A, Figure A-1, Table A-1 and Appendix B, Table B-1. The null hypothesis (H_0) for the TST statistical approach is: Mean discharge IWC response $\leq 0.75 \times$ Mean control response. A test result that rejects this null hypothesis is reported as “Pass.” A test result that does not reject this null hypothesis is reported as “Fail.” The relative “Percent Effect” at the discharge IWC is defined and reported as: $((\text{Mean control response} - \text{Mean discharge IWC response}) \div \text{Mean control response}) \times 100$. This is a t-test (formally Student’s t-Test), a statistical analysis comparing two sets of replicate observations -

in the case of WET, only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is “Pass” or “Fail”). The Welch’s t-test employed by the TST statistical approach is an adaptation of Student’s t-test and is used with two samples having unequal variances.

- 5.5.2. To comply with the Median Monthly Effluent Limit (MMEL) for chronic toxicity, up to three independent toxicity tests may be conducted during a calendar month. If the initial toxicity test, conducted in the beginning of the month, results in “Fail” at the IWC, then the Valencia WRP shall initiate up to two additional chronic aquatic toxicity tests in the remainder of the month, to determine compliance with the MMEL. If the second test conducted in the month is also a “Fail,” then that constitutes a violation of the MMEL. However, if the second and third tests result in a “Pass,” then the discharge is in compliance with the MMEL.
- 5.5.3. If the effluent toxicity test does not meet all test acceptability criteria (TAC) and all required test conditions specified in the referenced test method *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013) (See Table E.4 for TAC below), then the Permittee must re-sample and re-test within 14 days. Deviations from recommended test conditions, specified in the referenced test method *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013), must be evaluated on a case-by-case to determine the validity of test results. The Discharger shall consider the degree of the deviation and the potential or observed impact of the deviation on the test results in consultation with Los Angeles Water Board staff before rejecting or accepting a test result as valid, and shall report the results of the validity determination with supporting evidence for that decision in their monthly report.

Table E-4. USEPA Test Methods and Test Acceptability Criteria

Species & USEPA Test Method Number	Test Acceptability Criteria (TAC)
Fathead Minnow, <i>Pimephales promelas</i> , Larval Survival and Growth Test Method 1000.0 (Table 1 of the test method, above)	80% or greater survival in controls; average dry weight per surviving organism in control chambers equals or exceeds 0.25 mg. (required)
Daphnid, <i>Ceriodaphnia dubia</i> , Survival and Reproduction Test Method 1002.0. (Table 3 of the test method, above)	80% or greater survival of all control organisms and an average of 15 or more young per surviving female in the control solutions. 60% of surviving control females must produce three broods. (required)
Green Alga, <i>Selenastrum capricornutum</i> , Growth Toxicity Test Method 1003.0. (Table 3 of the test method, above)	Mean cell density of at least 1×10^6 cells/mL in the controls; and variability (CV%) among control replicates less than or equal to 20%. (required)

- 5.5.4. Dilution water and control water, including brine controls, shall be laboratory water prepared and used as specified in the test methods manual. If dilution water and control water is different from test organism culture water, then a second control using culture water shall also be used.
- 5.5.5. When preparing samples for toxicity testing, in addition to the required monitoring for conductivity, etc., it is recommended that total alkalinity and total hardness be measured in the undiluted effluent, receiving water, dilution water, and culture water (following the WET methods manual), as well as the major geochemical ions (see Mount et al., 2018).
- 5.5.6. Monthly reference toxicant testing is sufficient. All reference toxicant test results should be reviewed and reported using EC25. EC25 is a point estimate of the toxicant concentration that would cause an observable adverse effect (e.g., death, immobilization, or serious incapacitation) in 25 percent of the test organisms.
- 5.5.7. The Permittee shall perform toxicity tests on final effluent samples. Chlorine and ammonia shall not be removed from the effluent sample prior to toxicity testing, unless explicitly authorized under this section of the Monitoring and Reporting Program and the rationale is explained in the Fact Sheet (Attachment F).

5.6. Preparation of an Initial Investigation TRE Work Plan

The Permittee shall prepare and submit a copy of the Permittee's initial investigation TRE work plan to the Executive Officer of the Los Angeles Water Board for approval within 90 days of the effective date of this permit. If the Executive Officer does not disapprove the work plan within 60 days, the work plan shall become effective. The Permittee shall use USEPA manual EPA/833B-99/002 (municipal) as guidance, or most current version, or EPA manual *Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations (EPA/600/2-88/070, April 1989)*. At a minimum, the TRE Work Plan must contain the provisions in Attachment G. This work plan shall describe the steps that the Permittee intends to follow if toxicity is detected. At minimum, the work plan shall include:

- 5.6.1. A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
- 5.6.2. A description of the Facility's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in the operation of the Facility.
- 5.6.3. If a TIE is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor).

5.7. Toxicity Reduction Evaluation (TRE) Process

A TRE is required when toxicity is persistent: if the Permittee has any combination of two or more MDEL or MMEL violations within a single calendar month or within two successive calendar months. In addition, if other information indicates toxicity (e.g., results of additional monitoring, results of monitoring at a higher concentration than the IWC, fish kills, intermittent recurring toxicity), then Executive Officer of the Los Angeles Water Board may require a TRE. The Discharger shall conduct a TRE in accordance with a TRE Work

Plan as approved by Los Angeles Water Board. Routine monitoring shall continue during a TRE and TST results ("Pass" or "Fail", "Percent Effect") for chronic toxicity tests shall be reported as effluent compliance monitoring results for the chronic toxicity MDEL and MMEL. During the TRE process, the major ions (Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , SO_4^{2-} , and $\text{HCO}_3^-/\text{CO}_3^{2-}$), shall be analyzed for in effluent IWC, dilution water, and culture water used for toxicity testing. Those results shall be reported in the corresponding monitoring report.

5.7.1. Preparation and Implementation of Detailed TRE Work Plan. The Permittee shall immediately initiate a TRE using, according to the type of treatment facility, USEPA manual *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA/833/B-99/002, 1999) or EPA manual *Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations* (EPA/600/2-88/070, April 1989) and, within 30 days, submit to the Executive Officer a Detailed TRE Work Plan, which shall follow the TRE Work Plan revised as appropriate for this toxicity event. It shall include the following information, and comply with additional conditions set by the Executive Officer:

- a. Further actions by the Permittee to investigate, identify, and correct the causes of toxicity.
- b. Actions the Permittee will take to mitigate the effects of the discharge and prevent the recurrence of toxicity.
- c. A schedule for these actions, progress reports, and the final report.

5.7.2. TIE Implementation. The Permittee may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test method and, as guidance, USEPA manuals: *Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures* (EPA/600/6-91/003, 1991); *Chronic TIE Manual: Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I* (EPA/600/6-91/005F, 1992); *Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/080, 1993); *Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/081, 1993); and *Marine Toxicity Identification Evaluation (TIE): Phase I Guidance Document* (EPA/600/R-96-054, 1996). The TIE should be conducted on the species demonstrating the most sensitive toxicity response.

5.7.3. Many recommended TRE elements parallel required or recommended efforts for source control, pollution prevention, and stormwater control programs. TRE efforts should be coordinated with such efforts. As toxic substances are identified or characterized, the Permittee shall continue the TRE by determining the sources and evaluating alternative strategies for reducing or eliminating the substances from the discharge. All reasonable steps shall be taken to reduce toxicity to levels consistent with toxicity evaluation parameters.

5.7.4. The Permittee shall continue to conduct routine effluent monitoring and MMEL compliance monitoring while the TIE and/or TRE process is taking place. Additional TRE work plans are not required once a TRE has begun.

5.7.5. The Los Angeles Water Board recognizes that toxicity may be episodic and identification of causes and reduction of sources of toxicity may not be successful in all cases. However, TRE shall be carried out in accordance with the Executive Officer-approved TRE Work Plan.

5.7.6. The Board may consider the results of any TIE/TRE studies in an enforcement action.

5.8. Reporting

The Self-Monitoring Report (SMR) shall include a full laboratory report for each toxicity test. This report shall be prepared using the format and content of the test methods manual chapter called Report Preparation, including:

- 5.8.1. The valid toxicity test results for the TST statistical approach, reported as “Pass” or “Fail” and “Percent Effect” at the chronic toxicity IWC for the discharge, using *Ceriodaphnia dubia*. All toxicity test results (whether identified as valid or otherwise) conducted during the calendar month shall be reported on the SMR due date specified in Table E-7.
- 5.8.2. A summary of water quality measurements for each toxicity test (e.g., pH, dissolved oxygen, temperature, conductivity, total hardness, salinity, chlorine, and ammonia).
- 5.8.3. The statistical analysis used in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010) Appendix A, Figure A-1, Table A-1, and Appendix B, Table B-1.
- 5.8.4. TRE/TIE results. The Executive Officer shall be notified no later than 30 days from completion of each aspect of TRE/TIE analyses. Prior to the completion of the final TIE/TRE report, the Permittee shall provide status updates in the monthly monitoring reports, indicating which TIE/TRE steps are underway and which steps have been completed.
- 5.8.5. Statistical program (e.g., TST calculator, CETIS, etc.) output results, including graphical plots, for each toxicity test.
- 5.8.6. Tabular data and graphical plots clearly showing the laboratory’s performance for the reference toxicant, for each solution, for the previous 20 tests and the laboratory’s performance for the control mean, control standard deviation, and control coefficient of variation, for each solution, for the previous 12-month period.
- 5.8.7. Any additional QA/QC documentation or any additional chronic toxicity-related information, upon request from the Los Angeles Water Board Chief Deputy Executive Officer or the Executive Officer.

5.9. Ammonia Removal

- 5.9.1. Except with prior approval from the Executive Officer of the Los Angeles Water Board, ammonia shall not be removed from bioassay samples. The Permittee must demonstrate the effluent toxicity is caused by ammonia because of increasing test pH when conducting the toxicity test. It is important to distinguish the potential toxic effects of ammonia from other pH sensitive chemicals, such as certain heavy metals, sulfide, and cyanide. The following may be steps to demonstrate that the

toxicity is caused by ammonia and not other toxicants before the Executive Officer would allow for control of pH in the test.

- a. There is consistent toxicity in the effluent and the maximum pH in the toxicity test is in the range to cause toxicity due to increased pH.
- b. Chronic ammonia concentrations in the effluent are greater than 4 mg/L total ammonia.
- c. Conduct graduated pH tests as specified in the toxicity identification evaluation methods. For example, mortality should be higher at pH 8 and lower at pH 6.
- d. Treat the effluent with a zeolite column to remove ammonia. Mortality in the zeolite treated effluent should be lower than the non-zeolite treated effluent. Then add ammonia back to the zeolite-treated samples to confirm toxicity due to ammonia.

5.9.2. When it has been demonstrated that toxicity is due to ammonia because of increasing test pH, pH may be controlled using appropriate procedures which do not significantly alter the nature of the effluent, after submitting a written request to the Los Angeles Water Board, and receiving written permission expressing approval from the Executive Officer of the Los Angeles Water Board.

5.10. Chlorine Removal

Except with prior approval from the Executive Officer of the Los Angeles Water Board, chlorine shall not be removed from bioassay samples. Chlorine may be removed from the Valencia WRP effluent bioassay samples in the laboratory in order to simulate the dichlorination process at the facility.

6. LAND DISCHARGE MONITORING REQUIREMENTS (NOT APPLICABLE)

7. RECYCLING MONITORING REQUIREMENTS (NOT APPLICABLE)

8. RECEIVING WATER MONITORING REQUIREMENTS

8.1. Monitoring Location RSW-001, RSW-002, and RSW-003

The Permittee shall monitor Santa Clara River at RSW-001U, RSW-002D, and RSW-003D as follows:

Table E-5. Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Total flow	cfs	calculation	monthly	a
Turbidity	NTU	grab	monthly	b
Total residual chlorine	mg/L	grab	monthly	b
<i>E. coli</i>	MPN/100ml or CFU/100ml	grab	monthly	b
Temperature	°F	grab	monthly	b
pH	pH units	grab	monthly	b
Settleable solids	mL/L	grab	monthly	b

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Total suspended solids	mg/L	grab	monthly	b
BOD ₅ 20°C	mg/L	grab	monthly	b
Oil and grease	mg/L	grab	quarterly	b
Dissolved oxygen	mg/L	grab	monthly	b
Conductivity	µmho/cm	grab	monthly	b
Total Dissolved Solids	mg/L	grab	monthly	b
Sulfate	mg/L	grab	monthly	b
Chloride	mg/L	grab	monthly	b
Boron	mg/L	grab	monthly	b
Ammonia nitrogen	mg/L	grab	monthly	b
Nitrate nitrogen	mg/L	grab	monthly	b
Nitrite nitrogen	mg/L	grab	monthly	b
Organic nitrogen	mg/L	calculated	monthly	b
Total Kjeldahl nitrogen (TKN)	mg/L	grab	monthly	b
Total nitrogen	mg/L	calculated	monthly	b
Total phosphorus	mg/L	grab	monthly	b
Orthophosphate-P	mg/L	grab	monthly	b
Surfactants (MBAS)	mg/L	grab	quarterly	b
Surfactants (CTAS)	mg/L	grab	quarterly	b
Total hardness (CaCO ₃)	mg/L	grab	monthly	b
Chronic toxicity <i>Ceriodaphnia dubia</i> Survival and reproduction endpoints	Pass or Fail, (TST) and Percent Effect	grab	quarterly	b & c
Antimony	µg/L	grab	semiannually	b
Arsenic	µg/L	grab	quarterly	b
Beryllium	µg/L	grab	semiannually	b
Cadmium	µg/L	grab	semiannually	b
Chromium III	µg/L	calculated	semiannually	b
Chromium VI	µg/L	grab	semiannually	b
Copper	µg/L	grab	quarterly	b
Lead	µg/L	grab	semiannually	b
Mercury	µg/L	grab	quarterly	b & i
Nickel	µg/L	grab	semiannually	b
Selenium	µg/L	grab	monthly	b
Silver	µg/L	grab	semiannually	b

Parameter	Units	Sample Type	Minimum Sampling Frequency	Notes
Thallium	µg/L	grab	semiannually	b
Zinc	µg/L	grab	semiannually	b
Iron	µg/L	grab	monthly	b
Cyanide	µg/L	grab	monthly	b
Total trihalomethanes (TTHMs)	µg/L	grab	monthly	b
Methyl tert-butyl-ether (MTBE)	µg/L	grab	Semiannually	e
Perchlorate	µg/L	grab	semiannually	e
1,2,3-Trichloropropane	µg/L	grab	semiannually	e
1,4-Dioxane	µg/L	grab	semiannually	e
Diazinon	µg/L	grab	quarterly	b & f
TCDD equivalents	pg/L	grab	quarterly	b & g
PCBs as aroclors	µg/L	grab	annually	b & d
PCBs as congeners	pg/L	grab	annually	b & d
Remaining USEPA priority pollutants excluding asbestos and PCBs	µg/L	grab	semiannually	b & h

Footnotes for Table E-5

- When conditions at receiving water stations RSW-001 (R-C), RSW-002D (R-D), or RSW-003D (R-E) prevent accurate measurement of flow, the flow may be quantitatively estimated and reported.
- Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Los Angeles Water Board or State Water Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Attachment 4 of the SIP, the analytical method with the lowest ML must be selected.
- The Permittee shall conduct toxicity testing using *Ceriodaphnia dubia* as the test species, as outlined in section 5 of this MRP. For the *Ceriodaphnia dubia* reproduction endpoint, the median monthly effluent limitation (MMEL) summary result shall be reported as "Pass" or "Fail" and the maximum daily single result shall be reported as "Pass" or "Fail" and "% Effect." The *Ceriodaphnia dubia* reproduction endpoint shall be reported as "Pass" or "Fail" and "% Effect". The *Ceriodaphnia dubia* survival endpoint, shall be reported as "% Effect." If the chronic aquatic toxicity routine monitoring test results in a "Fail" at the instream waste concentration (IWC), then the Valencia WRP shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the "Fail" at the IWC. If the first chronic MMEL compliance test results in a "Fail" at the IWC, then the second MMEL compliance test is not necessary because the "Fail" results from the first two tests constitutes a violation of the chronic toxicity MMEL. Receiving water and effluent toxicity testing shall be performed on the same day as close to concurrently as possible.

- d. PCBs as aroclors shall be analyzed using USEPA method 608. PCBs as congeners shall be analyzed using method 1668c. USEPA recommends that until the USEPA proposed method 1668c is incorporated into 40 CFR 136, permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608 for monitoring data, reported as aroclor results, that will be used for determining compliance with WQBELs (if applicable) and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes.
- e. Emerging chemicals include 1,4-dioxane (USEPA 8270M test method), perchlorate (USEPA 314 test method, or USEPA method 331 if a detection limit of less than 6 µg/L is achieved), 1,2,3-trichloropropane (USEPA 504.1, 8260B test method, or USEPA 524.2 in SIM mode), and methyl tert-butyl ether (USEPA 8260B test method or USEPA method 624 if a detection level of less than 5 µg/L is achieved, and if the Permittee received ELAP certification to run USEPA method 624).
- f. Diazinon sampling shall be conducted concurrently with the chronic toxicity receiving water quality sampling.
- g. In accordance with the SIP, the Permittee shall conduct monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the effluent and in the receiving water. The Permittee shall use the appropriate Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i) concentration analytical result (Ci) and their corresponding Toxicity Equivalence Factor (TEFi), (i.e., $TEQ_i = C_i \times TEF_i$). The Discharger shall report all measured values of individual congeners, including data qualifiers. TCDD equivalents concentration shall be reported as the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

- h. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.
- i. The mercury effluent samples shall be analyzed using EPA method 1631E, per 40 CFR part 136, with a quantification level lower than 0.5 ng/L to analyze total mercury. If an alternative method with an equivalent or more sensitive method detection limit is approved in 40 CFR part 136, the Discharger may use that method in lieu of USEPA Method 1631E.

End of Footnotes for Table E-5

Receiving water samples shall not be taken during or within 48-hours following the flow of rainwater runoff into the Santa Clara River. Sampling may be rescheduled at receiving water stations if weather and/or flow conditions would endanger personnel collecting receiving water samples. The monthly monitoring report shall note such occasions.

8.2. Receiving Groundwater Monitoring Requirements

- 8.2.1. The Discharger shall maintain groundwater sampling by monitoring the groundwater aquifer at station RGW-001 (Well No. 4N/17W-14Q3) and submitting the results together with the effluent monitoring reports, for the following pollutants:

Table E-6. Groundwater Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Note
Nitrate-N + Nitrite-N	mg/L	grab	semiannually	a
Total dissolved solids (TDS)	mg/L	grab	semiannually	a

Parameter	Units	Sample Type	Minimum Sampling Frequency	Note
Chloride	mg/L	grab	semiannually	a
Sulfate	mg/L	grab	semiannually	a
Boron	mg/L	grab	semiannually	a
Chromium III	µg/L	calculation	semiannually	a
Chromium IV	µg/L	grab	semiannually	a
Total Chromium	µg/L	grab	semiannually	a
Priority pollutants excluding asbestos	µg/L	grab	semiannually	a
MTBE	µg/L	grab	semiannually	b
Perchlorate	µg/L	grab	semiannually	b
1,4-Dioxane	µg/L	grab	semiannually	b
1,2,3-Trichloropropane	µg/L	grab	semiannually	b
Total organic carbon	mg/L	grab	semiannually	c

Footnotes for Table E-6

- a. Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Los Angeles Water Board or State Water Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Attachment 4 of the SIP, the analytical method with the lowest ML must be selected.
- b. Emerging chemicals include 1,4-dioxane (USEPA 8270M test method), perchlorate (USEPA 314 test method, or USEPA method 331 if a detection limit of less than 6 µg/L is achieved), 1,2,3-trichloropropane (USEPA 504.1, 8260B test method, or USEPA 524.2 in SIM mode), and methyl tert-butyl ether (USEPA 8260B test method or USEPA method 624 if a detection level of less than 5 µg/L is achieved, and if the Permittee received ELAP certification to run USEPA method 624).
- c. Total organic carbon (TOC) monitoring may serve as a surrogate for other pollutants. Determinations of TOC can provide valuable diagnostic evidence of the extent of groundwater contamination by organic compounds. (Michael J. Barcelona, *Groundwater* , v22, no.1. Jan-Feb 1984)

End of Footnotes for Table E-6

9. OTHER MONITORING REQUIREMENTS

9.1. Watershed Monitoring

- 9.1.1. The goals of the Watershed-wide Monitoring Program for the Santa Clara River Watershed are to:
 - a. Determine compliance with receiving water limits.
 - b. Monitor trends in surface water quality.
 - c. Ensure protection of beneficial uses.

- d. Provide data for modeling contaminants of concern.
 - e. Characterize water quality including seasonal variation of surface waters within the watershed.
 - f. Assess the health of the biological community.
 - g. Determine mixing dynamics of effluent and receiving waters in the estuary.
- 9.1.2. To achieve the goals of the Watershed-wide Monitoring Program, the Permittee shall undertake the responsibilities delineated under an approved watershed-wide monitoring plan in the implementation of the Watershed-wide Monitoring Program for the Santa Clara River, which was approved by the Los Angeles Water Board on July 3, 2012.
- 9.1.3. In coordination with interested stakeholders in the Santa Clara River Watershed, the Permittee shall conduct instream bioassessment monitoring once a year, during the spring/summer period (unless an alternate sampling period is approved by the Executive Officer) and include an analysis of the community structure of the instream macroinvertebrate assemblages, the community structure of the instream algal assemblages, chlorophyll a and biomass for instream algae, and physical habitat assessment at the random monitoring stations designated by the Santa Clara River Watershed Monitoring Program. Over time, bioassessment monitoring will provide a measure of the physical condition of the waterbody and the integrity of its biological communities.
- a. The bioassessment program shall include an analysis of the community structure of the instream macroinvertebrate, algal assemblages, algal biomass, and physical habitat assessment at monitoring stations RSW-001U and RSW-002D.
- This program shall be implemented by appropriately trained staff. Alternatively, a professional subcontractor qualified to conduct bioassessments may be selected to perform the bioassessment work for the Permittee. Analyses of the results of the bioassessment monitoring program, along with photographs of the monitoring site locations taken during sample collection, shall be submitted in the corresponding annual report. If another stakeholder, or interested party in the watershed subcontracts a qualified professional to conduct bioassessment monitoring during the same season and at the same location as specified in the MRP, then the Permittee may, in lieu of duplicative sampling, submit the data, a report interpreting the data, photographs of the site, and related QA/QC documentation in the corresponding annual report.
- b. The Permittee must provide a copy of their Standard Operation Procedures (SOPs) for the Bioassessment Monitoring Program to the Los Angeles Water Board upon request. The document must contain step-by-step field, laboratory and data entry procedures, as well as related QA/QC procedures. The SOP must also include specific information about each bioassessment program including assessment program description, its organization and the responsibilities of all its personnel; assessment project description and objectives; qualifications of all personnel; and the type of training each member has received.

- c Field sampling must conform to the SOPs established in the Surface Water Ambient Monitoring Program's (SWAMP) Standard Operating Procedures for the Collection of Field Data for Bioassessment of California Wadable Streams: Benthic Macroinvertebrates, Algae and Physical Habitat. Field crews shall be trained on aspects of the protocol and appropriate safety issues. All field data and sample Chain of Custody (COC) forms must be examined for completion and gross errors. Field inspections shall be planned with random visits and shall be performed by the Permittee or an independent auditor. These visits shall report on all aspects of the field procedure with corrective action occurring immediately.
- d. A taxonomic identification laboratory shall process the biological samples that usually consist of subsampling organisms, enumerating and identifying taxonomic groups and entering the information into an electronic format. The Los Angeles Water Board may require QA/QC documents from the taxonomic laboratories and examine their records regularly. Intra-laboratory QA/QC for subsampling, taxonomic validation and corrective actions shall be conducted and documented. Biological laboratories shall also maintain reference collections, vouchered specimens (the Permittee may request the return of their sample voucher collections) and remnant collections. The laboratory should participate in an (external) laboratory taxonomic validation program at a recommended level of 10% or 20%. External QA/QC may be arranged through the California Department of Fish and Wildlife's Aquatic Bioassessment Laboratory located in Rancho Cordova, California.

9.1.4. The Executive Officer of the Los Angeles Water Board may modify the Monitoring and Reporting Program to accommodate the watershed-wide monitoring.

9.2. Tertiary Filter Treatment Bypasses

- 9.2.1. During any day that filters are bypassed, the Permittee shall monitor the effluent daily for BOD₂₀[°], suspended solids, settleable solids, and oil and grease, until it is demonstrated that the filter "bypass" has not caused an adverse impact on the receiving water.
- 9.2.2. The Permittee shall maintain chronological log of tertiary filter treatment process bypasses, to include the following:
 - a. Date and time of bypass start and end;
 - b. Total duration time; and,
 - c. Estimated total volume bypassed.
- 9.2.3. The Permittee shall notify Los Angeles Water Board staff by telephone within 24 hours of the filter bypass event.
- 9.2.4. The Permittee shall submit a written report to the Los Angeles Water Board, according to the corresponding monthly self-monitoring report schedule. The report shall include, at a minimum, the information from the chronological log. Results from the daily effluent monitoring, required by 9.2.1. above, shall be verbally reported to the Los Angeles Water Board as the results become available and submitted as part of the monthly SMR.

9.3. Monitoring of Volumetric Data for Wastewater and Recycled Water

The State Water Board adopted “Water Quality Control Policy for Recycled Water” (Recycled Water Policy) on December 11, 2018 and the Recycled Water Policy became effective on April 8, 2019. The Recycled Water Policy requires wastewater and recycled water dischargers to annually report monthly volumes of influent, wastewater produced, and effluent, including treatment level and discharge type. As applicable, dischargers are additionally required to annually report recycled water use by volume and category of reuse. The State Water Board issued a Water Code section 13267 and 13383 Order, Order WQ 2019-0037 EXEC, on July 24, 2019 to amend MRPs for all permits of NPDES, WDRs, WRRs, Master Recycling, and General WDRs. Annual reports are due by April 30 of each year, and the report must be submitted to GeoTracker. This Order implements the Recycled Water Policy by incorporating the volumetric monitoring reporting requirements in accordance with section 3 of the [Recycled Water Policy](https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/121118_7_final_amendment_oal.pdf) (https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/121118_7_final_amendment_oal.pdf). The State Water Board’s Order WQ 2019-0037 EXEC will no longer be applicable to the Discharger upon the effective date of this Order.

9.3.1. Influent: The Discharger shall monitor monthly total volume of wastewater collected and treated by the wastewater treatment plant.

9.3.2. Production: The Discharger shall monitor monthly volume of wastewater treated, specifying level of treatment.

9.3.3. Discharge: The Discharger shall monitor monthly volume of treated wastewater discharged to specific water bodies as categorized in the section 3.2.3 of the Recycled Water Policy. The level of treatment shall also be specified.

9.3.4. Reuse: The Discharger shall monitor monthly volume of recycled water distributed, and annual volume of treated wastewater distributed for beneficial use in compliance with California Code of Regulations, Title 22 in each of the use categories specified in section 3.2.4 of the Recycled Water Policy.

10. REPORTING REQUIREMENTS

10.1. General Monitoring and Reporting Requirements

10.1.1. The Permittee shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

10.1.2. If there is no discharge during any reporting period, the report shall so state.

10.1.3. Each monitoring report shall contain a separate section titled *Summary of Non-Compliance* which discusses the compliance record and the corrective actions taken or planned that may be needed to bring the discharge into full compliance with waste discharge requirements. This section shall clearly list all non-compliance with discharge requirements, as well as all excursions of effluent limitations.

10.1.4. The Permittee shall inform the Los Angeles Water Board well in advance of any proposed construction activity that could potentially affect compliance with applicable requirements.

10.2. Self-Monitoring Reports (SMRs)

10.2.1. The Permittee shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.

10.2.2. The Permittee shall report in the SMR the results for all monitoring specified in this MRP under sections 3 through 9. The Permittee shall submit monthly, quarterly, semiannual, and annual SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Permittee samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this MRP using approved analytical methods, the results of those analyses shall be included in the report. These results shall be reflected in the calculation of the average used in demonstrating compliance with limitations set forth in this Order.

10.2.3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-7. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On	Monitoring Period	SMR Due Date
Continuous	Order effective date	All	Submit with monthly SMR
Daily	Order effective date	(Midnight thru 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	Submit with monthly SMR
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	Submit with monthly SMR
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	By the 15 th day of the third month after the month of sampling
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	June 15 September 15 December 15 March 15

Sampling Frequency	Monitoring Period Begins On	Monitoring Period	SMR Due Date
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 thru June 30 July 1 thru December 31	September 15 March 15
Annually	January 1 following (or on) permit effective date	January 1 thru December 31	April 30
Annually (volumetric reporting)	Order effective date	January 1 thru December 31	April 30
Annually (Pretreatment Program)	January 1 following (or on) permit effective date	January 1 thru December 31	April 30

10.2.4. Reporting Protocols. The Permittee shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR part 136.

The Permittee shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- a. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.
- c. For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.
- d. Sample results less than the laboratory's MDL shall be reported as "Not Detected," or "ND".
- e. Permittees are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

10.2.5. Compliance Determination. Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above and section 7 of this Order. For purposes of reporting and administrative enforcement by the Los Angeles Water Board and State Water Board, the Permittee shall be deemed out of compliance with effluent limitations if the concentration of the priority

pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

10.2.6. Multiple Sample Data. When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Permittee shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure:

- a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
- b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

10.2.7. The Permittee shall submit SMRs in accordance with the following requirements:

- a. The Permittee shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Permittee is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Permittee shall electronically submit the data in a tabular format as an attachment.
- b. The Permittee shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.

10.3. Discharge Monitoring Reports (DMRs)

10.3.1. DMRs are U.S. EPA reporting requirements. The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic DMR submittal shall be in addition to electronic SMR submittal. Information about electronic DMR submittal is available at the [DMR website](http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring) at:
http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring.

10.4. Other Reports

10.4.1. The Permittee shall report the results of any special studies, chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions – section 6.3. The Permittee shall submit reports in compliance with SMR reporting requirements described in subsection 10.2. above.

10.4.2. Annual Summary Report

By April 30 of each year, the Permittee shall submit an annual report containing a discussion of the previous year's influent/effluent analytical results and receiving water monitoring data. The annual report shall contain an overview of any plans for upgrades to the treatment plant's collection system, the treatment processes, or the outfall system. The Permittee shall submit annual report to the Los Angeles Water Board in accordance with the requirements described in subsection 10.2.7 above.

10.4.3. Each annual monitoring report shall contain a separate section titled *Reasonable Potential Analysis* which discusses whether or not reasonable potential was triggered for pollutants which do not have a final effluent limitation in the NPDES permit. This section shall contain the following statement: "The analytical results for this sampling period did/ did not trigger reasonable potential." If reasonable potential was triggered, then the following information should also be provided:

- a. A list of the pollutant(s) that triggers reasonable potential.
- b. The Basin Plan or California Toxics Rule (CTR) criteria that was exceeded for each given pollutant.
- c. The concentration of the pollutant(s).
- d. The test method used to analyze the sample.
- e. The date and time of sample collection.

10.4.4. The Permittee shall submit to the Los Angeles Water Board, together with the monitoring report required by this Order, a list of all chemicals and proprietary additives which could affect this waste discharge, including quantities of each. Any subsequent changes in types and/or quantities shall be reported promptly.

10.4.5. The Los Angeles Water Board requires the Permittee to file with the Los Angeles Water Board, within 90 days after the effective date of this Order, a technical report on preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. The technical report should:

- a. Identify the possible sources of accidental loss, untreated waste bypass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks, and pipes should be considered.
- b. Evaluate the effectiveness of present facilities and procedures and state when they become operational.
- c. Describe facilities and procedures needed for effective preventive and contingency plans.
- d. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule contingent interim and final dates when they will be constructed, implemented, or operational.

10.4.6. Climate Change Effects Vulnerability Assessment and Mitigation Plan:

The Permittee shall consider the impacts of climate change as they affect the operation of the treatment facility due to flooding, wildfires, or other climate-related changes. The Permittee shall develop a Climate Change Effects Vulnerability Assessment and Mitigation Plan (Climate Change Plan) to assess and manage climate change-related effects that may impact the wastewater treatment facility's operation, water supplies, its collection system, and water quality, including any projected changes to the influent water temperature and pollutant concentrations, and beneficial uses. For facilities that discharge to the ocean including desalination plants, the Climate Change Plan shall also include the impacts from sea level rise. The Climate Change Plan is due 12 months after the effective date of this Order.

10.4.7. Annual Volumetric Reporting of Wastewater and Recycled Water

The Discharger shall electronically submit annual volumetric reports to the State Water Board by April 30 each year covering data collected during the previous calendar year using the [State Water Board's GeoTracker website](https://geotracker.waterboards.ca.gov) (geotracker.waterboards.ca.gov) under a site-specific global identification number. The annual volumetric report shall include information specified in section 9.3, above. A report upload confirmation from the GeoTracker shall be included in the annual report, which shall be submitted into CIWQS, by the report due date to demonstrate compliance with this reporting requirement.

10.4.8. Annual Pretreatment Reporting

The Permittee shall electronically submit annual pretreatment reports to the Los Angeles Water Board and to USEPA Region 9 by April 30 each year, covering data collected during the previous calendar year, in accordance with Pretreatment Reporting Requirements (Attachment H).

10.4.9. Semiannual and Annual Upper Santa Clara River Chloride TMDL Reporting

The Discharger shall electronically submit progress reports to the Los Angeles Water Board on a semiannual basis for Task 4 of the TMDL and on an annual basis for Task 3 of the TMDL, referenced in section 6.3.8 of the WDR for Order No. R4-2022-0174. Progress reports shall include, but not be limited to, supporting documentation that tasks were completed by the deadline.

ATTACHMENT F. FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in section 2.2 of this Order, the Los Angeles Water Board incorporates this Fact Sheet as findings of the Los Angeles Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Permittees in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Permittee. sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Permittee.

1. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	4A190107023
Discharger	Santa Clarita Valley Sanitation District (SCVSD, Discharger, or Permittee)
Name of Facility	Valencia Water Reclamation Plant including its associated wastewater collection system and outfalls
Facility Address	28185 The Old Road Santa Clarita, CA 91355 Los Angeles County
Facility Contact, Title and Phone	Lysa Gaboudian, Supervising Engineer, (562) 908-4288 x2811
Authorized Person to Sign and Submit Reports	Lysa Gaboudian, Supervising Engineer, (562) 908-4288 x2811
Mailing Address	1955 Workman Mill Road, Whittier, CA 90601
Billing Address	Same as Mailing Address
Type of Facility	POTW
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	Yes
Recycling Requirements	Producer/User
Facility Permitted Flow	21.6 MGD
Facility Design Flow	21.6 MGD
Watershed	Santa Clara River
Receiving Water	Santa Clara River
Receiving Water Type	Inland surface water

- 1.1. The Santa Clarita Valley Sanitation District (hereinafter SCVSD, Discharger, or Permittee) owns and operates a Publicly Owned Treatment Works (POTW) comprised of the Valencia Water Reclamation Plant (hereinafter Valencia WRP or Facility) and its associated wastewater collection system and outfalls.

For the purposes of this Order, references to the “Permittee” or “Discharger” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Permittee herein.

- 1.2. The Facility discharges wastewater to the Santa Clara River, a water of the United States and State of California. The Permittee was previously regulated by Order R4-2015-0071 and NPDES Permit Number CA0054216 adopted on April 9, 2015 and taking effect on May 3, 2015. This Order expired on April 30, 2020.

Regulations at 40 CFR section 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. Accordingly, Table 3 of this Order limits the duration of the discharge authorization. However, pursuant to California Code of Regulations, title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Discharger complies with all federal NPDES requirements for continuation of expired permits. The Permittee filed a report of waste discharge and submitted an application for reissuance of its WDRs and NPDES permit on November 1, 2019. The application was deemed incomplete on January 2, 2020. SCVSD submitted supplemental information on January 18, January 22, and March 27, 2020. The application was deemed complete on April 20, 2020. A site visit was conducted in March 2022, to observe operations and collect additional data to develop permit limitations and conditions. The terms and conditions of the current NPDES order have been automatically continued and remain in effect until new WDRs and NPDES permit are adopted pursuant to this Order. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

- 1.3. Applicable, state law requires dischargers to file a petition with the State Water Board, Division of Water Rights (DWR) and receive approval for any change in the point of discharge, place of use, or purpose of use of treated wastewater that decreases the flow in any portion of a watercourse. On October 21, 1993, the State Water Board DWR approved wastewater change petition Number WW0011, which allows the Permittee to reduce the treated effluent discharge for the purpose of irrigation. SCVSD has not filed a recent wastewater use petition. However, On October 4, 2021, Santa Clarita Valley Water Agency (SCV Water), the distributor and purveyor of recycled water in the Santa Clarita service area, contacted the State Water Board DWR to request approval to use water from the New Drop Program for irrigation purposes. On November 9, 2021, the State Water Board DWR notified SCV Water that a wastewater change petition was not needed for its New Drop Program because the New Drop Project would not decrease Santa Clara River flow. The New Drop Project would rely on “new flows” produced from new development in the SCV Water service area that have not previously been discharged to the Santa Clara River. These “new flows” referred to as New Drop water will be provided as influent to the Valencia WRP. Recycled water generated from New Drop sources will be utilized from the Valencia WRP to initially provide, during early phases of development at Five Point Valencia, up to up to 3,092 acre-feet per year for irrigation use to new homes, commercial-industrial, retail-restaurants, and hotels-

schools in the SCV Water service area. SCV Water is currently enrolled under the State Water Board's Order WQ 2016-0068-DDW, Water Reclamation Requirements for Recycled Water Use (General Recycled Water Order) to use recycled water for dust control purposes. However, they plan on preparing a Title 22 Engineering Report that would be submitted to the State Water Board Division of Drinking Water for approval of the following additional uses of recycled water: landscape irrigation, dual-plumbed irrigation, and landscape impoundments. SCV Water proposes to complete the recycled water irrigation projects under this request by 2030. Once the State Water Board DDW approves the Title 22 Engineering Report, SCV Water will submit a Notice of Intent to the Los Angeles Water Board, to revise their enrollment under the General Recycled Water Order. The State Water Board retains separate jurisdictional authority to enforce any applicable requirements under Water Code section 1211. This is not an NPDES permit requirement.

2. FACILITY DESCRIPTION

2.1. Description of Wastewater and Biosolids Treatment and Controls

2.1.1. The Valencia WRP, a tertiary wastewater treatment plant located at 28185 The Old Road, Santa Clarita, CA 91355, currently receives wastewater from the City of Santa Clarita and unincorporated areas of Los Angeles County. The wastewater is a mixture of domestic wastewater and industrial wastewater that is pre-treated pursuant to 40 CFR part 403. Valencia WRP has a design capacity of 21.6 MGD and serves an estimated population of 185,119 people.

The Valencia WRP is part of an integrated network of facilities, known as the SCVSD. The SCVSD currently incorporates the Valencia WRP and the Saugus WRP, which are connected by interceptors and trunk sewers. The Valencia WRP will temporarily treat the sewage generated by the first 6,000 dwelling units of Newhall Ranch, until such time as the Newhall Ranch WRP is built, as described in the amendment to the Valencia WRP NPDES permit, R4-2009-0074-A01, adopted by the Los Angeles Water Board on December 5, 2013. Once the Newhall Ranch WRP is built and operational, the biosolids generated at the Newhall Ranch WRP will be sent to the Valencia WRP for processing until such time as the Newhall Ranch WRP has its own solids handling facilities. The SCVSD system allows for the diversion of influent flows from the Saugus WRP to the Valencia WRP if so desired.

The interim Valencia-Newhall wastewater treatment scenario is described in further detail in a separate Los Angeles Water Board Order No. R4-2012-0139, *Clean Water Act Section 401 Water Quality Certification and Waste Discharge Requirements (401 WDRs) for Newhall Land and Farming Company* (File No. 11-168), adopted by the Los Angeles Water Board on September 14, 2012. Order No. R4-2012-0139, requires that: "For purposes of further treating wastewater (to a chloride level of 100 mg/l or less for up to 6000 equivalent dwelling units) from Newhall Ranch that will be sent to the Valencia WRP, Newhall Land, or its successor, shall complete construction of interim chloride and demineralization facilities to the satisfaction of the Los Angeles Water Board prior to discharging sewage from Newhall Land to the Valencia WRP or other publicly owned treatment works. The interim chloride and demineralization facilities shall be sufficient to

ensure that any wastewater discharge attributable to Newhall Ranch does not result in discharge to the Santa Clara River of effluent containing chloride in concentrations exceeding 100 mg/L. If sewage from Newhall Land does not already meet the chloride limit of 100 mg/L, an equivalent volume of effluent shall be removed from the combined Newhall/Valencia partially treated waste stream and shall be treated at the interim chloride and demineralization facility to meet 100 mg/L chloride prior to discharge.” Newhall Land will install temporary equipment and/or install facilities, at the Valencia WRP and/or adjacent to the Valencia WRP, to reduce chloride concentrations, as required by Los Angeles Water Board Order No. R4-2012-0139. Refer to the multiple Process Flow Diagrams in Attachment C to see the multiple treatment scenarios. Newhall may discharge the permeate (high quality product water) from the chloride reducing facility to (1) the Valencia WRP sewer system, authorized under a separate industrial user permit issued by SCVSD’s pretreatment program staff or (2) upstream of the UV disinfection process at the Valencia WRP.

- 2.1.2. The Discharger has undertaken a full evaluation of local limits for the SCVSD in the Santa Clara River Watershed. The most recent local limits evaluation was submitted to the Los Angeles Water Board on October 29, 2015. In that evaluation, SCVSD found that the existing limits were fully protective of the SCVSD system. However, a re-evaluation will be necessary and the Permittee shall submit to the Los Angeles Water Board, within 180 days of the effective date of Order No. R4-2022-0174, a revised local limits evaluation for Los Angeles Water Board approval.
- 2.1.3. Treatment at the Valencia WRP consists of influent screening, primary sedimentation, primary effluent equalization, activated sludge biological treatment with nitrification and denitrification, secondary sedimentation, inert media filtration, disinfection (UV and/or chloramination), and dechlorination (as needed). Treated wastewater discharged to Santa Clara River is dechlorinated when chloramines are used as the disinfecting agent.
- 2.1.4. Hypochlorite and ammonia are still used to disinfect treated effluent at the Valencia WRP prior to discharging the effluent to the Santa Clara River. SCVSD plans on using chloramines as the disinfecting agent for its recycled water program, which is regulated under separate Waste Discharge Requirements that serve as Water Reclamation Requirements. However, once the UV disinfection system is optimized, UV disinfection will replace the chloramine disinfection system for portion of the effluent that is discharged to the Santa Clara River. Disinfecting agent is added to the treated effluent prior to the filters to destroy bacteria, pathogens and viruses, and to minimize algal growth in the filters. Additional disinfectant may be dosed prior to the serpentine chlorine contact chamber. Prior to discharge, sulfur dioxide and/or sodium bisulfite is added to the treated effluent to remove residual chlorine.
- 2.1.5. Biosolids from both the Valencia and the Saugus WRPs are processed at the Valencia WRP and hauled offsite. Waste activated sludge from both the Saugus and Valencia WRPs is thickened by dissolved air flotation. Thickened waste activated sludge and primary sludge are combined, digested anaerobically, dewatered and then reused through composting, although the permittee may also dispose of

biosolids in a landfill or by direct land application. Attachment C contains schematics of the Valencia WRP wastewater flow.

- 2.1.6. The Permittee has constructed a biological nutrient removal system with a nitrogen denitrification process (NDN) in order to achieve compliance with the ammonia Basin Plan objectives. The system was completed and has been in operation since September 2003. SCVSD is making progress towards completing the UV disinfection, Micro Filtration, and Reverse Osmosis (MF/RO) capital improvement projects. in accordance with the milestones established in Time Schedule Order No. R4-2019-0055-A02, adopted by the Los Angeles Water Board on July 8, 2021. SCVSD began constructing the MF/RO treatment units and should complete construction by December 31, 2022.

2.2. Discharge Points and Receiving Waters

The Valencia WRP discharges tertiary-treated effluent to the Santa Clara River, a water of the United States, upstream of the Santa Clara River Estuary. Treated effluents are discharged from the plant to surface waters at the following discharge points:

Discharge Point 001 (approximate coordinates: 34.430546°N, 118.592546°W): Discharge to Reach 5 of the Santa Clara River, above the Estuary, and within the Santa Clara River Watershed.

Discharge Point 002 (approximate coordinates: 34.430075°N, 118.592208°W): Under high stream conditions, treated effluent is discharged through this discharge point. During dry season (May 1 – October 31), the primary sources of water flow in Santa Clara River, downstream of the discharge point, are the Valencia WRP effluent and other NPDES-permitted discharges, including urban runoff conveyed through the municipal separate storm sewer systems (MS4). Stormwater and dry weather urban runoff from the MS4 are regulated under an NPDES permit, *Waste Discharge Requirements for Municipal Storm Water and NPDES permit for Municipal Separate Storm Sewer System (MS4) Discharges within the Coastal Watersheds of Los Angeles and Ventura Counties*, Order Number R4-2021-0105, NPDES Permit Number CAS004004.

The Santa Clara River is unlined at the point of discharge. Groundwater recharge occurs incidentally in these unlined areas of the Santa Clara River where the underlying sediments may be transmissive to water as well as pollutants.

The Santa Clara River Watershed is the largest river system in southern California that remains in a relatively natural state and drains about 1,200 square miles; this is a high-quality natural resource for much of its approximately 100 miles length. The river originates in the northern slope of the San Gabriel Mountains in Los Angeles County, traverses Ventura County, and flows into the Pacific Ocean halfway.

Extensive patches of high-quality riparian habitat are present along the length of the river and its tributaries. An endangered fish, the unarmored stickleback, is resident in the river. One of the largest of the Santa Clara River's tributaries, Sespe Creek, is designated a wild trout stream by the State of California and supports significant spawning and rearing habitat. Sespe Creek is also designated a wild and scenic river. Piru and Santa Paula Creeks, which are tributaries to the Santa Clara River, also support good habitats for

steelhead. In addition, the river serves as an important wildlife corridor. A lagoon exists at the mouth of the river and supports a large variety of wildlife.

2.3. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

There was no discharge from Discharge Point 002. Effluent limitations contained in the existing Order for discharges from Discharge Point 001 (Monitoring Locations EFF-001 and EFF-002) and representative monitoring data, collected between May 2015 and July 2021, are as follows:

Table F-2. Historic Effluent Limitations and Monitoring Data

Constituent	Unit	Current AMEL	Current AWEL	Current MDEL	Max. Monthly Conc.	Max. Weekly Conc.	Max. Daily Conc.
BOD ₅ 20°C	mg/L	20	30	45	<0.6	--	11.7
Removal Percentage for BOD	%	85%	--	--	≥99%	--	--
Total Suspended Solids (TSS)	mg/L	15	40	45	<2.5	---	<2.5
Removal Percentage for TSS	%	85%	--	--	≥99%	--	--
Oil and Grease	mg/L	10	---	15	<0.8	---	1.7
Settleable Solids	mL/L	0.1	---	0.3	<0.1	---	<0.1
pH	S.U.	6.5	--	8.5	6.9	--	7.9
Temperature	°F	--	--	86	87	88	89
Total Residual Chlorine	mg/L	---	---	0.1	<0.03	---	0.2
Total Dissolved Solids	mg/L	1,000	---	---	694	---	879
Chloride	mg/L	100	---	230	147	---	147
Sulfate	mg/L	400	---	---	226	---	226
Boron	mg/L	1.5	---	---	0.46	---	0.68
Nitrate-N + Nitrite-N	mg/L	6.8	---	---	4.79	---	4.4
Nitrite-N	mg/L	0.9	---	0.9	<0.0012	---	0.177
Nitrate-N	mg/L	6.8	---	---	---	---	4.4
Ammonia-N	mg/L	1.75	---	5.2	0.887	---	1.39
MBAS	mg/L	0.5	---	---	<0.08	---	0.12
Copper	µg/L	12	---	39	1.8	---	3.84
Cyanide	µg/L	4.7	---	7.0	3.0	---	6.4
2,3,7,8-TCDD	pg/L	0.014	---	0.028	0.6	---	2.1
Selenium	µg/L	4.5	---	6.8	0.35	---	0.67
Bis(2-ethylhexyl) phthalate	µg/L	4	---	---	0.51	---	2
Iron	µg/L	300	---	---	91	---	140
Total Trihalomethanes	µg/L	80	---	---	35	---	61.5
Total Coliform	CFU/100 mL	23	2.2	240	1	≤1	61

Constituent	Unit	Current AMEL	Current AWEL	Current MDEL	Max. Monthly Conc.	Max. Weekly Conc.	Max. Daily Conc.
Chronic Toxicity	Pass or Fail, Percent Effect	Pass	--	Pass or Percent effect ≤ 50	Fail (3 times)	--	Percent Effect of 25%

2.4. Compliance Summary

The following table lists the Facility's exceedances of effluent limitations in Order R4-2015-0071 that occurred during the period from May 2015 to October 2021.

Table F-3. List of Non-Compliance

Date of Occurrence	Description of Exceedance
11/28/2015	Failed the Chronic Toxicity Monthly Median Effluent Limitation with the Ceriodaphnia dubia species, for the Reproduction endpoint
12/17/2015	Failed the Chronic Toxicity Monthly Median Effluent Limitation with the Ceriodaphnia dubia species, for the Reproduction endpoint

Monitoring data indicate that the Permittee has consistently complied with the final effluent limitations of Order Number R4-2015-0071. However, there were violations associated with the chronic toxicity final effluent limitation. These chronic toxicity exceedances were intermittent and the Discharger was unable to determine the cause of the toxicity. The effluent was back in compliance in January 2016. The Discharger also had non-reportable results for required monitoring in November 2020 and in August 2021 because quality assurance/quality control failures invalidated results for effluent bis(2-ethylhexyl) phthalate and influent and effluent BOD, respectively. Make-up samples were collected and successfully analyzed and reported in subsequent reporting periods.

2.5. Planned Changes

SCVSD plans on meeting the TMDL-based chloride final effluent limitation by upgrading its disinfection system, designing, and constructing additional treatment equipment, in accordance with the deadlines established in TSO No. R4-2019-0055 and its amendments. The UV technology will be designed to disinfect the treated effluent, lower the amounts of disinfection byproducts produced, as well as reduce chloride concentrations by approximately 7 mg/L.

3. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

3.1. Legal Authorities

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to

section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA) and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit authorizing the Discharger to discharge into waters of the United States at the discharge locations described in Table 2 subject to the WDRs in this Order.

3.2. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of chapter 3 of CEQA, (commencing with section 21100) of division 13 of the Public Resources Code.

3.3. State and Federal Laws, Regulations, Policies, and Plans

3.3.1. Water Quality Control Plan

The Water Quality Control Plan for the Los Angeles Region (Basin Plan) designates beneficial uses, establishes water quality objectives (WQOs), and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan.

Beneficial uses applicable to Santa Clara River and Santa Clara River Estuary are as follows:

Table F-4a. Basin Plan Beneficial Uses – Receiving Waters

Discharge Point	Hydrologic Unit Code (HUC)	Receiving Water Name	Beneficial Use(s)
001 & 002	180701020403 (formerly Calwater Hydro Unit 403.51)	Santa Clara River Reach 5 (Blue Cut gaging station to West Pier Hwy 99)	<u>Existing:</u> IND; PROC; AGR; GWR; FRSH; REC-1; REC-2; RARE; WARM; WILD; and, wetland habitat (WET), <u>Potential:</u> p*MUN
001 & 002	180701020403 (Hydro unit 403.41)	Santa Clara River Reach 4B (Piru Creek to Blue Cut gaging station)	<u>Existing:</u> IND, PROC, AGR, GWR, FRSH, REC-1, REC-2, RARE, MIGR, WARM, WILD, WET <u>Potential:</u> p*MUN
001 & 002	180701020403 (formerly Calwater Hydro Unit 403.31)	Santa Clara River reach 4A (A Street to Piru Creek)	<u>Existing:</u> IND, PROC, AGR, GWR, FRSH, REC-1, REC-2, RARE, MIGR, WARM, WILD, and WET, <u>Potential:</u> p*MUN

Discharge Point	Hydrologic Unit Code (HUC)	Receiving Water Name	Beneficial Use(s)
001 & 002	180701020903, 180701020902 &180701020802 (Hydro unit 403.21)	Santa Clara River Reach 3 (Freeman Diversion to Santa Paula Creek, Santa Paula Creek to Sespe Creek, & Sespe Creek to A Street, Fillmore)	<u>Existing:</u> IND, PROC, AGR, GWR, FRSH, REC-1, REC-2, RARE, MIGR, WARM, WILD, and WET, <u>Potential:</u> p*MUN
001 & 002	180701020904 & 180701020903 (Hydro unit 403.11)	Santa Clara River Reach 2 (Highway 101 bridge to Ellsworth Barranca & Ellsworth Barranca to Freeman Diversion)	<u>Existing:</u> IND, PROC, AGR, GWR, FRSH, REC-1, REC-2, RARE, MIGR, WARM, COLD, WILD, and WET, <u>Potential:</u> p*MUN
001 & 002	180701020904 (Hydro unit 403.11)	Santa Clara River Reach 1 (Estuary to Highway 101 bridge)	<u>Existing:</u> IND, PROC, AGR, GWR, FRSH, REC-1, REC-2, RARE, MIGR, WARM, COLD, WILD, and WET, <u>Potential:</u> p*MUN

Footnotes for Table F-4

- a. The potential municipal and domestic supply (MUN) beneficial use for the water body is consistent with the Sources of Drinking Water Policy (page 5-13 of the Basin Plan). However, the Los Angeles Water Board has only conditionally designated the MUN beneficial use in anticipation of further evaluation. Therefore, the Los Angeles Water Board is not establishing effluent limitations for the protection of the potential MUN at this time.

End of Footnotes for Table F-4

Beneficial uses of the receiving groundwaters are as follows:

Table F-4b. Basin Plan Beneficial Uses – Groundwaters

Receiving Water Name	MUN	IND	PROC	AGR	AQUA
Santa Clara River Valley East					
Mint Canyon (Department of Water Resources (DWR) Basin No. 4-4.07)	Existing	Existing	Existing	Existing	---

Receiving Water Name Santa Clara River Valley East	MUN	IND	PROC	AGR	AQUA
South Fork (DWR Basin 4-4.07)	Existing	Existing	Existing	Existing	---
Placerita Canyon (DWR Basin 4-4.07)	Existing	Existing	Existing	Existing	---
Bouquet and San Francisquito Canyons (DWR Basin 4-4.07)	Existing	Existing	Existing	Existing	---
Castaic Valley (DWR Basin 4-4.07)	Existing	Existing	Existing	Existing	---
Saugus Aquifer (DWR Basin 4-4.07)	Existing	---	---	---	---

3.3.2. National Toxics Rule (NTR) and California Toxics Rule (CTR)

USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain federal water quality criteria for priority pollutants.

3.3.3. State Implementation Policy

On March 2, 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Los Angeles Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.

3.3.4. Alaska Rule

On March 30, 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards become effective for CWA purposes (40 CFR section 131.21, 65 Federal Register 24641 (April 27, 2000)). Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.

3.3.5. Stringency of Requirements for Individual Pollutants.

This Order contains both technology-based effluent limitations (TBELs) and water quality-based effluent limitations (WQBELs) for individual pollutants. The TBELs consist of restrictions on BOD₅20°C, and TSS and percent removal of BOD₅20°C and TSS. Restrictions on BOD₅20°C, and TSS are discussed in section 4.2.2 of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are carried over from the previous permit.

WQBELs have been scientifically derived to implement water quality objectives (WQOs) that protect beneficial uses. Both the beneficial uses and the WQOs have been approved pursuant to federal law and are the applicable federal water quality standards. All beneficial uses and WQOs contained in the Basin Plan and statewide water quality control plans were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any WQOs and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA and the applicable water standards for purposes of the CWA. The final effluent limitations for these pollutants are described in additional detail in section 4.3.2 of the Fact Sheet.

3.3.6. Antidegradation Policy

Federal regulation 40 CFR section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal antidegradation policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16 (*Statement of Policy with Respect to Maintaining the Quality of the Waters of the State*). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Los Angeles Water Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies. The permitted discharges are consistent with the antidegradation provisions of 40 CFR section 131.12 and State Water Board Resolution 68-16. This Order is consistent with antidegradation policies, as discussed in section 4.4.2 of the Fact Sheet.

3.3.7. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. The effluent limitations in this Order are at least as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. This Order complies with anti-backsliding provisions.

3.3.8. Endangered Species Act Requirements

This Order prohibits any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (ESA) (Fish and Game Code, sections 2050 to 2097) or the Federal ESA (16 USC sections 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The Permittee is responsible for meeting all requirements of the applicable ESA.

3.3.9. Water Rights

Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a surface or subterranean stream, the Permittee must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under Water Code section 1211.

3.3.10. Domestic Water Quality

In compliance with Water Code section 106.3, it is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes.

3.3.11. Water Recycling

In accordance with statewide statutes and policies concerning water reclamation, (e.g., CWC sections 13000 and 13550-13557, State Water Board Resolution No. 77-1, Policy with Respect to Water Reclamation in California, and State Water Board Resolution Nos. 2009-0011, 2013-0003, and 2018-0057 (Recycled Water Policy)), the Los Angeles Water Board strongly encourages, wherever practical, water recycling, water conservation, and beneficial use of stormwater and dry-weather urban runoff. The Permittee shall investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or capture and treatment of dry-weather urban runoff, and stormwater on a permissive basis, for beneficial reuse. In 2020, a total of 5.05 million gallons per year and an average of 0.42 million gallons per month of Valencia WRP tertiary-treated water was delivered into recycled water distribution systems for reuse. All recycled water used during the year was used only for those purposes specified in the requirements. Recycled water was delivered to reuse sites that overlie the Santa Clara River Valley East Groundwater Basin.

During 2020, the Santa Clarita Valley water Agency (SCV Water) provided these quantities of recycled water, produced at the Valencia WRP, for landscape irrigation to the following locations: 0.334 MGD to the Tournament Players Club Golf Course, 0.023 MGD to The Old Road medians, 0.014 MGD to Magic Mountain Parkway medians and the Old Road medians, 0.005 MGD to Media Center Drive within the Entrada development, and 0.022 MGD to Entertainment Drive within the Entrada development.

The Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.

The State Water Board adopted “Water Quality Control Policy for Recycled Water” (Recycled Water Policy) on February 3, 2009 and amended it most recently on December 11, 2018. The most recent amendments became effective on April 8, 2019. The Recycled Water Policy requires wastewater and recycled water dischargers to annually report monthly volumes of influent, wastewater produced, and effluent, including treatment level and discharge type. As applicable, dischargers are additionally required to annually report recycled water use by volume and category of reuse. The State Water Board issued a Water Code section 13267 and 13383 Order, Order WQ 2019-0037 EXEC, on July 24, 2019 to amend MRPs for all permits of NPDES, WDRs, WRRs, Master Recycling, and General WDRs. Annual reports are due by April 30 of each year, and the report must be submitted to GeoTracker. This Order implements the Recycled Water Policy by incorporating the volumetric monitoring reporting requirements in accordance with section 3 of the [Recycled Water Policy](https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/12_1118_7_final_amendment_oal.pdf) (https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/12_1118_7_final_amendment_oal.pdf). The State Water Board’s Order WQ 2019- 0037 EXEC will no longer be applicable to the Discharger upon the effective date of this Order.

3.3.12. Monitoring and Reporting

40 CFR section 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code section 13383 authorizes the Los Angeles Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP) establishes monitoring and reporting requirements to implement federal and state requirements. This MRP is provided in Attachment E.

3.3.13. Sewage Sludge/Biosolids Requirements

Section 405 of the CWA and implementing regulations at 40 CFR part 503 require that producers of sewage sludge/biosolids meet certain reporting, handling, and use or disposal requirements. USPEA has not delegated the authority to regulate biosolids/sludge under 40 CFR part 503 to the state of California. The Permittee is responsible for meeting all applicable requirements of 40 CFR part 503 that are under USEPA’s enforcement authority.

3.3.14. Pretreatment Requirements

The application of pretreatment requirements is monitored by the Discharger and the permit will be reopened when additional pretreatment requirements are determined to be applicable to the discharge. The Permittee has developed and is implementing a Pretreatment Program that was previously approved by USEPA. This Order requires continuous implementation of the approved Pretreatment Program. The Sanitation District reported in the 2020 Pretreatment Annual Report that there were 303 Categorical Industrial User (CIU) permittees, 822 significant industrial user (SIU) permittees, and 1,312 other industrial uses in the Sanitation District’s entire Pretreatment Program. In the Report of Waste Discharge (ROWD), SCVSD reported that the Valencia WRP receives wastewater from 11 SIUs and 10 CIUs. Any change to the Pretreatment Program shall be reported to the Los Angeles Water Board in writing and shall not become effective until approved by the Executive Officer in accordance

with procedures established in 40 CFR § 403.18. The Discharger shall comply with requirements contained in Attachment H – Pretreatment Reporting Requirements.

3.3.15. Mercury Provisions

The State Water Board adopted “*Part 2 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California- Tribal and Subsistence Fishing Beneficial Uses and Mercury Provisions*” (Mercury Provisions) through Resolution Number 2017-0027, which was approved by the California Office of Administrative Law (OAL) on June 28, 2017 and became effective upon U.S. EPA approval on July 14, 2017. The Mercury Provisions established one narrative and four numeric water quality objectives for mercury and three new beneficial use definitions, implemented through NPDES permits issued pursuant to CWA section 402, waste discharge requirements, or waivers of waste discharge requirements. The Provisions included implementation provisions for individual non-stormwater NPDES permits for municipal and industrial dischargers; stormwater discharges including MS4 discharges and discharges regulated by the Industrial General Permit (NPDES Number CAS000001); mine site remediation; nonpoint source discharges; dredging activities; and wetland projects.

The Mercury Provisions contain provisions that apply to POTWs and individual industrial discharges. The Mercury Provisions converted the fish tissue-based water quality objectives to water column values, denoted as “C”. The implementation section of the Mercury Provisions requires the application of section 1.3 of the SIP with modifications to determine whether a discharge has reasonable potential to cause or contribute to an exceedance of the water column concentration for mercury and the development of effluent limitations for mercury based on the water quality objective applicable to the receiving water in accordance with chapter IV.D.2.b in Mercury Provisions. (See section 4.3.3 of Fact Sheet for SIP procedures).

The Mercury Provisions convert the fish tissue-based water quality objectives into water column values to be used for reasonable potential analysis and development of effluent limitations. The objective for Santa Clara River, which is a flowing water body, is 12 nanograms per liter (ng/L) for total mercury. The annual averages of effluent sample testing results ranged from 0.4 ng/L to 5.9 ng/L during the monitoring period from May 2015 through July 2021. According to the Mercury Provisions, a water quality-based effluent limitation is not required unless the highest observed annual average effluent mercury concentration is greater than the applicable objective (water column concentration, 12 ng/L). Since the data indicated that there is no reasonable potential to cause or contribute to an excursion above the water quality standard, no effluent limitations for mercury are established in this Order. However, monitoring requirements for mercury in the effluent and receiving water are included in Attachment E with the new detection limit of 0.5 ng/L, which the Mercury Provisions specify as a quantification limit for the water samples.

3.3.16. Bacteria Provisions

The State Water Board adopted “*Part 3 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California- Bacteria Provisions and Water Quality Standards Variance Policy*” (ISWEBE Bacteria Provisions) through

Resolution Number 2018-0038, which was approved by OAL on February 4, 2019 and became effective upon USEPA approval on March 22, 2019. The Bacteria Provisions establish *Escherichia coli* (*E. coli*) as the sole indicator of pathogens in freshwater. These *E. coli* water quality objectives supersede any numeric water quality objectives for bacteria for the protection of the REC-1 beneficial use in the Los Angeles Water Board Basin Plan prior to the effective date of the Bacteria Provisions, except in certain circumstances, such as where there are site-specific numeric water quality objectives for bacteria. Further, where there is a TMDL to implement prior bacteria objectives, these TMDLs remain in effect.

This Order includes permit effluent limitations based on Title 22 disinfected tertiary treated wastewater for the protection of human health. These effluent limitations based on Title 22 disinfection requirements are more stringent than the Bacteria Provision effluent limitations. They are also consistent with the applicable WLA for the discharge in the Santa Clara River Bacteria TMDL. Therefore, the limitations based on the Bacteria Provisions are not used in this Order, and the Title 22 based effluent limitations are retained. In addition, USEPA states in its document "*NPDES Water Quality Based Permit Limits for Recreational Water Quality Criteria (2015)*" that it expects the direct application of criteria values at the end-of-pipe approach where the objective is applied directly as permit limits at the discharge point. Since the effluent limitations are applied at the discharge point (end-of-pipe) based on Title 22, which are more stringent than the ISWEBE Bacteria Provisions, additional receiving water limitations are not established.

3.3.17. Toxicity Provisions

Beginning in May 2013 the Los Angeles Water Board began incorporating into the NPDES permits for POTWs and industrial facilities numeric water quality objectives for both acute and chronic toxicity, using the Test of Significant Toxicity (TST), and a program of implementation to control toxicity. As explained later in the Fact Sheet, this approach is a preferred statistical method because it provides a higher confidence in results classifying in-waste stream concentrations as toxic or non-toxic and it is supported by US EPA. This methodology is used in the existing permit and would be carried over into this permit. On December 1, 2020, the State Water Board adopted statewide numeric water quality objectives for both acute and chronic toxicity, using the test of Significant Toxicity (TST) statistical approach, and a program of implementation to control toxicity, which are collectively known as the Toxicity Provisions. On October 5, 2021, the State Water Board adopted a resolution rescinding the December 1, 2020 establishment of Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California and confirming that the Toxicity Provisions were adopted as a State Policy for Water Quality Control, for all inland surface waters, enclosed bays, estuaries, and coastal lagoons of the state, regardless of their status as waters of the United States. The Toxicity Provisions establish a uniform regulatory approach to provide consistent protection of aquatic life beneficial uses, and protect aquatic habitats and life from the effects of known and unknown toxicants. The Provisions are not yet in effect. The Provisions will take effect upon approval by the California Office of Administrative Law for purposes of state law and upon review and approval by the U.S. Environmental Protection Agency for purposes of federal law.

3.3.18. Impaired Water Bodies on CWA 303(d) List

The State Water Board adopted the California 2014 and 2016 Integrated Report based on a compilation of the Los Angeles Water Boards' Integrated Reports. These Integrated Report contain both the Clean Water Act (CWA) section 305(b) water quality assessment and section 303(d) list of impaired waters. In developing the Integrated Reports, the Water Boards solicit data, information, and comments from the public and other interested persons. On October 3, 2017, the State Water Board approved the CWA section 303(d) List portion of the State's 2014 and 2016 Integrated Report (State Water Board Resolution Number 2017-0059). On April 06, 2018, USEPA approved California's 2014 and 2016 Integrated Report. The CWA section [303\(d\) list](#) can be found at the following link:

http://www.waterboards.ca.gov/water_issues/programs/tmdl/integrated2014_2016.shtml

Santa Clara River and the Santa Clara River Estuary are on the 303(d) list. The following are the identified pollutants impacting the receiving water:

Santa Clara River Creek Reach 5 (Blue Cut gaging station to West Pier Hwy 99 Bridge) (was named Santa Clara River Reach 7 on 2002 303(d) list) - Hydrologic unit 403.51 or 40351000, USGS, HUC Watershed boundary dataset HUC 18070102

Pollutants: Chloride, Indicator Bacteria, Iron and trash.

Santa Clara River Reach 3 (Freeman Diversion to A Street) -- Hydrologic unit 403.31 or 40331000, USGS HUC/ Watershed boundary dataset HUC 18070103

Pollutants: Chloride, indicator bacteria, selenium, total dissolved solids, toxicity and trash.

Santa Clara River Reach 1 (Estuary to Hwy 101 Bridge) -- Hydrologic unit 403.11 or 40311000, USGS HUC/ Watershed boundary dataset HUC 18070103

Pollutants: Dissolved oxygen, toxicity, trash, and pH.

Santa Clara River Estuary -- Hydrologic unit 403.11 or 40311000, USGS HUC/Watershed boundary dataset HUC 18070103

Pollutants: Ammonia, Chem A, indicator bacteria, toxaphene, and toxicity, where Chem A (abbreviation for chemical group A) is a suite of bio-accumulating pesticides that includes aldrin, dieldrin, chlordane, endrin, heptachlor, heptachlor epoxide, hexachlorocyclohexane (HCH) (including lindane), endosulfan, and toxaphene.

3.4. Other Plans, Policies and Regulations

3.4.1. Climate Change Adaptation and Mitigation

On March 7, 2017, the State Water Board adopted a resolution in recognition of the challenges posed by climate change that requires a proactive approach to climate change in all State Water Board actions, including drinking water regulation, water quality protection, and financial assistance (Resolution Number 2017-0012). The resolution lays the foundation for a response to climate change that is integrated into all State Water Board actions, by giving direction to the State Water Board divisions

and encouraging coordination with the Los Angeles Water Board. The Los Angeles Water Board also adopted “A Resolution to Prioritize Actions to Adapt to and Mitigate the Impacts of Climate Change on the Los Angeles Region’s Water Resources and Associated Beneficial Uses” (Resolution Number R18-004) on May 10, 2018. The resolution summarizes the steps taken so far to address the impacts of climate change within the Los Angeles Water Board’s programs and lists a series of steps to move forward. These include the identification of potential regulatory adaptation and mitigation measures that could be implemented on a short-term and long-term basis by each of the Los Angeles Water Board’s programs to take into account, and assist in mitigating where possible, the effects of climate change on water resources and associated beneficial uses. This Order contains provisions to require planning and actions to address climate change impacts in accordance with both the State and Los Angeles Water Board’ resolutions.

The Permittee shall develop a Climate Change Effects Vulnerability Assessment and Management Plan (Climate Change Plan) and submit the Climate Change Plan to the Los Angeles Water Board for the Executive Officer’s approval no later than 12 months after the effective date of this Order. The Climate Change Plan shall include an assessment of short and long term vulnerabilities of the facility and operations as well as plans to address vulnerabilities of collection systems, facilities, treatment systems, and outfalls for predicted impacts in order to ensure that facility operations are not disrupted, compliance with permit conditions is achieved, and receiving waters are not adversely impacted by discharges. Control measures shall include, but are not limited to, emergency procedures, contingency plans, alarm/notification systems, training, backup power and equipment, and the need for planned mitigations to ameliorate climate-induced impacts including, but not limited to, changing influent and receiving water quality and conditions, as well as the impact of rising sea level (where applicable), wildfires, storm surges and back-to-back severe storms, which are expected to become more frequent.

3.4.2. Sources of Drinking Water Policy

On May 19, 1988, the State Water Board adopted Resolution Number 88-63, *Sources of Drinking Water Policy*, which established a policy that all surface and groundwaters, with limited exemptions, are suitable or potentially suitable for municipal and domestic supply. To be consistent with State Water Board’s Resolution 88-63, on March 27, 1989, the Los Angeles Water Board adopted Resolution Number 89-03, *Incorporation of Sources of Drinking Water Policy into the Water Quality Control Plans (Basin Plans) – Santa Clara River Basin (4A)/ Los Angeles River Basin (4B)*.

This Order is consistent with the existing Basin Plan.

3.4.3. Title 22 of the California Code of Regulations (CCR Title 22)

The California Department of Public Health (CDPH) established primary and secondary maximum contaminant levels (MCLs) for inorganic, organic, and radioactive contaminants in drinking water. These MCLs are codified in Title 22. The Basin Plan (chapter 3) incorporates Title 22 primary MCLs by reference. This incorporation by reference is prospective, including future changes to the incorporated provisions as the changes take effect. Title 22 primary MCLs have been used as bases for effluent

limitations in WDRs and NPDES permits to protect groundwater recharge (GWR) beneficial use when that receiving groundwater is designated as MUN. Also, the Basin Plan specifies that “Ground waters shall not contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.” Since Reach 5 of the Santa Clara River and downstream reaches have designated GWR beneficial uses, this Order establishes effluent limitations based on primary MCLs of CCR Title 22 to protect these uses. In addition, this Order contains title 22 based total coliform effluent limitations to protect other beneficial uses and reuse of the treated water.

3.4.4. Secondary Treatment Regulations

40 CFR part 133 establishes the minimum levels of effluent quality to be achieved by secondary treatment. These limitations, established by USEPA, are incorporated into this Order, except where more stringent limitations are required by other applicable plans, policies, or regulations or to prevent backsliding.

3.4.5. Stormwater

CWA section 402(p), as amended by the Water Quality Act of 1987, requires NPDES permits for stormwater discharges. Pursuant to this requirement, in 1990, USEPA promulgated 40 CFR part 122.26 that established requirements for stormwater discharges under an NPDES program. To facilitate compliance with federal regulations, in November 1991, the State Water Board issued a statewide general permit, General NPDES Permit Number CAS000001 and Waste Discharge Requirements for Discharges of Stormwater Associated with Industrial Activities. General NPDES Permit Number CAS000001 has been amended and reissued several times since 1991, and most recently on November 6, 2018. The latest amendment became effective on July 1, 2020.

On June 4, 1992, the Sanitation Districts of Los Angeles County filed an initial Notice of Intent (NOI) to comply with the requirements of the general permit and was assigned this Waste Discharge Identification Number 4 191007154 for the IGP. On August 16, 2019, the Permittee submitted a new NOI to comply with the requirements of the new Industrial Stormwater General Permit (IGP). The Permittee developed and currently implements a Stormwater Pollution Prevention Plan (SWPPP), to comply with the State Water Board’s General NPDES permit Number CAS000001.

3.4.6. Sanitary Sewer Overflows (SSOs)

The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311, 1342). The Permittee must comply with State Water Board Order Number 2006-0003-DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems*, (SSS WDRs) as amended by State Water Board Order Number WQ 2013-0058-exec and any subsequent order updating these requirements. These statewide WDRs require public agencies that own or operate sanitary sewer systems with greater than one mile of sewer lines to enroll for coverage and comply with requirements to develop and implement sewer system management plans and report all SSOs to the State Water Board’s online SSOs database. The Permittee

enrolled in the SSS WDRs in 2006, and the collection systems of the Permittee are covered under the SSS WDRs. This NPDES permit also contains requirements pertaining to the Permittee's collection system. The Permittee must properly operate and maintain its collection system (40 CFR section 122.41 (e)), report any non-compliance (40 CFR section 122.41(1)(6) and (7)), and mitigate any discharge from the collection system in violation of this NPDES permit (40 CFR section 122.41(d)).

The requirements contained in this Order sections 6.3.3.b (Spill Cleanup Contingency Plan section), 6.3.4 (Construction, Operation and Maintenance Specifications section), and 6.3.6 (Spill Reporting Requirements section) are consistent with the requirements of the SSS WDRs. The Los Angeles Water Board recognizes that there may be some overlap between these NPDES permit provisions and SSS WDRs requirements, related to the collection systems. The requirements of the SSS WDRs are considered the minimum thresholds (see Finding 11 of State Water Board Order Number 2006-0003-DWQ). To encourage efficiency, the Los Angeles Water Board will accept the documentation prepared by the Permittees under the SSS WDRs for compliance purposes as satisfying the requirements in sections 6.3.3.b, 6.3.4, and 6.3.6, provided the more stringent provisions contained in this NPDES permit are also addressed in the SSS WDRs submission. Pursuant to SSS WDRs, section D, provision 2(iii) and (iv), the provisions of this NPDES permit supersede the SSS WDRs, for all purposes, including enforcement, to the extent the requirements may be deemed duplicative.

The requirements of this Order are more stringent than the SSS WDRs because in addition to the SSS WDR requirements, this NPDES permit requires water quality monitoring of the receiving water when the spill reaches the surface water.

3.4.7. Watershed Management

This Los Angeles Water Board has been implementing a Watershed Management Approach (WMA) to address water quality protection in the Los Angeles Region. Information about watersheds in the region can be obtained at the [Los Angeles Water Board's website](http://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/watershed/index.shtml) at http://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/watershed/index.shtml. The WMA emphasizes cooperative relationships between regulatory agencies, the regulated community, environmental groups, and other stakeholders in the watershed to achieve the greatest environmental improvements with the resources available.

This Order fosters the implementation of this approach by protecting beneficial uses in the watershed and requiring the Discharger to participate with other stakeholders in the development and implementation of a watershed-wide monitoring program. The Monitoring and Reporting Program (Attachment E) requires the Discharger to undertake the responsibilities delineated under an approved watershed-wide monitoring plan in the implementation of the Watershed-wide Monitoring Program for the Santa Clara River, which was approved by the Los Angeles Water Board on July 3, 2012.

3.4.8. Relevant TMDLs

Section 303(d) of the CWA requires states to identify water bodies that do not meet water quality standards and then to establish TMDLs for each waterbody for each pollutant of concern. TMDLs identify the maximum amount of pollutants that can be discharged to waterbodies without causing violations of water quality standards.

- a. **Santa Clara River Watershed Chloride TMDL** - On December 11, 2008, the Los Angeles Water Board adopted Resolution No. R4-2008-012, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Adopt Site Specific Chloride Objectives and to Revise the Upper Santa Clara River Chloride TMDL*, which proposed to incorporate site specific objectives (SSOs) for chloride in Reaches 4B, 5, and 6 of the Santa Clara River and the groundwater basins underlying those reaches. Resolution No. R4-2008-012 was approved by State Water Board, OAL, and USEPA on December 9, 2009, February 2, 2010, and April 14, 2010, respectively.

On October 9, 2014, the Los Angeles Water Board adopted Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate an Averaging Period for Chloride Water Quality Objectives in Reaches 4B, 5, 6; Incorporate New Site Specific Objectives for Chloride in Reach 5 and 6; and Revise the Total Maximum Daily Load for Chloride in the Upper Santa Clara River (Resolution No. R14-010), which:

- Revised the TMDL compliance deadline from May 4, 2015 to July 1, 2019, to provide time for permitting, design, construction, and start-up of the new facilities.
- Revised the chloride water quality objective for the upper Santa Clara River to include a 3-month averaging period rather than the current application as an instantaneous maximum concentration.
- Revised the chloride water quality objectives in Reaches 5 and 6 and the corresponding TMDL WLAs for the Saugus and Valencia WRPs in order to eliminate the need to pump reverse osmosis product water from the Valencia WRP to the Saugus WRP, to meet 100 mg/L as a flow-weighted 3-month average in the discharge from both plants. The chloride water quality objectives for Reach 5 upstream of the Valencia WRP and Reach 6 were revised from 100 mg/L to 150 mg/L and a corresponding modification was made to the waste load allocation for the Saugus WRP. The Valencia WRP is now assigned a variable waste load allocation less than 100 mg/L as a 3-month rolling average, which would allow the Saugus WRP to discharge up to 150 mg/L as a 3-month rolling average, while still meeting the numeric target of 100 mg/L as a 3-month rolling average in Reach 5 immediately downstream of the Valencia WRP. The TMDL included interim milestones to ensure that the facilities needed to attain flow-weighted WLAs are constructed in time for the Saugus and Valencia WRPs to attain the final WLAs.

Resolution No. R14-010 was approved by the State Water Board, OAL, and USEPA on December 16, 2014, March 18, 2015, April 28, 2015, respectively.

- b. Santa Clara River Nutrients TMDL** - On August 7, 2003, the Los Angeles Water Board adopted Resolution No. 2003-011, *Amendment to the Basin Plan for the Los Angeles Region to Include a TMDL for Nitrogen Compounds in the Santa Clara River (Nitrogen Compounds TMDL)*. The TMDL was approved from State Water Board, OAL, and USEPA on November 19, 2003, February 27, 2004, and March 18, 2004, respectively. The *Santa Clara River Nitrogen Compounds TMDL* became effective on March 23, 2004.
- c. Santa Clara River Bacteria TMDL** - On July 8, 2010, the Los Angeles Water Board adopted Resolution No. R10-006. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on October 4, 2010, December 19, 2011, and January 13, 2012, respectively. This TMDL assigns Waste Load Allocations (WLAs) to water reclamation plants, municipal separate storm sewer system (MS4) discharges, and other discharges. WLAs for the Valencia WRP are a 7-day median of 2.2 MPN/100 mL and a Daily Maximum of 235 MPN/100 mL for E. coli. However, the existing Title 22-based bacteria limitations for total coliform are retained because they are more stringent than the TMDL WLAs for E.coli. Therefore, separate WLA-based bacteria limitations are not established.

4. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source Permittees to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in 40 CFR section 122.44(a) requires that permits include applicable TBELs and standards; and 40 CFR section 122.44(d) requires that permits include WQBELs to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

The variety of potential pollutants found in the Facility discharges presents a potential for aggregate toxic effects to occur. Whole effluent toxicity (WET) is an indicator of the combined effect of pollutants contained in the discharge. Chronic toxicity is a more stringent requirement than acute toxicity. Therefore, chronic toxicity is considered a pollutant of concern for protection and evaluation of narrative Basin Plan Water Quality Objectives for toxicity.

4.1. Discharge Prohibitions

Effluent and receiving water limitations in this Order are based on the CWA, Basin Plan, State Water Board's plans and policies, USEPA guidance and regulations, and best practicable waste treatment technology. This Order authorizes the discharge of tertiary-treated wastewater from Discharge Point 001. It does not authorize any other types of discharges.

4.2. Technology-Based Effluent Limitations (TBELs)

4.2.1. Scope and Authority

Technology-based effluent limits require a minimum level of treatment for industrial/municipal point sources based on currently available treatment technologies while allowing the Permittee to use any available control techniques to meet the

effluent limits. The 1972 CWA required POTWs to meet performance requirements based on available wastewater treatment technology. Section 301 of the CWA established a required performance level--referred to as "secondary treatment", which all POTWs were required to meet by July 1, 1977. More specifically, section 301(b)(1)(B) of the CWA required that USEPA develop secondary treatment standards for POTWs as defined in section 304(d)(1). Based on this statutory requirement, USEPA developed national secondary treatment regulations which are specified in 40 CFR part 133. These technology-based regulations apply to all POTWs and identify the minimum level of effluent quality to be attained by secondary treatment in terms of BOD₅20°C, and TSS.

4.2.2. Applicable TBELs

This Facility is subject to the technology-based regulations for the minimum level of effluent quality attainable by secondary treatment in terms of BOD₅20°C, and TSS. The principal design parameters for wastewater treatment plants are the daily BOD₅20°C and TSS loading rates and the corresponding removal rate of the system. In applying 40 CFR part 133 for weekly and monthly average BOD₅20°C and TSS limitations, the application of tertiary treatment processes results in the ability to achieve lower levels of BOD₅20°C and TSS than the secondary standards. This Facility is also subject to TBELs contained in similar NPDES permits, for similar facilities, based on the treatment level achievable by tertiary-treated wastewater treatment systems. In addition to the average weekly and average monthly effluent limitations, daily maximum effluent limitations for BOD₅20°C and TSS are include in the Order to ensure that the treatment works are not organically overloaded and operate in accordance with design capabilities. Valencia WRP can meet these limitations with the existing treatment processes in place at their facility. Further, mass-based effluent limitations are based on a design flow rate of 21.6 MGD. (40 C.F.R. §122.45(b)(1), (f).). The removal efficiency for BOD₅20°C and TSS is set at the minimum level attainable by secondary treatment technology. The following Table summarizes the TBELs applicable to the Facility:

Table F-5. Summary of TBELs

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily
BOD ₅ 20°C	mg/L	20	30	45
BOD ₅ 20°C	lbs/day	3,600	5,400	8,100
TSS	mg/L	15	40	45
TSS	lbs/day	2,700	7,200	8,100
Removal Efficiency for BOD ₅ 20°C and TSS	%	≥85	--	--

Additional reasoning upon which the foregoing limits are based is set forth below:

BOD₅20°C and TSS

BOD₅20°C is a measure of the quantity of the organic matter in the water and, therefore, the water's potential for becoming depleted in dissolved oxygen. As organic

degradation takes place, bacteria and other decomposers use the oxygen in the water for respiration. Unless there is a steady resupply of oxygen to the system, the water will quickly become depleted of oxygen. Adequate dissolved oxygen levels are required to support aquatic life. Depressions of dissolved oxygen can lead to anaerobic conditions resulting in odors, or, in extreme cases, fish kills.

40 CFR part 133 describes the minimum level of effluent quality attainable by secondary treatment, for BOD₅20°C and TSS, as:

- The 30-day average shall not exceed 30 mg/L, and
- The 7-day average shall not exceed 45 mg/L.

Valencia WRP provides tertiary treatment so the BOD₅20°C and TSS limits in this Order are more stringent than secondary treatment requirements and are based on Best Professional Judgement (BPJ). The Facility achieves solids removals that are better than secondary-treated wastewater by filtering the effluent. Those limits were all included in the previous permit and the Valencia WRP has been able to meet these limits. Accordingly, these limits are carried over here in this Order.

In addition to having mass-based and concentration-based effluent limitations for BOD₅20°C and TSS, the Valencia WRP also has a percent removal requirement for these two constituents. In accordance with 40 CFR sections 133.102(a)(3) and 133.102(b)(3), the 30-day average percent removal shall not be less than 85 percent. Percent removal is defined as a percentage expression of the removal efficiency across a treatment plant for a given pollutant parameter, as determined from the 30-day average values of the raw wastewater influent pollutant concentrations to the Facility and the 30-day average values of the effluent pollutant concentrations for a given time period.

4.3. Water Quality-Based Effluent Limitations (WQBELs)

4.3.1. Scope and Authority

CWA section 301(b) and 40 CFR section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains more stringent requirements than technology-based requirements, including secondary-treatment requirements, that are necessary to meet applicable water quality standards. The rationale for these requirements is discussed beginning in section 4.3.2. of this Fact Sheet.

40 CFR section 122.44(d)(1)(i) requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard in the receiving water, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as

provided in section 122.44(d)(1)(vi). WQBELs must also be consistent with the assumptions and requirements of TMDL WLAs approved by USEPA.

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable WQOs and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

4.3.2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

The Basin Plan establishes the beneficial uses for surface water bodies in the Los Angeles region. The beneficial uses of the Santa Clara River and Santa Clara River Estuary affected by the discharge have been described previously in this Fact Sheet.

The Basin Plan also specifies narrative and numeric WQOs applicable to surface water as described below:

a. pH

The hydrogen ion activity of water (pH) is measured on a logarithmic scale, ranging from 0 to 14. While the pH of “pure” water at 25°C is 7.0, the pH of natural waters is usually slightly basic. Minor changes from natural conditions can harm aquatic life. In accordance with 40 CFR § 133.102(c), the effluent values for pH shall be maintained within the limits of 6.0 to 9.0 unless the POTW demonstrates that (1) inorganic chemicals are not added to the waste stream as part of the treatment process; and (2) contributions from industrial sources do not cause the pH of the effluent to be less than 6.0 or greater than 9.0. The effluent limitation for pH in this Order requiring that the wastes discharged shall at all times be within the range of 6.5 to 8.5 is taken from the Basin Plan which reads “the pH of inland surface waters shall not be depressed below 6.5 or raised above 8.5 as a result of waste discharge.” The pH effluent limitations are based on the Basin Plan WQOs, which are more stringent than the TBEL. These effluent limitations have been carried over from the previous Order.

b. Settleable Solids

Excessive deposition of sediments can destroy spawning habitat, blanket benthic (bottom dwelling) organisms, and abrade the gills of larval fish. The limits for settleable solids are based on the Basin Plan narrative, “Waters shall not contain suspended or settleable material in concentrations that cause nuisance or adversely affect beneficial uses.” The numeric limits are empirically based on results obtained from the settleable solids 1-hour test, using an Imhoff cone.

It is impracticable to use a 7-day average limitation, because short-term spikes of settleable solid levels that would be permissible under a 7-day average scheme would not be adequately protective of all beneficial uses. The monthly average and daily maximum limits were both included in the previous permit and the Valencia WRP has been able to meet both limits. These effluent limitations have been carried over from the previous permit.

c. Oil and Grease

Oil and grease are not readily soluble in water and form a film on the water surface. Oily films can coat birds and aquatic organisms, which can impact respiration and thermal regulation, and cause death. Oil and grease can also cause nuisance conditions (odors and taste), are aesthetically unpleasant, and can restrict a wide variety of beneficial uses. The limits for oil and grease are based on the Basin Plan narrative WQO, "Waters shall not contain oils, greases, waxes, or other materials in concentrations that result in a visible film or coating on the surface of the water or on objects in the water, that cause nuisance, or that otherwise adversely affect beneficial uses."

The numeric limits are empirically based on concentrations at which an oily sheen becomes visible in water. It is impracticable to use a 7-day average limitation, because spikes that occur under a 7-day average scheme could cause a visible oil sheen. A 7-day average scheme would not be sufficiently protective of beneficial uses. The monthly average and the daily maximum limits were included in the previous permit and the Valencia WRP has been able to meet both limits. These effluent limitations have been carried over from the previous permit.

d. Residual Chlorine

Disinfection of wastewaters with chlorine produces a chlorine residual. Chlorine and its reaction products are toxic to aquatic life. The limit for residual chlorine is based on the Basin Plan narrative WQO, "Chlorine residual shall not be present in surface water discharges at concentrations that exceed 0.1 mg/L and shall not persist in receiving waters at any concentration that causes impairment of beneficial uses."

It is impracticable to use a 7-day average or a 30-day average limitation, because it is not as protective as of beneficial uses as a daily maximum limitation is. Chlorine is very toxic to aquatic life and short-term exposures of chlorine may cause fish kills. Thus, the maximum daily effluent limitations for residual chlorine have been carried over from the previous permit.

e. TDS, Sulfate, and Boron

The limitations for total dissolved solids, chloride, sulfate, and boron are based on Basin Plan Table 3-10, for the Santa Clara River watershed (between West Pier Highway 99 and Blue Cut Gauging Station), are as follows: TDS = 1000 mg/L; Sulfate = 400 mg/L, and Boron = 1.5 mg/L. It is practicable to express these limits as monthly averages, since they are not expected to cause acute effects on beneficial uses.

Limits based upon the Basin Plan Objectives have been included in this Order because, based upon Best Professional Judgment, these constituents are always present in potable water which is the supply source of the wastewater entering the Treatment Plant. They may be present in concentrations which meet California drinking water standards but exceed the Basin Plan Objectives. Therefore, limitations are warranted to protect the beneficial uses of the receiving water. The limitations are continued herein.

f. Chloride

The chloride final effluent limitation is based on Resolution No. R14-010, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate an Averaging Period for Chloride Water Quality Objectives in Reaches 4B, 5, 6; Incorporate new Site Specific Objectives for Chloride in Reach 5 and 6; and revise the Total Maximum Daily Load for Chloride in the Upper Santa Clara River*, as described in section 3.5.8.a of this Fact Sheet.

g. Methylene Blue Activated Substances (MBAS) and Cobalt Thiocyanate Active Substances (CTAS)

The existing permit effluent limitation of 0.5 mg/L for MBAS was developed based on the Basin Plan incorporation of Title 22, Drinking Water Standards. The effluent limitation for MBAS is included to protect the existing GWR beneficial use that is designated for the surface receiving waters downstream of the discharge as well as the existing MUN beneficial use of the underlying groundwater basins. The Saugus Aquifer and alluvium aquifers serve as an existing source of potable water for the Santa Clarita Valley Water Agency's service area. Water from the soft-bottomed Santa Clara River incidentally recharges the underlying Groundwater Basins. Section 1.3, Step 7 of the SIP lists the type of information that can be used with the permit writer's "best professional judgement" to determine RP. Page 7 of the SIP states, "Information that may be used to aid in determining if a water quality-based effluent limitation is required includes: the facility type, the discharge type, solids loading analysis, lack of dilution, history of compliance problems, potential toxic impact of discharge, fish tissue residue data, water quality beneficial uses of the receiving water, CWA 303(d) listing of the pollutant, the presence of endangered or threatened species or critical habitat, and other information."

The MBAS limit also protects the recreational, aquatic life, and wildlife beneficial uses of the surface receiving water downstream of the discharge against foam and implements the Basin Plan WQO for floating material. Volume 44, No. 179 of the Federal Register (on page 53467) explains that foaming is a characteristic of water which has been contaminated by the presence of detergents and similar substances. The 0.5 mg/L limit for foaming agents is based on the fact that at higher concentrations, the water may exhibit undesirable taste and foaming properties.

Santa Clarita Valley Water Agency relies on their groundwater as a source of potable water supply for its residents. Groundwater constitutes 25% of the water provided by Santa Clarita Valley Water Agency to its customers. Since the groundwater is an important component of the drinking water supply for the service area, using the MCL as an effluent limitation for this facility is necessary to ensure the quality of the groundwater is protected.

Given the nature of the Facility which accepts domestic wastewater into the sewer system and treatment plant, and the characteristics of the pollutants discharged, the discharge has reasonable potential to exceed both the numeric MBAS WQO and the narrative WQO for the prohibition of floating material such as foams and scums. Therefore, an effluent limitation is required.

Cobalt thiocyanate active substances (CTAS) are monitored in the same way as MBAS. The presence or absence of CTAS during sampling assists permit writers

and the Permittee in diagnosing the source of floating materials, such as foam or scum, which are prohibited by the Basin Plan when they cause nuisance or adversely affect beneficial uses. There is no limitation or compliance requirement for CTAS because it has no established water quality objective.

h. Total Inorganic Nitrogen ($\text{NO}_2 + \text{NO}_3$ as N)

Total inorganic nitrogen is the sum of Nitrate-nitrogen and Nitrite-nitrogen. High nitrate levels in drinking water can cause health problems in humans. Infants are particularly sensitive and can develop methemoglobinemia (blue-baby syndrome). Nitrogen is also considered a nutrient. Excessive amounts of nutrients can lead to other water quality impairments. Excessive growth of algae and/or other aquatic plants can degrade water quality. Algal blooms sometimes occur naturally, but they are often the result of excess nutrients (i.e., nitrogen, phosphorus) from waste discharges or nonpoint sources. These algal blooms can lead to problems with tastes, odors, color, and increased turbidity and can depress the dissolved oxygen content of the water, leading to fish kills. Floating algal scum and algal mats are also an aesthetically unpleasant nuisance.

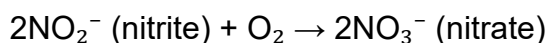
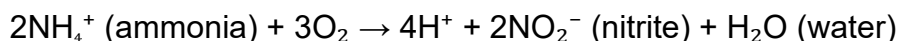
The effluent limits for biostimulatory substances are based on the Basin Plan narrative WQO, "Waters shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses," and other relevant information to arrive at a mass based-limit intended to be protective of the beneficial uses, pursuant to 40 CFR section 122.44(d). Total inorganic nitrogen will be the indicator parameter intended to control algae, pursuant to 40 CFR section 122.44(d)(1)(vi)(C).

The total inorganic nitrogen ($\text{NO}_2\text{-N} + \text{NO}_3\text{-N}$) water quality objective in the Basin Plan is 10 mg/L for the Santa Clara River Watershed (between West Pier Highway 99 and Blue Cut Gauging Station). However, the Santa Clara River Nitrogen Compounds TMDL has been in effect since March 23, 2004. Therefore, this permit includes a total inorganic nitrogen final effluent limitation of 6.8 mg/L, consistent with the Santa Clara River Nitrogen Compounds TMDL and the Waste Load Allocation which was assigned to the Valencia WRP.

Since the TMDL does not specify any mass-based WLA for nitrate plus nitrite as nitrogen, mass-based limitations are not included for $\text{NO}_2\text{-N} + \text{NO}_3\text{-N}$.

i. Nitrite as Nitrogen and Nitrate as Nitrogen

The effluent limit for nitrite as nitrogen ($\text{NO}_2\text{-N}$) of 0.9 mg/L is based on the Santa Clara River Watershed Nitrogen Compound TMDL Waste Load Allocation which was assigned to the Valencia WRP. The effluent limit for nitrate as nitrogen ($\text{NO}_3\text{-N}$) of 6.8 mg/L is based on the Santa Clara River Watershed Nitrogen Compound TMDL Waste Load Allocation which was assigned to the Valencia WRP. Since the TMDL does not specify any mass-based WLA for nitrate as nitrogen or nitrite as nitrogen, mass based limitations are not included for either of the two constituents.



j. Total Ammonia

a. Water Quality Objectives

Ammonia is a pollutant routinely found in the wastewater effluent of POTWs. Ammonia exists in two forms – un-ionized ammonia (NH_3) and the ammonium ion (NH_4^+). They are both toxic, but the neutral, un-ionized ammonia species (NH_3) is much more toxic, because it is able to diffuse across the epithelial membranes of aquatic organisms much more readily than the charged ammonium ion. The form of ammonia is primarily a function of pH, but it is also affected by temperature and other factors. Additional impacts can also occur as the oxidation of ammonia lowers the dissolved oxygen content of the water, further stressing aquatic organisms. Oxidation of ammonia to nitrate may lead to groundwater impacts in areas of recharge. There is groundwater recharge in these reaches. Ammonia also combines with chlorine (often both are present in POTW treated effluent discharges) to form chloramines – persistent toxic compounds that extend the effects of ammonia and chlorine downstream.

The Basin Plan includes one-hour, 30-day average, and four-day objectives for ammonia nitrogen in inland surface waters that are freshwaters. The freshwater one-hour average objective is dependent on pH and fish species (salmonids present or absent), but not temperature. It is assumed that salmonids may be present in waters designated in the Basin Plan as “COLD” or “MIGR” and that salmonids are absent in waters not designated in the Basin Plan as “COLD” or “MIGR,” in the absence of additional information to the contrary. The freshwater 30-day average objective is dependent on pH, temperature, and the presence or absence of early life stages of fish (ELS). Early life stages of fish are presumptively present and must be protected at all times of the year unless the water body is listed under the ELS “Absent” condition in Table 3-5 of the Basin Plan or unless a site-specific study is conducted, which justifies applying the ELS absent condition or a seasonal ELS present condition. A watershed may have some reaches and tributaries with ELS present conditions and others with ELS absent conditions. Implementation actions to achieve applicable ammonia objectives must implement downstream objectives. The freshwater four-day average objective is 2.5 times the 30-day average objective.

The Basin Plan specifies implementation procedures to translate water quality objectives described previously into effluent limitations. For the one-hour objective, the freshwaters where the discharge points lie do not have COLD or MIGR beneficial uses. Therefore, salmonids are assumed to be absent in all receiving waters and the applicable condition under Table 3-1 of the Basin Plan is applied. The Basin Plan includes site-specific 30-day average objectives that include the Santa Clara River, Reach 5 (West Pier Hwy 99 to Blue Cut gauging station) that receive discharges from the Facility as shown in Table 3-4.

Applicable Ammonia WQO – The Valencia WRP discharges into the Santa Clara River Watershed, which is governed by the Santa Clara River Watershed Nitrogen Compound TMDL (Resolution No. 03-011). Therefore, the monthly average effluent limit of 1.75 mg/L and the daily maximum effluent limit of 5.2 mg/L for ammonia as

nitrogen (NH₃-N) are based on the Santa Clara River Watershed Nitrogen Compound TMDL Waste Load Allocations which were assigned to the Valencia WRP.

Table F-6. Summary of Ammonia Nitrogen Effluent Limitations for Discharge Points 001 and 002

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily
Ammonia Nitrogen	mg/L	1.75	--	5.2

k. Bacteria Indicator

Total coliform bacteria are used to indicate the likelihood of pathogenic bacteria in surface waters. Given the nature of the facility, a wastewater treatment plant, pathogens are likely to be present in the effluent in cases where the disinfection process is not operating adequately. As such, this Order contains the following effluent limitations:

- The 7-day median number of coliform organisms at some point in the treatment process must not exceed a Most Probable Number (MPN) or Colony Forming Units (CFU) of 2.2 per 100 milliliters,
- The number of coliform organisms must not exceed an MPN or CFU of 23 per 100 milliliters in more than one sample within any 30-day period, and
- No sample shall exceed an MPN of CFU of 240 total coliform bacteria per 100 milliliters.

These disinfection-based effluent limitations for coliform are for human health protection and are consistent with requirements established by the California Department of Public Health. These limits for coliform must be met at the point of the treatment train immediately following disinfection, as a measure of the effectiveness of the disinfection process.

l. Temperature

The Basin Plan contains the following water quality objective for temperature:

The natural receiving water temperature of all regional waters shall not be altered unless it can be demonstrated to the satisfaction of the Regional Board that such alteration in temperature does not adversely affect beneficial uses. Alterations that are allowed must meet the requirements below.

For waters designated WARM, water temperature shall not be altered by more than 5°F above the natural temperature. At no time shall these WARM-designated waters be raised above 80°F as a result of waste discharges.

Temperature can adversely affect beneficial uses. The USEPA document, *Quality Criteria for Water 1986* [EPA 440/5-86-001, May 1, 1986], also referred to as the *Gold Book*, discusses temperature and its effects on beneficial uses, such as recreation and aquatic life.

- i. The Federal Water Pollution Control Administration in 1967 called temperature “a catalyst, a depressant, an activator, a restrictor, a stimulator, a controller, a killer, and one of the most important water quality characteristics to life in water.” The suitability of water for total body immersion is greatly affected by temperature. Depending on the amount of activity by the swimmer, comfortable temperatures range from 20°C to 30°C (68 °F to 86 °F).
- ii. Temperature also affects the self-purification phenomenon in water bodies and therefore the aesthetic and sanitary qualities that exist. Increased temperatures accelerate the biodegradation of organic material both in the overlying water and in bottom deposits which makes increased demands on the dissolved oxygen resources of a given system. The typical situation is exacerbated by the fact that oxygen becomes less soluble as water temperature increases. Thus, greater demands are exerted on an increasingly scarce resource which may lead to total oxygen depletion and obnoxious septic conditions. Increased temperature may increase the odor of water because of the increased volatility of odor-causing compounds. Odor problems associated with plankton may also be aggravated.
- iii. Temperature changes in water bodies can alter the existing aquatic community. Coutant (1972) has reviewed the effects of temperature on aquatic life reproduction and development. Reproductive elements are noted as perhaps the most thermally restricted of all life phases assuming other factors are at or near optimum levels. Natural short-term temperature fluctuations appear to cause reduced reproduction of fish and invertebrates.

The prior Order (Order Number R4-2015-0071) contained 86°F as a temperature effluent limitation. The Order stated that “[t]he temperature of wastes discharged shall not exceed 86°F except as a result of external ambient temperature.” This Order revises the temperature effluent limitation to 80°F to be consistent with the temperature water quality objectives in the Basin Plan, which is a new interpretation compared to the previous order.

The Facility’s temperature data during the previous permit term shows that effluent and receiving water temperatures exceed the newly interpreted 80°F temperature limitation, especially during summer months. Temperature data also exceeds the 86°F limitation in Order R4-2015-0071 primarily during the summer months and high ambient air temperatures. Since the facility can’t consistently comply with the 80°F newly interpreted temperature effluent limitation, the Discharger submitted a request for a compliance schedule to comply with the newly interpreted effluent and receiving water limitations for temperature. This Order provides a compliance schedule with tasks.

m. Turbidity

Turbidity is an expression of the optical property that causes light to be scattered in water due to particulate matter such as clay, silt, organic matter, and microscopic organisms. Turbidity can result in a variety of water quality impairments. The effluent limitation for turbidity is based on the Basin Plan and section 60301.320 of Title 22, chapter 3, “Filtered Wastewater” of the CCR. To protect water contact recreation, the Order limits turbidity of the wastewater such that it does not exceed: (a) a daily

average of 2 Nephelometric turbidity units (NTU); (b) 5 NTU more than 5 percent of the time (72 minutes) during any 24-hour period; and (c) 10 NTU at any time.”

n. Radioactivity

Radioactive substances are generally present in natural waters in extremely low concentrations. Mining or industrial activities increase the amount of radioactive substances in waters to levels that are harmful to aquatic life, wildlife, or humans. Section 301(f) of the CWA contains the following statement with respect to effluent limitations for radioactive substances, “Notwithstanding any other provisions of this Act, it shall be unlawful to discharge any radiological, chemical, or biological warfare agent, any high-level radioactive waste, or any medical waste, into the navigable waters.” Chapter 5.5 of the CWC contains a similar prohibition under section 13375, which reads as follows: “The discharge of any radiological, chemical, or biological warfare agent into the waters of the state is hereby prohibited.” However, rather than an absolute prohibition on radioactive substances, Los Angeles Water Board staff have set the following effluent limit for radioactivity: “Radioactivity of the wastes discharged shall not exceed the limits specified in Title 22, Chapter 15, Article 5, sections 64442 and 64443, of the CCR, or subsequent revisions.” The limit is based on the Basin Plan incorporation of Title 22, CCR, *Drinking Water Standards*, by reference, to protect beneficial use. Therefore, this Order retains the limitation for radioactivity.

o. Iron

In 1979, USEPA established a secondary MCL for iron. Volume 44, No. 179 of the Federal Register (at page 53467) explains that iron is a highly objectionable constituent of water supplies. It imparts a brownish discoloration and a bitter or astringent taste to drinking water. At 1.0 mg/l, a substantial number of people will note the bitter astringent taste of iron. Also, at this concentration level the staining problems associated with iron will be pronounced, thus making the water unpleasant to the consumer and unsatisfactory for most industries. The Basin Plan incorporates MCLs as WQOs by reference prospectively.

The iron secondary MCL is used to protect the existing GWR beneficial use in surface water and the MUN beneficial use in the underlying groundwater basin. The iron limit also protects the industrial process supply (PROC), industrial service supply (IND), recreational, aquatic life, and wildlife beneficial uses of the surface receiving water downstream of the discharge against the nuisances associated with color and undesirable tastes, and implements the Basin Plan water quality objective for color and taste.

The Valencia WRP has reasonable potential to contribute to an exceedance of the water quality objective because if the receiving water concentration was 7,120 µg/L, which exceeds the 300 µg/L secondary MCL by more than twenty-one times, and therefore a limit is necessary. As depicted in the process schematic, the Valencia WRP adds ferric sulfate to the treatment process to enhance coagulation of particulate matter in the plant influent, so it introduces iron into the treatment plant. Since the discharge has reasonable potential to cause or contribute to an

exceedance, a limit for iron, based on the 300 µg/L MCL, is prescribed for the Valencia WRP.

p. Total trihalomethanes

The permit effluent limitation of 80 µg/L for TTHMs was developed based on the Basin Plan incorporation of Title 22, Drinking Water Standards. The effluent limitation for TTHMs is included to protect the existing GWR beneficial use that is designated for the surface receiving waters downstream of the discharge as well as the existing MUN beneficial use of the underlying groundwater basin. Water from the soft-bottomed Santa Clara River incidentally recharges the underlying groundwater basin. The aquifers listed on Table F-4 are an existing source of potable water for the citizens of Santa Clarita. Section 1.3, Step 7 of the SIP lists the type of information, which under the permit writer's "best professional judgment," can be used to determine RP. The SIP, at page 7, states: "Information that may be used to aid in determining if a water quality-based effluent limitation is required includes: the facility type, the discharge type, solids loading analysis, lack of dilution, history of compliance problems, potential toxic impact of discharge, fish tissue residue data, water quality and beneficial uses of the receiving water, CWA 303(d) listing for the pollutant, the presence of endangered or threatened species or critical habitat, and other information." Valencia WRP has reasonable potential to contribute to an exceedance of the water quality objective based on the Technical Support document reasonable potential procedure and because the Valencia WRP forms TTHMs as a byproduct of the chlorine disinfection process that is used at the facility. Since the discharge has reasonable potential to cause or contribute to an exceedance, a limit for TTHMs, based on the 80 µg/L criteria, is prescribed for the Valencia WRP.

4.3.3. CTR and SIP

This Order implements the CTR and SIP, which specify numeric objectives for toxic substances and the procedures whereby these objectives are to be implemented. The procedures include those used to conduct reasonable potential analysis (RPA) to determine the need for effluent limitations for priority pollutants. The U.S. EPA's *Technical Support Document for Water Quality-based Toxics Control* (TSD) specifies the procedures to conduct reasonable potential analyses for non-priority pollutants.

4.3.4. Determining the Need for WQBELs

The Los Angeles Water Board developed the WQBEL for chloride, ammonia as nitrogen, nitrite plus nitrate as nitrogen, nitrite nitrogen, and nitrate nitrogen based upon Total Maximum Daily Loads (TMDLs). This effluent limitation was established on the basis of the wasteload allocations established for Valencia WRP. The Los Angeles Water Board developed water quality-based effluent limitations for these pollutants pursuant to 40 CFR section 122.44(d)(1)(vii), which does not require or contemplate a separate reasonable potential analysis at the permitting stage. Similarly, the SIP at section 1.3 recognizes that a separate reasonable potential analysis at the permitting stage is not necessary if a TMDL has been developed.

In accordance with section 1.3 of the SIP, and noting the exceptions above, during the development of this Order, the Los Angeles Water Board conducted a reasonable

potential analysis for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the permit. The Los Angeles Water Board analyzed effluent data to determine if a pollutant in a discharge has a reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that demonstrate reasonable potential, numeric WQBELs are required. The RPA considers water quality criteria from the CTR and NTR, and, when applicable, water quality objectives specified in the Basin Plan. To conduct the RPA, the Los Angeles Water Board staff identified the maximum effluent concentration (MEC) and maximum background concentration in the receiving water for each constituent, based on data provided by the Permittee. The monitoring data cover the period from May 2015 to July 2021.

Section 1.3 of the SIP provides the procedures for determining reasonable potential to exceed applicable water quality criteria and objectives. The SIP specifies three triggers to complete an RPA:

Trigger 1 – If the MEC is greater than or equal to the CTR water quality criteria or applicable objective (C), a limitation is needed.

Trigger 2 – If background water quality (B) > C and the pollutant is detected in the effluent; a limitation is needed.

Trigger 3 – If other related information such as CWA 303(d) listing for a pollutant, discharge type, compliance history, then best professional judgment is used to determine that a limit is needed.

Sufficient effluent and ambient data are needed to conduct a complete RPA. If data are not sufficient, the Permittee will be required to gather the appropriate data for the Los Angeles Water Board to conduct the RPA. Upon review of the data, and if the Los Angeles Water Board determines that WQBELs are needed to protect the beneficial uses, this Order will be reopened for appropriate modification.

The RPA was performed for the priority pollutants regulated in the CTR for which data are available. Based on the RPA, pollutants that demonstrate reasonable potential to cause or contribute to an exceedance of the CTR criteria are Selenium, Cyanide, and 2,3,7,8 TCDD. The discharge also had reasonable potential to cause or contribute to an exceedance of the Basin Plan WQO for total trihalomethanes and iron. For iron, the receiving water concentration B was greater than C. For total trihalomethanes, the RPA procedure in USEPA's *Technical Support Document for Water Quality-based Toxics Control* (EPA/505/2-90-001, March 1991 (Technical Support Document) projected that the WQO would be exceeded in the receiving water. RPA for pollutants with WLA assigned by TMDLs are based on Tier 3, other information available, namely the TMDL for the pollutant. The following Table summarizes results from the RPA.

Table F-7. Summary of Reasonable Potential Analysis (Note a)

CTR No.	Constituent	Applicable C	MEC pg/L	B for DP 001	Need Limitation for DP 001?	Reason for RPA for DP 001
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10	Selenium	5	0.67	22	Yes	B≥C & detected in effluent
14	Cyanide	5.2 µg/L	6.4	3.8	Yes	MEC≥C
16	2,3,7,8-TCDD	0.014 pg/L	2.1	0.98	Yes	MEC≥C, B≥C
	Iron	300 µg/L	140	7,120	Yes	B≥C
	TTHMs	80 µg/L	61.5	0.17	Yes	TSD RPA

Footnotes for Table F-7:

- a. B: Maximum Detected Receiving Water
C: Water Quality (WQ) Criteria
CTR: California Toxics Rule
DP: Discharge Point
MEC: Maximum Effluent Concentration

End of Footnotes for Table F-7

4.3.5. WQBEL Calculations

- a. Calculation Options.** Once RPA has been conducted using either the TSD or the SIP methodologies, WQBELs are calculated. Alternative procedures for calculating WQBELs include:
- Use WLA from applicable TMDL.
 - Use a steady-state model to derive MDELs and AMELs.
 - Where sufficient data exist, use a dynamic model which has been approved by the State Water Board.
- b. Santa Clara River Chloride TMDL.** Chapter 7-6 of the Basin Plan, Upper Santa Clara River Chloride TMDL, describes that the final conditional WLAs for chloride (found on page 7-71 of the Basin Plan) for the Saugus and Valencia WRPs shall apply only when flow-weighting projects are in operation by the SCVSD according to the implementation section on page 7-73 of the Basin Plan. If these flow-weighting conditions are not met, WLAs for each plant shall be based on water quality objectives for chloride of 100 mg/L as a 3- month rolling average.

The 100 mg/L 3- month rolling average final effluent chloride limitation applies on the effective date of this Order. The Discharger is subject to a separate TSO until the UV disinfection and MF/RO capital improvement projects are completed because the Valencia WRP effluent data demonstrates that the Facility's discharge is unable to consistently comply with the final WLA-based limitations.

Santa Clara River Nitrogen Compounds TMDL. Chapter 7-9 of the Basin Plan, Santa Clara River Nitrogen Compounds TMDL, contains the final concentration-based WLAs allocated to major point sources of ammonia and nitrite plus nitrite nitrogen in the Santa Clara River, including the Saugus and Valencia WRPs. The permit contains final effluent limitations for the pollutants consistent with the final WLAs. These final effluent limitations apply on the effective date of this Order. The

Valencia WRP effluent data demonstrates that the Facility's discharge is able to comply with the final WLA-based limitations.

- d. SIP Calculation Procedure.** section 1.4 of the SIP requires the step-by-step procedure to "adjust" or convert CTR numeric criteria into AMELs and MDELs, for toxics.

Step 3 of section 1.4 of the SIP (page 8) lists the statistical equations that adjust CTR criteria for effluent variability.

Step 5 of section 1.4 of the SIP (page 10) lists the statistical equations that adjust CTR criteria for averaging periods and exceedance frequencies of the criteria/objectives. This section also reads, "For this method only, maximum daily effluent limitations shall be used for POTWs in place of average weekly limitations.

Sample calculation for 2,3,7,8-TCDD:

Step 1. Identify applicable water quality criteria.

From May 18, 2000 California Toxics Rule (CTR), we can obtain the Criterion Maximum Concentration (CMC) and the Criterion Continuous Concentration (CCC).

Freshwater Aquatic Life Criteria:

CMC = Not Applicable (NA) pg/L (CTR page 31712, column B1) and

CCC = NA pg/L (CTR page 31712, column B2); and

Human Health Criteria for Organisms only = 0.014 pg/L (CTR page 31712, column D2).

Step 2. Calculate effluent concentration allowance (ECA)

ECA = Criteria in CTR, since no dilution is allowed.

Step 3. Determine long-term average (LTA) discharge condition

Calculate CV:

$CV = \text{Standard Deviation} / \text{Mean} = 0.6$

Find the ECA Multipliers by calculating them using equations on SIP page 8. When $CV = 0.6$, then:

$ECA \text{ Multiplier}_{\text{acute}} = 0.321$ and

$ECA \text{ Multiplier}_{\text{chronic}} = 0.527$

$LTA_{\text{acute}} = ECA_{\text{acute}} \times ECA \text{ Multiplier}_{\text{acute}}$

$= NA \text{ pg/L} \times 0.321 = NA \text{ pg/L}$

$LTA_{\text{chronic}} = ECA_{\text{chronic}} \times ECA \text{ Multiplier}_{\text{chronic}}$

$= NA \text{ pg/L} \times 0.527 = NA \text{ pg/L}$

Step 4. Select the lowest LTA

In this case, both LTA_{chronic} and LTA_{acute} are NA, therefore lowest LTA is NA.

Step 5. Calculate the Average Monthly Effluent Limitation (AMEL) & Maximum Daily Effluent Limitation (MDEL) for AQUATIC LIFE

Find the multipliers. Find CV and n (frequency of sample collection per month). If effluent samples are collected 4 times a month or less, then $n = 4$. CV was determined to be 0.6 in a previous step.

AMEL Multiplier = 1.55

MDEL Multiplier = 3.11

AMEL aquatic life = lowest LTA (from Step 4) x AMEL Multiplier

= NA pg/L x 1.55 = NA pg/L

MDEL aquatic life = lowest LTA (from Step 4) x MDEL Multiplier

= NA pg/L x 3.11 = NA pg/L

Step 6. Find the Average Monthly Effluent Limitation (AMEL) & Maximum Daily Effluent Limitation (MDEL) for HUMAN HEALTH

Find factors. Given CV = 0.6 and $n = 4$.

For AMEL human health limit, there is no factor.

The MDEL/AMEL human health factor = 2.01

AMEL human health = ECA = 0.014 pg/L

MDEL human health = ECA x MDEL/AMEL factor

= 0.014 pg/L x 2.01 = 0.028 pg/L

Step 7. Compare the AMELs for Aquatic life and Human health and select the lowest. Compare the MDELs for Aquatic life and Human health and select the lowest

Lowest AMEL = 0.014 pg/L (Based on human health protection)

Lowest MDEL = 0.026 pg/L (Based on human health protection)

e. Impracticability Analysis

Federal NPDES regulations contained in 40 CFR section 122.45(d), states that, for POTWs, all permit limitations, standards, and prohibitions, including those to achieve water quality standards, shall, unless impracticable, be stated as average weekly and average monthly discharge limitations.

As stated by USEPA in its long-standing guidance for developing WQBELs, average limitations alone are not practical for limiting acute, chronic, and human health toxic effects (See, section 5.2.3 of USEPA's Technical Support Document for Water Quality-based Toxics Control (EPA/505/2-90-001, March 1991)).

For example, a POTW sampling for a toxicant to evaluate compliance with a 7-day average limitation could fully comply with this average limit, but still be discharging toxic effluent on one, two, three, or up to four of these seven days and not be meeting 1-hour average acute criteria or 4-day average chronic criteria

Similarly, a 7-day average alone would not protect one, two, three, or four days of discharging pollutants in excess of the acute and chronic criteria.

For these reasons, USEPA recommends daily maximum and 30-day average limits for regulating toxics in all NPDES discharges. For the purposes of protecting the acute effects of discharges containing toxicants, daily maximum limitations have been established in this NPDES permit for certain priority pollutants. Thirty-day (or monthly) average limitations have been established for priority pollutants that cause chronic or long-term impacts because they are carcinogenic, bioaccumulative and/or endocrine disruptors.

f. Mass-based limits. 40 CFR section 122.45(f)(1) requires that, except under certain conditions, or for certain pollutants, all permit limits, standards, or prohibitions be expressed in terms of mass units. 40 CFR section 122.45(f)(2) allows the permit writer, at its discretion, to express limits in additional units (e.g., concentration units). The regulations mandate that, where limits are expressed in more than one unit, the permittee must comply with both.

Generally, mass-based limits ensure that proper treatment, and not dilution, is employed to comply with the final effluent concentration limits. Concentration-based effluent limits, on the other hand, discourage the reduction in treatment efficiency during low-flow periods and require proper operation of the treatment units at all times. In the absence of concentration-based effluent limits, a permittee would be able to increase its effluent concentration (i.e., reduce its level of treatment) during low-flow periods and still meet its mass-based limits. To account for this, this Order includes mass and concentration limits for some constituents.

Table F-8. Summary of WQBELs for Pollutants for Discharge Points 001 & 002

Parameters	Units	Average Monthly	Maximum Daily	Notes
Temperature	°F	--	80	b
Total Coliform	MPN or CFU/ 100 mL	23	240	c
Oil and Grease	mg/L	10	15	--
Oil and Grease	lbs/day	1,800	2,700	a
Settleable Solids	ml/L	0.1	0.3	--
Total Residual Chlorine	mg/L	--	0.1	--
Total Residual Chlorine	lbs/day	--	18	a
Combined Radium-226 and Radium-228	pCi/L	5	--	--
Gross Alpha particle activity	pCi/L	15	--	--
Uranium	pCi/L	20	--	--
Gross Beta/photon emitters	millirem/year	4	--	--

Parameters	Units	Average Monthly	Maximum Daily	Notes
Strontium-90	pCi/L	8	--	--
Tritium	pCi/L	20,000	--	--
Ammonia Nitrogen	mg/L	1.75	5.2	--
Nitrate + Nitrite (as N)	mg/L	6.8	--	--
Nitrite (as N)	mg/L	0.9	--	--
Methylene blue active substances (MBAS)	mg/L	0.5	--	--
Iron	µg/L	300	--	--
Iron	lbs/day	16	--	--
Total trihalomethanes	µg/L	80	--	--
Total trihalomethanes	lbs/day	4.3	28	a
2,3,7,8-TCDD (Dioxin)	pg/L	0.014	0.028	--
2,3,7,8-TCDD (Dioxin)	lbs/day	2.5E-09	5E-09	a
Chronic Toxicity	Pass or Fail, (TST), Percent Effect	Pass	Pass (TST) or Percent Effect <50% (survival endpoint)	d & e

Footnotes for Tables F-8

- The mass-based effluent limitations are based on the plant design flow rate of 21.6 MGD and are calculated as follows: Flow (mgd) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations shall be the only applicable effluent limitations.
- An interim effluent limitation for temperature is included in section 4.1.2 of this Order for the duration of the compliance schedule.
- The wastes discharged to water courses shall be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if: (1) the median number of total coliform bacteria at some point in the treatment process does not exceed a 7-day median of 2.2 Most Probable Number (MPN) or Colony Forming Units (CFU) per 100 milliliters utilizing the bacteriological results of the last seven (7) days for which an analysis has been completed, (2) the number of total coliform bacteria does not exceed 23 MPN or CFU per 100 milliliters in more than one sample within any 30-day period, and (3) no sample shall exceed 240 MPN or CFU of total coliform bacteria per 100 milliliters. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
- The average monthly result is compared to the Median Monthly Effluent Limitation (MMEL) and shall be reported as "Pass" or "Fail." The maximum daily result is compared to Maximum Daily Effluent Limitation (MDEL) and shall be reported as "Pass" or "Fail" and "% Effect." If the chronic aquatic toxicity routine monitoring test conducted in the beginning of the month results in a "Fail" at the instream waste concentration (IWC), then the Valencia WRP shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the "Fail" at the IWC. If the first chronic MMEL compliance test results in a "Fail" at the IWC, then the second MMEL compliance

test is not necessary because the “Fail” results from the first two tests would constitute a violation of the chronic toxicity MMEL.

- e. A numeric WQBEL is established because effluent data showed that there was reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The Chronic Toxicity final effluent limitation is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objectives. These final effluent limitations are established using current USEPA guidance in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, June/2010) and [EPA Regions 8, 9, and 10 Toxicity Training Tool](#) (January 2010), <https://www.epa.gov/sites/production/files/documents/ToxTrainingTool10Jan2010.pdf>.

End of Footnotes for Tables F-8

4.3.6. Whole Effluent Toxicity (WET)

Whole effluent toxicity (WET) testing protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth. Chronic toxicity is a more stringent requirement than acute toxicity. A chemical at a low concentration can have chronic effects but no acute effects until it gets to the higher level. Therefore, chronic toxicity is considered a pollutant of concern for protection and evaluation of narrative Basin Plan objectives for toxicity.

Because of the nature of industrial discharges into the POTW sewer shed, it is possible that toxic constituents could be present in the influent of Valencia WRP, or could have synergistic or additive effects. As previously stated in this permit, Valencia WRP receives wastewater from 11 Significant Industrial User (SIU) permittees. Los Angeles Water Board staff determined that, pursuant to the SIP, reasonable potential exists for chronic toxicity. Thus, this Order carried over the chronic toxicity effluent limitation from the existing permit.

In the past, the State Water Board reviewed circumstances warranting a numeric chronic toxicity effluent limitation for POTWs when there is reasonable potential. (See SWRCB/OCC Files A-1496 & A-1496(a) [Los Coyotes/Long Beach Petitions].) On September 16, 2003, at a public hearing, the State Water Board adopted Order No. WQO 2003-0012 deferring the issuance of numeric chronic toxicity effluent limitation for POTWs until a subsequent Phase of the SIP is adopted. In the meantime, the State Water Board replaced the numeric chronic toxicity limit with a narrative effluent limitation and a 1 TUC trigger, in the Long Beach and Los Coyotes WRP NPDES permits.

Since the State Water Board adopted the Los Coyotes Order in 2003, USEPA published two new guidance documents with respect to chronic toxicity; and the Los Angeles Water Board adopted numerous NPDES permits for POTWs and industrial facilities incorporating the Test of Significant Toxicity (TST) statistical approach based effluent limitations for chronic toxicity and has adopted numerous permits containing numeric chronic toxicity effluent limitations including this Facility. Because the Los Coyotes Order explicitly “declined to make a determination ... regarding the propriety of the final numeric effluent limitations for chronic toxicity...,” (Los Coyotes Order, p. 9), and because of the differing facts before the Los Angeles Water Board in 2014 as

compared to the facts that were the basis for the Los Coyotes Order in 2003, the Los Angeles Water Board concludes that the Los Coyotes Order does not require inclusion of narrative rather than numeric effluent limitations for chronic toxicity. Further, the Los Angeles Water Board finds that numeric effluent limitations for chronic toxicity are necessary, feasible, and appropriate. Thus, Order Number R4-2015-0071, which served as the NPDES permit for the Valencia WRP, contained numeric chronic toxicity final effluent limitations expressed as Pass/ Fail and % Effect using the TST statistical approach.

Use of the TST statistical approach is consistent promulgated requirements of the test method, such as specified biological and laboratory procedures (see sections below for a description of U.S. EPA Method Update Rule and U.S. EPA's response). As has been affirmed by U.S. EPA, the TST statistical approach can be used with current U.S.EPA methods that require testing multiple concentrations of effluent. This ATP would provide significant cost savings for those existing permittees upon approval by USEPA.

Because effluent data exhibited reasonable potential to cause or contribute to an exceedance of the water quality objective, and consistent with the previous Order, this Order contains numeric chronic toxicity effluent limitations. Compliance with the chronic toxicity requirement contained in this Order shall be determined in accordance with section 7.10. of this Order. This Order contains a reopener to allow the Los Angeles Water Board to modify the permit, if necessary, to make it consistent with any new policy, law, or regulation. On December 1, 2020, the State Water Board adopted statewide numeric water quality objectives for both acute and chronic toxicity, using the TST, and a program of implementation to control toxicity, which are collectively known as the Toxicity Provisions. On October 5, 2021 the State Water Board adopted a resolution rescinding the December 1, 2020 establishment of Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California and confirming that the Toxicity Provisions were adopted as a State Policy for Water Quality Control, for all inland surface waters, enclosed bays, estuaries, and coastal lagoons of the state, regardless of their status as waters of the United States. The Provisions will take effect upon approval by the California Office of Administrative Law for purposes of state law and upon approval by the U.S. Environmental Protection Agency for purposes of federal law.

For this Order, chronic toxicity in the discharge is evaluated using the TST hypothesis testing statistical approach. Chronic toxicity limitations are expressed as "Pass" for the MMEL and "Pass" and "<50 Percent Effect" for each MDEL individual results. The chronic toxicity effluent limitations are as stringent as necessary to protect the statewide Water Quality Objective for aquatic chronic toxicity.

The prior permit had based the WET final effluent limitations on the following:

- a. In January 2010, USEPA published a guidance document titled *EPA Regions 8, 9 and 10 Toxicity Training Tool*, which among other things discusses permit limit expression for chronic toxicity. The document acknowledges that NPDES regulations at 40 CFR section 122.45(d) requires that all permit limitations be expressed, unless impracticable, as an AWEL and an AMEL for POTWs. Following

section 5.2.3 of the Technical Support Document (TSD), the use of an AWEL is not appropriate for WET. In lieu of an AWEL for POTWs, USEPA recommends establishing an MDEL for toxic pollutants and pollutants in water quality permitting, including WET. This is appropriate for two reasons. The basis for the average weekly requirement for POTWs derives from secondary treatment regulations and is not related to the requirement to assure achievement of water quality standards (WQS). Moreover, an average weekly requirement comprising of up to seven daily samples could average out daily peak toxic concentrations for WET and therefore, the discharge's potential for causing acute and chronic effects would be missed. It is impracticable to use an AWEL, because short-term spikes of toxicity levels that would be permissible under the 7-day average scheme would not be adequately protective of all beneficial uses. The MDEL is the highest allowable value for the discharge measured during a calendar day or 24-hour period representing a calendar day. The AMEL is the highest allowable value for the average of daily discharges obtained over a calendar month. For WET, this is the average of individual WET test results for that calendar month. However, in cases where a chronic mixing zone is not authorized, USEPA Regions 9 and 10 continue to recommend that the AMEL for chronic WET should be expressed as a median monthly limit (MMEL).

- b. In June 2010, USEPA published a guidance document titled *Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June 2010), in which they recommend the following: "Permitting authorities should consider adding the TST statistical approach to their implementation procedures for analyzing valid WET data for their current NPDES WET Program." The TST statistical approach is another statistical option for analyzing valid WET test data. Use of the TST statistical approach does not result in any changes to EPA's WET test methods. section 9.4.1.2 of the USEPA's *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R-02/013, 2002), recognizes that, "the statistical methods in this manual are not the only possible methods of statistical analysis." The TST statistical approach can be applied to acute (survival) and chronic (sublethal) endpoints and is appropriate to use for both freshwater and marine EPA WET test methods.
- c. USEPA's WET testing program and acute and chronic WET methods rely on the measurement result for a specific test endpoint, not upon achievement of specified concentration-response patterns to determine toxicity. USEPA's WET methods do not require achievement of specified effluent or ambient concentration-response patterns prior to determining that toxicity is present. For more details, see *Supplementary Information in support of the Final Rule establishing WET test methods* at 67 Fed. Reg. 69952, 69963 (November 19, 2002). Nevertheless, USEPA's acute and chronic WET methods require that effluent and ambient concentration-response patterns generated for multi-concentration acute and chronic toxicity tests be reviewed - as a component of test review following statistical analysis - to ensure that the calculated measurement result for the toxicity test is interpreted appropriately. (EPA-821-R-02-012, section 12.2.6.2; EPA-821-R-02-013, section 10.2.6.2.). In 2000, USEPA provided guidance for such reviews to ensure that test endpoints for determining toxicity based on the statistical approaches

utilized at the time the guidance was written (NOEC, LC50's, IC25's) were calculated appropriately (EPA 821-B-00-004).

- d. USEPA designed its 2000 guidance as a standardized step-by step review process that investigates the causes for 10 commonly observed concentration-response patterns and provides for the proper interpretation of the test endpoints derived from these patterns for NOECs, LC50s, and IC25s, thereby reducing the number of misclassified test results. The guidance provides one of three determinations based on the review steps: (1) that calculated effect concentrations are reliable and should be reported, (2) that calculated effect concentrations are anomalous and should be explained, or (3) that the test was inconclusive and should be repeated with a newly collected sample. The standardized review of the effluent and receiving water concentration-response patterns provided by USEPA's 2000 guidance decreased discrepancies in data interpretation for NOEC, LC50, and IC25 test results, thereby lowering the chance that a truly nontoxic sample would be misclassified and reported as toxic.
- e. Appropriate interpretation of the measurement result from USEPA's TST statistical approach (pass/fail) for effluent and receiving water samples is, by design, independent from the concentration-response patterns of the toxicity tests for those samples. Therefore, when using the TST statistical approach, application of USEPA's 2000 guidance on effluent and receiving waters concentration-response patterns will not improve the appropriate interpretation of TST results as long as all Test Acceptability Criteria and other test review procedures - including those related to Quality Assurance for effluent and receiving water toxicity tests, reference toxicity tests, and control performance (mean, standard deviation, and coefficient of variation) - described by the WET test methods manual and TST guidance, are followed. The 2000 guidance may be used to identify reliable, anomalous, or inconclusive concentration-response patterns and associated statistical results to the extent that the guidance recommends review of test procedures and laboratory performance already recommended in the WET test methods manual. The guidance does not assist in interpreting single-concentration (IWC) and control statistical t-tests and does not apply to the statistical assumptions on which the TST is based. The Los Angeles Water Board will not consider a concentration-response pattern as sufficient basis to determine that a TST t- test result for a toxicity test is anything other than valid, absent other evidence. In a toxicity laboratory, unexpected concentration-response patterns should not occur with any regular frequency and consistent reports of anomalous or inconclusive concentration-response patterns or test results that are not valid will require an investigation of laboratory practices.
- f. Any Data Quality Objectives or Standard Operating Procedure used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent or receiving water toxicity test measurement results from the TST statistical approach which include a consideration of concentration-response patterns and/or PMSDs must be submitted for review by the Los Angeles Water Board, in consultation with USEPA and the State Water Board's Quality Assurance Officer and Environmental Laboratory Accreditation Program (40 CFR section 122.41(h)). As described in the bioassay laboratory audit directives to the San Jose Creek

Water Quality Laboratory from the State Water Resources Control Board dated August 7, 2014, and from the USEPA dated December 24, 2013, the PMSD criteria only apply to compliance for NOEC and the sublethal endpoints of the NOEC, and therefore are not used to interpret TST results.

4.4. Final Effluent Limitation Considerations

4.4.1. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR part 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. The effluent limitations in this Order are as stringent as those in the prior permit, Order Number R4-2015-0071, with the exception of the limitations for copper, bis(2-ethylhexyl)phthalate, applicable to the Santa Clara River discharges.

a. Copper and bis(2-ethylhexyl)phthalate: Attainment Water for Santa Clara River

The final effluent limitations for copper and bis(2-ethylhexyl)phthalate that were included in the prior order are removed in this Order for the Santa Clara River because the discharge did not show reasonable potential to cause or contribute to an exceedance of the applicable water quality criteria for these pollutants, based on the most recent monitoring data.

Section 303(d)(4)(B) of the CWA allows relaxation of effluent limitations where the quality of the receiving water equals or exceeds the levels necessary to protect the designated uses of the water or otherwise required by applicable water quality standards, if the revision is subject to and consistent with the State's Antidegradation Policy. The Santa Clara River is not impaired for copper or bis(2-ethylhexyl)phthalate because the concentrations of these pollutants in the receiving water do not exceed the applicable water quality standards in the water column. The monitoring data, collected from May 2015 to July 2021, showed that the copper concentrations ranged from 3.8 µg/L to 1.0 µg/L, below the CTR criterion of 20 µg/L, and bis(2-ethylhexyl)phthalate concentrations ranged from <0.16 µg/L to 2 µg/L, below the 4 µg/L MCL. As described below, relaxation or removal of effluent limitations for these pollutants is consistent with the state and federal antidegradation policies. Therefore, the exception to the prohibition on relaxation of effluent limitations found in section 303(d)(4)(B) allows the removal of these effluent limitations.

4.4.2. Antidegradation Policies

40 CFR section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal antidegradation policy. On October 28, 1968, the State Water Board established California's antidegradation policy when it adopted Resolution Number 68-16, *Statement of Policy with Respect to Maintaining the Quality of the Waters of the State*. Resolution Number 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. In State Water Board Order Number 86-17 and an October 7, 1987 guidance memorandum, the State Water Board interpreted Resolution Number 68-16 to be fully

consistent with the federal antidegradation policy contained in 40 CFR section 131.12. Similarly, CWA section 303(d)(4)(B) and 40 CFR section 131.12 require that all permitting actions be consistent with the federal antidegradation policy. Together, the state and federal antidegradation policies are designed to ensure that a water body will not be degraded as a result of the permitted discharge. The Los Angeles Water Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies.

Discharges permitted in this Order are consistent with the antidegradation provisions of 40 CFR part 131.12 and State Water Board Resolution Number 68-16 because the discharge will not degrade existing high-quality water. Effluent limitations for copper and bis(2-ethylhexyl)phthalate are not included in this Order because monitoring data demonstrated that there is no reasonable potential for the discharge to cause or contribute to an exceedance of the water quality standards for these pollutants.

The Los Angeles Water Board may modify the terms of this Order to prevent degradation of high-quality waters based on any change in the concentration of these constituents in the effluent or receiving water that indicates that a degradation of receiving water quality may occur. The treatment required by this Order is the best practicable treatment or control of the discharge necessary to assure that a pollution or nuisance will not occur and the highest water quality consistent with maximum benefit to the people of the State will be maintained.

4.4.3. Stringency of Requirements for Individual Pollutants

This Order contains both TBELs and WQBELs for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD₅20°, TSS, and percent removal of BOD₅20°C and TSS. Restrictions on BOD₅20°C, and TSS are discussed in section 4.2 of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements as necessary to meet water quality standards.

Water quality-based effluent limitations have been scientifically derived to implement WQOs that protect beneficial uses. Both the beneficial uses and the WQOs have been approved or established (in the case of CTR criteria) pursuant to federal law and are the applicable federal water quality standards. The scientific procedures for calculating the individual water quality-based effluent limitations for priority pollutants are based on the CTR and SIP. All beneficial uses and WQOs contained in the Basin Plan or statewide water quality control plans were approved under state law and submitted to and approved by U.S. EPA prior to May 30, 2000. Any WQOs and beneficial uses submitted to U.S. EPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR section 131.21(c)(1). With the exception of the limits for turbidity, and total coliform, this Order's restrictions on individual pollutants are collectively no more stringent than required to implement the requirements of the CWA and the applicable water quality standards for purposes of the CWA).

**Table F-9. Summary of Final Effluent Limitations for Discharge Points 001 & 002
(Santa Clara River)**

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Notes
BOD ₅ 20°C	mg/L	20	30	45	Existing/ Secondary treatment	--
BOD ₅ 20°C	lbs/day	3,600	5,400	8,100	Existing/ Secondary treatment	b
TSS	mg/L	15	40	45	Existing/ Secondary treatment	--
TSS	lbs/day	2,700	7,200	8,100	Existing/ Secondary treatment	b
Temperature	°F	--	--	80	Basin Plan	h
Total Coliform	MPN or CFU/100 mL	23	2.2	240	Title 22	c
Removal Efficiency for BOD ₅ 20°C and TSS	%	≥85	--	--	Existing/ Technology base	--
Oil and Grease	mg/L	10	--	15	Existing/ Basin Plan	--
Oil and Grease	lbs/day	1,800	--	2,700	Existing/ Basin Plan	b
Settleable Solids	ml/L	0.1	--	0.3	Existing/ Basin Plan	--
Total Residual Chlorine	mg/L	--	--	0.1	Existing/ Basin Plan	--
Total Residual Chlorine	lbs/day	--	--	18	Existing/ Basin Plan	b
Combined Radium- 226 and Radium 228	pCi/L	5	--	--	Existing/ Title 22	--
Gross Alpha particle activity (excluding radon and uranium)	pCi/L	15	--	--	Existing/ Title 22	--
Uranium	pCi/L	20	--	--	Existing/ Title 22	--
Gross Beta/photon emitters	millirem/ year	4	--	--	Existing/ Title 22	--

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Notes
Strontium-90	pCi/L	8	--	--	Existing/ Title 22	--
Tritium	pCi/L	20,000	--	--	Existing/ Title 22	--
Total dissolved solids (TDS)	mg/L	1,000	--	--	Existing/ Basin Plan	--
TDS	lbs/day	180,000	---	---	Existing/ Calculation	--
Sulfate	mg/L	400	---	---	Existing/ Basin Plan	--
Sulfate	lbs/day	72,000	--	--	Existing/ Calculation	--
Chloride	mg/L	100	---	---	Revised/ TMDL	Note d
Chloride	mg/L	Formula	--	230	Revised/ TMDL	Note e
Boron	mg/L	1.5	--	--	Existing/ Basin Plan	--
Boron	lbs/day	270	--	--	Existing/ Calculation	--
Methylene Blue Activated Substances (MBAS)	mg/L	0.5	--	--	Existing/ Basin Plan	--
MBAS	lbs/day	90	--	--	Existing/ Calculation	b
Ammonia Nitrogen	mg/L	1.75	--	5.2	Existing/ TMDL	--
Nitrate + Nitrite (as N)	mg/L	6.8	--	--	Existing/ TMDL	--
Nitrite (as N)	mg/L	0.9	--	--	Existing/ TMDL	--
Selenium	µg/L	4.5	--	6.8	SIP/CTR	--
Selenium	lbs/day	0.81	--	1.2	SIP/CTR	b
Cyanide	µg/L	4.4	--	8.1	SIP/CTR	--
Cyanide	lbs/day	0.79	--	1.5	SIP/CTR	b
Iron	µg/L	300	--	--	Existing/ Basin Plan	--
Iron	lbs/day	54	--	--	Existing/ Calculation	b
Total trihalomethanes (TTHMs)	µg/L	80	--	--	Existing/ Basin Plan	--

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Notes
TTHMs	lbs/day	14	--	--	Existing/ Calculation	b
2,3,7,8-TCDD (Dioxin)	pg/L	0.014	--	0.028	SIP/CTR	--
2,3,7,8-TCDD (Dioxin)	lbs/day	2.5E-09	--	5E-09	SIP/CTR	b
Chronic Toxicity <i>Ceriodaphnia dubia</i> Survival and reproduction endpoint	Pass or Fail (TST), Percent Effect	Pass	--	Pass (TST) or Percent Effect < 50% (survival endpoint)	Existing	f & g

Footnotes for Table F-9

- The effluent values for pH shall be maintained within the limits of 6.5 (instantaneous minimum) and 8.5 (instantaneous maximum).
- The mass-based limitations are based on the plant design flow rate of 21.6 MGD and are calculated as follows: Flow (MGD) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations shall be provide the only applicable effluent limitations.
- The wastes discharged to water courses shall at all times be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if (1) the median number of total coliform bacteria at some point in the treatment process does not exceed a 7-day median of 2.2 Most Probable Number (MPN) or Colony Forming Units (CFU) per 100 milliliters utilizing the bacteriological results of the last seven (7) days for which an analysis has been completed, (2) the number of total coliform bacteria does not exceed 23 MPN or CFU per 100 milliliters in more than one sample within any 30-day period, and (3) no sample shall exceed 240 MPN or CFU of total coliform bacteria per 100 milliliters. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
- The 100 mg/L 3-month rolling average is based on the water quality objective in the Basin Plan because the Permittee was not able to meet the July 1, 2019 deadline, specified in the Implementation section of the *Upper Santa Clara River Chloride TMDL*, Resolution No. 2008-012, for completing the ultra violet (UV) disinfection and microfiltration and reverse osmosis (MF/RO) capital improvement projects. On January 1, 2023, this limitation will be superseded by the limitation described below in Footnote e, upon the expiration of Time Schedule Order No. R4-2019-0055-A02 and the completion of the UV and MF/RO construction at the Valencia WRP.
- The following three-month rolling average effluent limitation is derived from the waste load allocation for chloride, as set forth in Resolution No. R14-010, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate an Averaging Period for Chloride Water Quality Objectives in Reaches 4B, 5 and 6; Incorporate New Site Specific Objectives for Chloride in Reaches 5 and 6; and Revise the Total Maximum Daily Load for Chloride in the Upper Santa Clara River*, adopted by the Los Angeles Water Board on October 9, 2014 and approved by the State Water Resources Control Board, the Office of Administrative Law (OAL) and USEPA on, December 16, 2014, March 18, 2015, and April 28, 2015, respectively:

$$C_{VAL,3mo.av} = \frac{1}{3} * \sum_{mi=1}^3 \left[\frac{Q_{SAU,mi}(100 - C_{SAU,mi})}{Q_{VAL,mi}} + 100 \right]$$

Where: $Q_{VAL,mi}$ = Valencia WRP monthly effluent flow in MGD;

$C_{SAU,mi}$ = Saugus WRP monthly effluent chloride level in mg/L;

$C_{VAL,mi}$ = Valencia WRP monthly effluent chloride level in mg/L;

$Q_{SAU,mi}$ and $Q_{VAL,mi}$ shall not exceed the design flow during dry-weather periods.

On January 1, 2023, this limitation will supersede the limitation described above in Footnote d, upon the expiration of Time Schedule Order No. R4-2019-0055-A02 and the completion of the UV and MF/RO construction at the Valencia WRP

- f. A numeric WQBEL is established because effluent data showed that there is reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The Chronic Toxicity final effluent limitation is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objectives. These final effluent limitations are established using current USEPA guidance in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June /2010) and [EPA Regions 8, 9, and 10 Toxicity Training Tool](#) (January 2010), <https://www.epa.gov/sites/production/files/documents/ToxTrainingTool10Jan2010.pdf>.
- g. The average monthly result is compared to the Median Monthly Effluent Limitation (MMEL) and shall be reported as "Pass" or "Fail." The maximum daily result is compared to Maximum Daily Effluent Limitation (MDEL) and shall be reported as "Pass" or "Fail" and "% Effect." If the chronic aquatic toxicity routine monitoring test conducted in the beginning of the month results in a "Fail" at the instream waste concentration (IWC), then the Valencia WRP shall complete a maximum of two MMEL compliance tests. The MMEL compliance tests shall be initiated within the same calendar month that the first routine monitoring test was initiated that resulted in the "Fail" at the IWC. If the first chronic MMEL compliance test results in a "Fail" at the IWC, then the second MMEL compliance test is not necessary because the "Fail" results from the first two tests would constitute a violation of the chronic toxicity MMEL.
- h. An interim effluent limitation for temperature is included in section 4.1.2 of this Order for the duration of the compliance schedule.

End of Footnotes for Table F-9

4.5. Interim Effluent Limitations

The Valencia WRP will be subject to a compliance schedule for the temperature 80°F final effluent limitation, as described further in section 6.2.7 of this Fact Sheet. Since that compliance schedule exceeds one year, the Order includes an 86°F interim effluent limitation for temperature, except as a result of external ambient temperature.

4.6. Land Discharge Specifications – Not Applicable

4.7. Recycling Specifications

On October 23, 2019, the Permittee submitted a feasibility study of recycling, conservation, and alternative disposal methods of wastewater. The Santa Clarita Valley Water Agency (SCV Water), formerly the Castaic Lake Water Agency (CLWA), is the regional importer and wholesaler of State Project water in the Santa Clarita Valley and has begun the implementation of a recycled water distribution system. Construction of a 10,000 gpm

pump station located adjacent to the Valencia WRP's chlorine contact tanks was completed in 1999; construction of a transmission pipeline southerly along The Old Road to Valencia Boulevard was completed in May 2002; recycled water deliveries at the Tournament Players Club golf course and several road medians began in September 2003; several other road medians were added to the system in 2010 and three sites at the Entrada development were connected in 2015. Most recently, SCV Water has completed construction of its Phase 2D extension on Westridge Parkway and Valencia Boulevard and is in the final permitting phase to accommodate service of several irrigation accounts along the alignment.

In June 2009, the former CLWA began investigating the feasibility of expanding the recycled water distribution system. Phase 2A of the Master Plan consists of a booster pump station, several thousand feet of pipelines and a storage reservoir. This system would deliver and estimated 511 AFY of recycled water to the 80-acre Central Park, the River Village and Bridgeport developments and assorted city landscaping; however, design has not yet begun and this project has been put on hold by SCVWA. Design on the 670 AFY Phase 2C (South End Project) is in its final stages. Phase 2C consists of a new transmission main from Valencia Blvd/The Old Road to Newhall Elementary School with the alignment running east on Valencia Blvd, south on Rockwell Canyon to McBean Parkway.

Five Point Valencia, formerly the Newhall Land and Farming Company, a major landowner in the Santa Clarita Valley, is developing a 15,000-acre residential/commercial development known as Newhall Ranch. A new sanitation district, the Newhall Ranch Sanitation District of Los Angeles County, has been formed to serve the wastewater needs of Newhall Ranch and is now officially a part of the Sanitation Districts of Los Angeles County. The future Newhall Ranch WRP is expected to produce approximately 3,750 AFY of recycled water to be reused for landscape irrigation and other approved uses within Newhall Ranch. Wastewater from the initial phases of development will be conveyed to the Valencia WRP until there is sufficient flow generated to operate the Newhall Ranch WRP (approximately 6,000 homes), at which time the flow will be redirected. The earliest phase of development is currently under construction. Model homes opened in April 2021.

In 2020, the Valencia WRP treated 13.67 MGD of influent and produced 13.61 MGD of tertiary treated effluent on average. An average of 0.42 MGD of Valencia WRP's tertiary-treated effluent was delivered to the recycled water distribution systems for non-potable reuse. Recycled water was delivered to reuse sites that overlie the Santa Clara River Valley East Groundwater Basin.

Applicable state law requires dischargers to file a petition with the State Water Board, Division of Water Rights (DWR) and receive approval for any change in the point of discharge, place of use, or purpose of use of treated wastewater that decreases the flow in any portion of a watercourse. On October 21, 1993, the State Water Board DWR approved wastewater change petition Number WW0011, which allows the Permittee to reduce the treated effluent discharge from the Valencia WRP for irrigation purposes.

SCVSD has not filed a recent wastewater use petition. However, on October 4, 2021, Santa Clarita Valley Water Agency (SCV Water), the distributor and purveyor of recycled water in the Santa Clarita service area, contacted the State Water Board DWR to request

approval to use water from the New Drop Program for irrigation purposes. On November 9, 2021, the State Water Board DWR notified SCV Water that a wastewater change petition was not needed for its New Drop Program because the New Drop Project would not decrease Santa Clara River flow. The New Drop Project would rely on “new flows” produced from new development in the SCV Water service area that have not previously been discharged to the Santa Clara River. These “new flows” referred to as New Drop water will be provided as influent to the Valencia WRP. Recycled water generated from New Drop sources will be utilized from the Valencia WRP to initially provide, during early phases of development at Five Point Valencia, up to up to 3,092 acre-feet per year for irrigation use to new homes, commercial-industrial, retail-restaurants, and hotels-schools in the SCV Water service area. SCV Water is currently enrolled under the State Water Board’s Order WQ 2016-0068-DDW, Water Reclamation Requirements for Recycled Water Use (General Recycled Water Order) to use recycled water for dust control purposes. However, they plan on preparing a Title 22 Engineering Report that would be submitted to the State Water Board Division of Drinking Water for approval of the following additional uses of recycled water: landscape irrigation, dual-plumbed irrigation, and landscape impoundments. SCV Water proposes to complete the recycled water irrigation projects under this request by 2030. Once the State Water Board DDW approves the Title 22 Engineering Report, SCV Water will submit a Notice of Intent to the Los Angeles Water Board, to revise their enrollment under the General Recycled Water Order. The State Water Board retains separate jurisdictional authority to enforce any applicable requirements under Water Code section 1211. This is not an NPDES permit requirement.

The Permittee shall continue to investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or beneficial use of stormwater and dry-weather urban runoff. The Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.

5. RATIONALE FOR RECEIVING WATER LIMITATIONS

5.1. Surface Water

Receiving water limitations are based on WQOs contained in the Basin Plan and applicable statewide water quality control plans and are a required part of this Order.

6. RATIONALE FOR PROVISIONS

6.1. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR section 122.42, are provided in Attachment D. The Permittee must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

40 CFR sections 122.41(a)(1) and (b) through (n) establish conditions that apply to all state issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 CFR allows the state to omit or modify conditions to impose more stringent requirements. In accordance

with 40 CFR section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR section 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

6.2. Special Provisions

6.2.1. Reopener Provisions

This provision is based on 40 CFR part 123. The Los Angeles Water Board may reopen the permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, new information based on the results of any special studies, conducted as required by this Order, or adoption of new regulations by the State Water Board or Los Angeles Water Board, including revisions to the Basin Plan.

6.2.2. Special Studies and Additional Monitoring Requirements

a. Antidegradation Analysis and Engineering Report for Any Proposed Plant Expansion. This provision is based on the State Water Board Resolution Number 68-16, which requires the Los Angeles Water Board in regulating the discharge of waste to maintain high quality waters of the state. The Permittee must demonstrate that it has implemented adequate controls (e.g., adequate treatment capacity) to ensure that high quality waters will be maintained. This provision requires the Permittee to clarify that it has increased plant capacity through the addition of new treatment system(s) to obtain alternative effluent limitations for the discharge from the treatment system(s). This provision requires the Permittee to report specific time schedules for the plant's projects. This provision requires the Permittee to submit a report to the Los Angeles Water Board for approval.

b. Operations Plan for Proposed Expansion. This provision is based on section 13385(j)(1)(D) of the Water Code and allows a time period not to exceed 90 days in which the Permittee may adjust and test the treatment system(s). This provision requires the Permittee to submit an Operations Plan describing the actions the Permittee will take during the period of adjusting and testing to prevent violations.

c. Treatment Plant Capacity. The treatment plant capacity study required by this Order shall serve as an indicator for the Los Angeles Water Board regarding Facility's increasing hydraulic capacity and growth in the service area.

6.2.3. Best Management Practices and Pollution Prevention

Pollutant Minimization Program (PMP). This provision is based on the requirements of section 2.4.5 of the SIP.

6.2.4. Construction, Operation, and Maintenance Specifications

This provision is based on the requirements of 40 CFR section 122.41(e) and the previous Order. 40 CFR section 122.41(e) also requires the operation of back-up or auxiliary facilities or similar systems when the operation is necessary to achieve compliance with the conditions of the permit. For proper and effective operation of such facilities or systems, routine maintenance and operational testing of emergency infrastructure/equipment is necessary. Major sewage spills can cause harm to

residents of the Los Angeles Region, such as the closure of beaches, and harm to wildlife and benthic life. The impact of any such incident to the receiving waters can be minimized or prevented if the operation of emergency infrastructure occurs unimpeded by operational challenges and in a timely fashion. Thus, this Order contains requirements for routine maintenance and operational testing of emergency infrastructure/equipment in section 6.3.4.d.

6.2.5. Special Provisions for Municipal Facilities (POTWs Only)

- a. Biosolids Requirements.** To implement CWA section 405(d), on February 19, 1993, USEPA promulgated 40 CFR part 503 to regulate the use and disposal of municipal sewage sludge. This regulation was amended on September 3, 1999. The regulation requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. It is the responsibility of the Permittee to comply with said regulations that are enforceable by USEPA, because California has not been delegated the authority to implement this program. The Permittee is also responsible for compliance with WDRs and NPDES permits for the generation, transport and application of biosolids issued by the State Water Board, other Los Angeles Water Boards, Arizona Department of Environmental Quality or USEPA, to whose jurisdiction the Facility's biosolids will be transported and applied.
- b. Pretreatment Requirements.** This Order contains pretreatment requirements consistent with applicable effluent limitations, national standards of performance, and toxic and performance effluent standards established pursuant to sections 208(b), 301, 302, 303(d), 304, 306, 307, 403, 404, 405, and 501 of the CWA, and amendments thereto. This Order contains requirements for the implementation of an effective pretreatment program pursuant to section 307 of the CWA; 40 CFR parts 35 and 403; and/or title 23, CCR section 2233.
- c. Filter Bypass Requirements.** Conditions pertaining to bypass are contained in Attachment D, section 1. Standard Provisions – Permit Compliance, subsection 7. The bypass or overflow of untreated or partially treated wastewater to waters of the State is prohibited, except as allowed under conditions stated in 40 CFR section 122.41(m) and (n). During periods of elevated, wet weather flows, a portion of the secondary treated wastewater is diverted around the tertiary filters as a necessary means to avoid loss of life, personal injury or severe property damage. There are no feasible alternatives to this diversion. These anticipated discharges are approved under the bypass conditions when all storage has been utilized and the resulting combined discharge of fully treated (tertiary) and partially treated (secondary) wastewater complies with the effluent and receiving water limitations in this Order. The ROWD constitutes notice of these anticipated bypasses.
- d. Spill Reporting Requirements.** This Order established a reporting protocol for how different types of spills, overflow or bypasses of raw or partially treated sewage from its collection system or treatment plant covered by this Order shall be reported to regulatory agencies.

The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (SSS WDRs) on May 2, 2006. The Monitoring and Reporting Requirements for the SSS WDRs were

amended by Water Quality Order WQ 2008-0002-EXEC on February 20, 2008. The SSS WDRs requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the SSS WDRs. The SSS WDRs requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the SSS WDRs contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch that the Permittee's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section 6.3.5. For instance, the 24-hour reporting requirements in this Order are not included in the SSS WDRs. The Permittee must comply with both the SSS WDRs and this Order. The Permittee and public agencies that are discharging wastewater into the Facility were required to obtain enrollment for regulation under the SSS WDRs by December 1, 2006.

In the past, the Los Angeles Water Board has experienced loss of recreational use in coastal beaches and in recreational areas as a result of major sewage spills. The SSS WDRs requirements are intended to prevent or minimize impacts to receiving waters as a result of spills.

6.2.6. Other Special Provisions (Not Applicable)

6.2.7. Compliance Schedules

In general, an NPDES permit must include final effluent and receiving surface limitations that are consistent with CWA section 301 and with 40 CFR. section 122.44(d). There are exceptions to this general rule. State Water Board adopted the *Policy for Compliance Schedules in National Pollutant Discharge Elimination System Permits* (Resolution 2008-0025), which is the governing policy for compliance schedules in NPDES permits (hereafter "Compliance Schedule Policy"). The Compliance Schedule Policy allows compliance schedules for new, revised, or newly interpreted water quality objectives or criteria, or in accordance with a TMDL. All compliance schedules must be as short as possible and may not exceed 10 years from the effective date of the adoption, revision, or new interpretation of the applicable water quality objective or criterion, unless a TMDL allows a longer schedule. Where a compliance schedule for a final effluent limitation exceeds 1 year, the Order must include interim numeric effluent limitations for that constituent or parameter, interim requirements and dates toward achieving compliance, and compliance reporting within 14 days after each interim date. The Order may also include interim requirements to control the pollutant, such as pollutant minimization and source control measures.

In accordance with the Compliance Schedule Policy and 40 CFR 122.47, a Discharger who seeks a compliance schedule must demonstrate additional time is necessary to implement actions to comply with a more stringent permit limitation. The Discharger must provide the following documentation as part of the application requirements:

- a. Diligent efforts have been made to quantify pollutant levels in the discharge and the sources of the pollutant in the waste stream, and the results of those efforts;

- b. Source control efforts are currently underway or completed, including compliance with any pollution prevention programs that have established;
- c. A proposed schedule for additional source control measures or waste treatment;
- d. Data demonstrating current treatment facility performance to compare against existing permit effluent limits, as necessary to determine which is the more stringent interim, permit effluent limit to apply if a schedule of compliance is granted;
- e. The highest discharge quality that can reasonably be achieved until final compliance is attained;
- f. The proposed compliance schedule is as short as possible, given the type of facilities being constructed or programs being implemented, and industry experience with the time typically required to construct similar facilities or implement similar programs; and
- g. Additional information and analyses to be determined by the Los Angeles Water Board on a case-by-case basis.

The permit limitations for temperature in effluent and receiving surface waters are more stringent than the limitations previously implemented. The new limitations are based on a new interpretation of temperature WQO established in the Basin Plan. The Discharger has complied with the application requirements in paragraph 4 of the Compliance Schedule Policy. Based on information submitted with the Report of Waste Discharge, self-monitoring reports, infeasibility analysis, and other miscellaneous submittals, it has been demonstrated to the satisfaction of the Los Angeles Water Board that the Discharger needs time to implement actions to comply with the new effluent and receiving water limitations for temperature. Therefore, this Order includes the following compliance schedule to comply with the newly interpreted effluent limitation for temperature.

Table F-10. Compliance Schedule & Milestone Dates

Task	Completion Date
Submit and Begin Implementation of Pollution Prevention Plan (PPP) for Source Control	August 1, 2022
Convene Technical Advisory Committee and Stakeholder Committee	October 1, 2022
Release the Request for Proposal to Retain Consultant to Evaluate Temperature Impacts in the Watershed and Management Options	February 28, 2023
Finalize the Technical Workplan	February 28, 2024
Prepare a Technical Workplan Progress Report	February 28, 2025
Complete Implementation of Technical Workplan	February 28, 2026
Notify Los Angeles Water Board of Selected Preferred Project and Identify Regulatory Approval Process and Present Results of Technical Workplan at Next Scheduled Board Meeting	April 1, 2026
Complete Preliminary Design	April 1, 2027
Complete Environmental Review	October 1, 2027
Complete Design of Preferred Project	July 1, 2028
Issue Notice to Proceed for Project Work	July 1, 2029
Complete Preferred Project	July 1, 2030

7. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code section 13383 also authorizes the Los Angeles Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The MRP of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this Facility.

7.1. Influent Monitoring

Influent monitoring is required:

- To determine compliance with the permit conditions for BOD₅ 20°C and suspended solids removal rates.
- To assess treatment plant performance.
- To assess the effectiveness of the Pretreatment Program.

- As a requirement of the .

7.2. Effluent Monitoring

The Permittee is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the MRP Attachment E. This provision requires compliance with the MRP, and is based on 40 CFR sections 122.44(i), 122.62, 122.63, and 124.5. The MRP is a standard requirement in almost all NPDES permits (including this Order) issued by the Los Angeles Water Board. In addition to containing definition of terms, it specifies general sampling/analytical protocols and the requirements of reporting spills, violation, and routine monitoring data in accordance with NPDES regulations, the Water Code, and Los Angeles Water Board policies. The MRP also contains sampling program specific for the Permittee's wastewater treatment plant. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with section 1.3 of the SIP, a periodic monitoring is required for all priority pollutants defined by the CTR, for which criteria apply and for which no effluent limitations have been established, to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

Monitoring for those pollutants expected to be present in the discharge from the Facility, will be required as set forth in the MRP and as required in the SIP. Semi-annual monitoring for priority pollutants in the effluent is required in accordance with the Pretreatment requirements.

The accelerated chronic toxicity monitoring, which served as an indicator of persistent toxicity, was not included in the 2022 permit, so that the Discharger may initiate a Toxicity Reduction Evaluation (TRE) sooner than in the 2015 permit. On occasions when the toxicity was intermittent, the accelerated monitoring step delayed the initiation of the TRE, such that when the TRE was initiated, the effluent no longer exhibited toxicity, and subsequently, the cause of toxicity could not be identified. In the 2021 permit, a TRE would be initiated following two consecutive chronic toxicity violations.

The monitoring for fecal coliform was removed because the monitoring results from *E. coli* testing would be indicative of the presence of fecal coliform.

Table F-11. Monitoring Frequency Comparison

Parameter	Monitoring Frequency (2015 Permit)	Monitoring Frequency (2022 Permit)
Total flow	continuous	continuous
Turbidity	continuous	continuous
Total residual chlorine	continuous	continuous
Total residual chlorine	daily	daily
Total coliform	weekly	weekly
Fecal coliform	weekly	deleted
<i>E. coli</i>	weekly	weekly
Temperature	weekly	weekly
pH	weekly	weekly

Parameter	Monitoring Frequency (2015 Permit)	Monitoring Frequency (2022 Permit)
Settleable solids	weekly	weekly
Total suspended solids (TSS)	weekly	weekly
BOD5 20°C	weekly	weekly
Oil and grease	quarterly	quarterly
Dissolved oxygen	monthly	monthly
Total dissolved solids	monthly	monthly
Sulfate	monthly	monthly
Chloride	monthly	monthly
Boron	monthly	monthly
Fluoride	annually	annually
Ammonia nitrogen	monthly	monthly
Nitrite nitrogen	monthly	monthly
Nitrate nitrogen	monthly	monthly
Organic nitrogen	monthly	monthly
Total Kjeldahl nitrogen	monthly	monthly
Total nitrogen	monthly	monthly
Total phosphorus	quarterly	quarterly
Orthophosphate -P	quarterly	quarterly
Surfactants (MBAS)	quarterly	quarterly
Surfactants (CTAS)	quarterly	quarterly
Total hardness (CaCO ₃)	monthly	monthly
Chronic toxicity <i>Ceriodaphnia dubia</i> Survival and reproduction endpoints	monthly	monthly
Radioactivity (Including gross alpha, gross beta, combined radium-226 and radium-228, tritium, strontium-90 and uranium)	semiannually	semiannually
TCDD equivalents	quarterly	quarterly
Benzo(a)Pyrene	semiannually	semiannually
Benzo(b)Fluoranthene	semiannually	semiannually
Benzo(k)Fluoranthene	semiannually	semiannually
Chrysene	semiannually	semiannually
Dibenzo(a,h)Anthracene	semiannually	semiannually
Indeno(1,2,3-cd)Pyrene	semiannually	semiannually
Antimony	semiannually	semiannually
Arsenic	monthly	quarterly
Beryllium	semiannually	semiannually

Parameter	Monitoring Frequency (2015 Permit)	Monitoring Frequency (2022 Permit)
Cadmium	semiannually	semiannually
Chromium III	semiannually	semiannually
Chromium VI	semiannually	semiannually
Total Chromium	semiannually	semiannually
Copper	monthly	quarterly
Lead	semiannually	semiannually
Mercury	monthly	quarterly
Nickel	quarterly	semiannually
Selenium	monthly	monthly
Silver	semiannually	quarterly
Thallium	semiannually	semiannually
Zinc	quarterly	quarterly
Cyanide	monthly	monthly
Bis(2-ethylhexyl)phthalate	monthly	quarterly
Dichlorobromomethane	monthly	monthly
Dibromochloromethane	monthly	monthly
Bromoform	monthly	monthly
Chloroform	monthly	monthly
Total trihalomethanes (TTHMs)	monthly	monthly
Iron	monthly	monthly
Perchlorate	semiannually	semiannually
1,4-Dioxane	semiannually	semiannually
1,2,3-Trichloropropane	semiannually	semiannually
Methyl tert-butyl-ether (MTBE)	semiannually	semiannually
PCBs as aroclors	annually	annually
PCBs as congeners	annually	annually
Chlorpyrifos	semiannually	annually
Diazinon	semiannually	annually
Remaining USEPA priority pollutants excluding asbestos and PCBs	semiannually	semiannually

This Order has maintained the same monitoring frequency in the previous permit for most of constituents except a few pollutants. Monitoring for fecal coliform was removed as a result of Basin Plan Resolution Number R10-005 that eliminates the limitation for fecal coliform. The monitoring for copper and bis(2-ethylhexyl)phthalate was decreased from monthly to quarterly because the effluent no longer had reasonable potential to exceed the criteria. The monitoring for chlorpyrifos and diazinon were decreased to annually because they were not detected.

7.3. Whole Effluent Toxicity Requirements

WET testing protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth. Chronic toxicity is a more stringent requirement than acute toxicity. A chemical at a low concentration can have chronic effects but no acute effects until it gets to the higher level. For this Order, chronic toxicity in the discharge is evaluated using USEPA's 2010 TST hypothesis testing statistical approach and is expressed as "Pass" or "Fail" for the median monthly summary results and "Pass" or "Fail" and "Percent Effect" for maximum daily chronic toxicity result. The chronic toxicity effluent limitations protect the narrative Basin Plan Water Quality Objective for chronic toxicity. The rationale for WET has been discussed extensively in section 4.3.6 of this Fact Sheet.

7.4. Receiving Water Monitoring

7.4.1. Surface Water

Receiving water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water.

7.4.2. Groundwater

Groundwater monitoring is required to characterize the water quality of the groundwater.

7.5. Other Monitoring Requirements

7.5.1. Watershed Monitoring and Bioassessment Monitoring

The goals of the Watershed-wide Monitoring Program including the bioassessment monitoring for the Santa Clara River Watershed are to:

- a. Determine compliance with receiving water limits.
- b. Evaluate progress in achieving numeric targets and waste load allocations in the Santa Clara River and Tributaries Metals TMDLs.
- c. Monitor trends in surface water quality.
- d. Ensure protection of beneficial uses.
- e. Provide data for modeling contaminants of concern.
- f. Characterize water quality including seasonal variation of surface waters within the watershed.
- g. Assess the health of the biological community.
- h. Determine mixing dynamics of effluent and receiving waters.

7.5.2. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), USEPA requires major and selected minor dischargers under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of

laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by USEPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to USEPA's DMR-QA Coordinator and Quality Assurance Manager.

8. CONSIDERATION OF NEED TO PREVENT NUISANCE AND WATER CODE SECTION 13241 FACTORS.

Some of the provisions/requirements in this Order are included to implement state law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations. As required by Water Code section 13263, the Los Angeles Water Board has considered the need to prevent nuisance and the factors listed in Water Code section 13241 in establishing the state law provisions/requirements. The Los Angeles Water Board finds, on balance, that the state law requirements in this Order are reasonably necessary to prevent nuisance and to protect beneficial uses identified in the Basin Plan, and the section 13241 factors are not sufficient to justify failing to protect those beneficial uses.

8.1. Need to prevent pollution or nuisance: In establishing effluent limitations in this Order, the Los Angeles Water Board has considered state law to prevent pollution or nuisance as defined in section 13050, subdivisions (l) and (m), of the Water Code. Many effluent limitations are also required to comply with narrative water quality objectives in the Basin Plan. These state requirements include, but are not limited to, spill prevention plans, operator certification, sanitary sewer overflow reporting, and requirements for standby or emergency power. All of these requirements are designed to prevent pollution or nuisance.

8.2. Past, present, and probable future beneficial uses of water: Chapter 2 of the Basin Plan identifies designated beneficial uses for water bodies in the Los Angeles Region. Beneficial uses of water relevant to this Order are also identified above in section 3.3.1. The Los Angeles Water Board has taken this factor into account in establishing effluent limitations in the Order and follows the anti-backsliding and antidegradation requirements (Fact Sheet sections 4.4.1 and 4.4.2). The limits herein protect the past, present and probable future beneficial uses of the water.

8.3. Environmental characteristics of the hydrographic unit under consideration, including the quality of water available thereto: The environmental characteristics are discussed in the Basin Plan, the Region's Watershed Management Initiative Chapter, and are also

available in State of the Watershed reports and the State's CWA section 303(d) List of impaired waters. The environmental characteristics of the hydrographic unit, including the quality of available water, will be improved by compliance with the requirements of this Order. Additional information on the [Santa Clara River watershed](https://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/watershed/) is available at: https://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/watershed/

- 8.4. Water quality conditions that could reasonably be achieved through the coordinated control of all factors which affect water quality in the area: The water quality standards necessary to protect beneficial uses of the waterbodies in the Santa Clara River Watersheds can reasonably be achieved through the coordinate control of all factors that affect water quality in the area. TMDLs have been developed (as required by the CWA) for many of the impairments in the receiving water. A number of Los Angeles Water Board programs and actions are in place to address the water quality impairments in the watershed, including regulation of point source municipal and industrial discharges with appropriate NPDES permits and non-point source discharges such as irrigated agriculture. All of these regulatory programs control the discharge of pollutants to surface and ground waters to prevent nuisance and protect beneficial uses. These regulatory programs have resulted in watershed solutions and have improved water quality. Generally, improvements in the quality of the receiving waters impacted by the permittee's discharges can be achieved by reducing the volume of discharges to receiving waters (e.g., through increased recycling), reducing pollutant loads through source control/pollution prevention, including operational source control such as pretreatment requirements and public education (e.g., disposal of pesticides, pharmaceuticals, and personal care products into the sewer) and product or materials elimination or substitution, and removing pollutants through treatment. The Los Angeles Water Board has taken this factor into account in establishing effluent limitations in the Order.
- 8.5. Economic considerations: The Permittee did not present any evidence regarding economic considerations related to this Order. However, the Los Angeles Water Board has considered the economic impact of requiring certain provisions pursuant to state law, and in conjunction with the applicable TMDLs incorporated into this Order. Any additional costs associated with complying with state law requirements are reasonably necessary to prevent nuisance and protect beneficial uses identified in the Basin Plan. Further, the loss of, or impacts to, beneficial uses would have a detrimental economic impact. Economic considerations related to costs of compliance are therefore not sufficient, in the Los Angeles Water Board's determination, to justify failing to prevent nuisance and protect beneficial uses.
- 8.6. Need for developing housing within the region: The Los Angeles Water Board does not anticipate that the state law requirements in this Order will adversely impact the need for housing in the area. The region generally relies on imported water to meet many of its water resource needs. Imported water makes up a vast majority of the region's water supply, with local groundwater, local surface water, and reclaimed water making up the remaining amount. This Order helps address the need for housing by controlling pollutants in discharges, which will improve the quality of local surface and ground water, as well as water available for recycling and re-use. This in turn may reduce the

demand for imported water thereby increasing the region's capacity to support continued housing development. A reliable water supply for future housing development is required by law, and with less imported water available to guarantee this reliability, an increase in local supply is necessary. Therefore, the potential for developing housing in the area will be facilitated by improved water quality.

8.7. Need to develop and use recycled water: The State Water Board's Recycled Water Policy requires the Los Angeles Water Boards to encourage the use of recycled water. In addition, as discussed immediately above, a need to develop and use recycled water exists within the region, especially during times of drought. To encourage recycling, the Permittee is required by this Order to continue to explore the feasibility of recycling to maximize the beneficial reuse of tertiary treated effluent. The Discharger shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal. The Discharger has recycled 0.42 MGD of their tertiary treated effluent on average in 2020 for non-potable reuse (section 4.7 supra.).

9. PUBLIC PARTICIPATION

The Los Angeles Water Board has considered the issuance of WDRs that will serve as an NPDES permit for Valencia WRP. As a step in the WDR adoption process, the Los Angeles Water Board staff has developed tentative WDRs and has encouraged public participation in the WDR adoption process.

9.1. Notification of Interested Parties

The Los Angeles Water Board notified the Permittee and interested agencies and persons of its intent to prescribe WDRs for the discharge and provided an opportunity to submit written comments and recommendations, and the public notice, the fact sheet, and the draft order were posted on the Los Angeles Water Board's home page at <https://www.waterboards.ca.gov/losangeles/> under the "Tentative Permits" heading. Permittee notification was provided by posting a copy of the notice at the entrance of the Valencia WRP, 28185 The Old Road, Santa Clarita, CA 91355. In addition, interested agencies and persons are notified through a transmittal email to the Discharger, being included in the email transaction, for the Los Angeles Water Board's intention to prescribe WDRs for the discharge.

The public had access to the agenda and any changes in dates and locations through the [Los Angeles Water Board's website](http://www.waterboards.ca.gov/losangeles/) at: <http://www.waterboards.ca.gov/losangeles/>.

9.2. Written Comments

Interested persons were invited to submit written comments concerning tentative amended WDRs as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the Los Angeles Water Board at the address above on the cover page of this Order, or by [email](mailto:losangeles@waterboards.ca.gov) submitted to losangeles@waterboards.ca.gov.

To be fully responded to by staff and considered by the Los Angeles Water Board, the written comments were due at the Los Angeles Water Board office by **5:00 p.m. on April 18, 2022**.

9.3. Public Hearing

The Los Angeles Water Board held a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: May 12, 2022

Time: 9:00 a.m.

Location: Las Virgenes Municipal Water District – Board Room, 4232 Las Virgenes Road #1994, Calabasas, CA 91302

A virtual platform is also available for those who want to join online. Please follow the directions provided in the agenda to register or to view the Board meeting.

Additional information about the location of the hearing and options for participating will be available 10 days before the hearing. Any person desiring to receive future notices about any proposed Board action regarding this Discharger, please contact Veronica Cuevas at veronica.cuevas@waterboards.ca.gov, to be included on the e-mail list.

Interested persons were invited to attend. At the public hearing, the Los Angeles Water Board heard testimony pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested in writing.

9.4. Review of Waste Discharge Requirements

Any person aggrieved by this action of the Los Angeles Water Board may petition the State Water Board to review the action in accordance with Water Code section 13320 and California Code of Regulations, title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., within 30 calendar days of the date of adoption of this Order at the following address, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or state holiday, the petition must be received by the State Water Board by 5:00 p.m. on the next business day:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

Or by [email](mailto:waterqualitypetitions@waterboards.ca.gov) at waterqualitypetitions@waterboards.ca.gov

For instructions on [how to file a petition for review](http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtm),
http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtm
|

Filing a petition does not automatically stay any of the requirements of this Order.

9.5. Information and Copying

The ROWD, other supporting documents, and comments received are on file and may be inspected at the address below by appointment between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Los Angeles Water Board at the address below or by calling (213) 576-6600.

Los Angeles Regional Water Quality Control Board
320 W. 4th Street, Suite 200

Los Angeles, CA 90013-2343

9.6. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Los Angeles Water Board, reference this facility, and provide a name, address, and phone number.

9.7. Additional Information

Requests for additional information or questions regarding this order should be directed to Veronica Cuevas via [email](mailto:Veronica.Cuevas@waterboards.ca.gov) at Veronica.Cuevas@waterboards.ca.gov

ATTACHMENT G. TOXICITY REDUCTION EVALUATION (TRE) WORK PLAN

1. Gather and Review Information and Data
 - 1.1. POTW Operations and Performance
 - 1.2. POTW Influent and Pretreatment Program
 - 1.3. Effluent Data, including Toxicity Results
 - 1.4. Sludge (Biosolids) Data
2. Evaluate Facility Performance
3. Conduct Toxicity Identification Evaluation (TIE)
4. Evaluate Sources and In-Plant Controls
5. Implement Toxicity Control Measures
6. Conduct Confirmatory Toxicity Testing

ATTACHMENT H. PRETREATMENT REPORTING REQUIREMENTS

The Santa Clarita Valley Sanitation District (Permittee) is required to submit annual Pretreatment Program Compliance Report (Report) to the Los Angeles Water Board and United States Environmental Protection Agency, Region 9 (USEPA). This Attachment outlines the minimum reporting requirements of the Report. If there is any conflict between requirements stated in this attachment and provisions stated in the Waste Discharge Requirements (WDR), those contained in the WDR will prevail.

1. PRETREATMENT REQUIREMENTS

- 1.1. The Permittee shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR part 403, including any subsequent regulatory revisions to part 403. Where part 403 or subsequent revision places mandatory actions upon the Permittee as Control Authority but does not specify a timetable for completion of the actions, the Permittee shall complete the required actions within six months from the issuance date of this Order or the effective date of the part 403 revisions, whichever comes later. For violations of pretreatment requirements, the Permittee shall be subject to enforcement actions, penalties, fines and other remedies by the USEPA or other appropriate parties, as provided in the Act. USEPA may initiate enforcement action against a nondomestic user for noncompliance with applicable standards and requirements as provided in the act.
- 1.2. The Permittee shall enforce the requirements promulgated under sections 307(b), 307(c), 307(d) and 402(b) of the Act with timely, appropriate and effective enforcement actions. The Permittee shall cause all nondomestic users subject to federal categorical standards to achieve compliance no later than the date specified in those requirements or, in the case of a new nondomestic user, upon commencement of the discharge.
- 1.3. The Permittee shall perform the pretreatment functions as required in 40 CFR part 403 including, but not limited to:
 - 1.3.1. Implement the necessary legal authorities as provided in 40 CFR section 403.8(f)(1);
 - 1.3.2. Enforce the pretreatment requirements under 40 CFR sections 403.5 and 403.6;
 - 1.3.3. Implement the programmatic functions as provided in 40 CFR section 403.8(f)(2);
and
 - 1.3.4. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR section 403.8(f)(3).
- 1.4. The Permittee shall submit annually a report to USEPA Pacific Southwest Region, and the State describing its pretreatment activities over the previous year. In the event the District is not in compliance with any conditions or requirements of this Order, then the District shall also include the reasons for noncompliance and state how and when the District shall comply with such conditions and requirements. This annual report shall cover operations from January 1 through December 31 and is due on April 30 of each year. The report shall contain, but not be limited to, the following information:
 - 1.4.1. A summary of analytical results from representative, flow proportioned, 24-hour composite sampling of the publicly owned treatment works (POTW) influent and effluent for those pollutants USEPA has identified under section 307(a) of the Act

which are known or suspected to be discharged by nondomestic users. This will consist of an annual full priority pollutant scan, with quarterly samples analyzed only for those pollutants detected in the full scan. The District is not required to sample and analyze for asbestos. Sludge sampling and analysis are covered in the sludge section of this Order. The District shall also provide any influent or effluent monitoring data for nonpriority pollutants which the District believes may be causing or contributing to interference or pass through. Sampling and analysis shall be performed with the techniques prescribed in 40 CFR part 136;

- 1.4.2. A discussion of Upset, Interference or Pass-Through incidents, if any, at the treatment plant which the District knows or suspects were caused by nondomestic users of the POTW system. The discussion shall include the reasons why the incidents occurred, the corrective actions taken and, if known, the name and address of the nondomestic user(s) responsible. The discussion shall also include a review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass through or interference;
- 1.4.3. An updated list of the District's significant industrial users (SIUs) including their names and addresses, and a list of deletions, additions and SIU name changes keyed to the previously submitted list. The District shall provide a brief explanation for each change. The list shall identify the SIUs subject to federal categorical standards by specifying which set(s) of standards are applicable to each SIU. The list shall also indicate which SIUs are subject to local limitations;
- 1.4.4. The District shall characterize the compliance status of each SIU by providing a list or table which includes the following information:
 - a. Name of the SIU;
 - b. Category, if subject to federal categorical standards;
 - c. The type of wastewater treatment or control processes in place;
 - d. The number of samples taken by the POTW during the year;
 - e. The number of samples taken by the SIU during the year;
 - f. For an SIU subject to discharge requirements for total toxic organics, whether all required certifications were provided;
 - g. A list of the standards violated during the year. Identify whether the violations were for categorical standards or local limits;
 - h. Whether the facility is in significant noncompliance (SNC) as defined at 40 CFR section 403.8(f)(2)(viii) at any time during the year; and
 - i. A summary of enforcement or other actions taken during the year to return the SIU to compliance. Describe the type of action, final compliance date, and the amount of fines and penalties collected, if any. Describe any proposed actions for bringing the SIU into compliance.
- 1.4.5. A brief description of any programs the POTW implements to reduce pollutants from nondomestic users that are not classified as SIUs;

- 1.4.6. A brief description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to, changes concerning the program's administrative structure, local limits, monitoring program or monitoring frequencies, legal authority, enforcement policy, funding levels, or staffing levels;
- 1.4.7. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases; and
- 1.4.8. A summary of activities to involve and inform the public of the program including a copy of the newspaper notice, if any, required under 40 CFR section 403.8(f)(2)(viii).

2. LOCAL LIMITS EVALUATION

2.1. In accordance with 40 CFR section 122.44(j)(2)(ii), the POTW shall provide a written technical evaluation of the need to revise local limits under 40 CFR section 403.5(c)(1) within 180 days following the effective date of Order No. R4-2022-0174.

3. SIGNATORY REQUIREMENTS AND REPORT SUBMITTAL

3.1. Signatory Requirements.

The annual report must be signed by a principal executive officer, ranking elected official or other duly authorized employee if such employee is responsible for the overall operation of the POTW. Any person signing these reports must make the following certification [40 CFR section 403.6(a)(2)(ii)]:

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

3.2. Report Submittal.

The Annual Pretreatment Report shall be submitted electronically using the State Water Board's [California Integrated Water Quality System \(CIWQS\) Program website](http://www.waterboards.ca.gov/ciwqs/index.html) (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS website will provide additional information for SMR/DMR submittal in the event there will be a planned service interruption for electronic submittal.

A copy of the Annual Pretreatment Report must be sent to USEPA electronically to the following [address](mailto:R9Pretreatment@epa.gov): R9Pretreatment@epa.gov

ATTACHMENT I. BIOSOLIDS USE AND DISPOSAL REQUIREMENTS

1. All biosolids generated by the Discharger shall be reused or disposed of in compliance with the applicable portions of:
 - 1.1. 40 C.F.R. Part 503: for biosolids that are land applied, placed in surface disposal sites (dedicated land disposal sites or monofills), or incinerated; 40 C.F.R. 503 Subpart B (land application) applies to biosolids placed on the land for the purpose of providing nutrients or conditioning the soil for crops or vegetation. 40 C.F.R. 503 Subpart C (surface disposal) applies to biosolids placed on the land for the purpose of disposal.
 - 1.2. 40 C.F.R. Part 258: for biosolids disposed of in Municipal Solid Waste landfills.
 - 1.3. 40 C.F.R. Part 257: for all biosolids disposal practices not covered under 40 C.F.R. Parts 258 or 503.
2. The Discharger is responsible for assuring that all biosolids from its facility are used or disposed of in accordance with 40 C.F.R. Part 503, whether the Discharger reuses or disposes of the biosolids itself or transfers them to another party for further treatment, reuse, or disposal. The Discharger is responsible for informing subsequent preparers, appliers, or disposers of the requirements they must meet under 40 C.F.R. Part 503.
3. Duty to mitigate: The Discharger shall take all reasonable steps to prevent or minimize any biosolids use or disposal which may adversely impact human health or the environment.
4. No biosolids shall be allowed to enter wetland or other waters of the United States.
5. Biosolids treatment, storage, and use or disposal shall not contaminate groundwater.
6. Biosolids treatment, storage, and use or disposal shall not create a nuisance such as objectionable odors or flies.
7. The Discharger shall assure that haulers who transport biosolids off-site for further treatment, storage, reuse, or disposal take all necessary measures to keep the biosolids contained.
8. If biosolids are stored for over two years from the time they are generated, the Discharger must ensure compliance with all the requirements for surface disposal under 40 C.F.R. Part 503 Subpart C, or must submit a written request to EPA with the information in Part 503.20(b), requesting permission for longer temporary storage.
9. Sewage sludge containing more than 50 mg/kg PCB's shall be disposed of in accordance with 40 C.F.R. Part 761.
10. Any off-site biosolids treatment, storage, use or disposal site operated by the Discharger within Region 4 (Los Angeles Region of RWQCB) that is not subject to its own Waste Discharge Requirements shall have facilities adequate to divert surface runoff from the adjacent area, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials in the disposal site to escape from the site. Adequate protection is defined as protected from at least a 100-year storm and from the highest tidal stage that may occur.

11. Inspection and Entry: The Regional Board, USEPA or an authorized representative thereof, upon the presentation of credentials, shall be allowed by the Discharger, directly or through contractual arrangements with their biosolids management contractors, to:

11.1. Enter upon all premises where biosolids are produced by the Discharger and all premises where Discharger biosolids are further treated, stored, used, or disposed, either by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal;

11.2. Have access to and copy any records that must be kept under the conditions of this permit or of 40 C.F.R. Part 503, by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal; and

11.3. Inspect any facilities, equipment (including monitoring and control equipment), practices, or operations used in the production of biosolids and further treatment, storage, use, or disposal by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal.

12. Monitoring shall be conducted as follows:

12.1. Biosolids shall be tested for the metals required in Part 503.16 (for land application) or Part 503.26 (for surface disposal), using the methods in "Test Methods for Evaluating Solids Waste, Physical/Chemical Methods" (SW-846), as required in Part 503.8(b)(4), at the following minimum frequencies:

<u>Volume (Dry metric tons/year)</u>	<u>Frequency</u>
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0 to 290	once per year
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290 to 1500	once per quarter
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1500 to 15000	once per 60 days
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> 15000	once per month
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For accumulated, previously untested biosolids, the Discharger shall develop a representative sampling plan, which addresses the number and location of sampling points, and collect representative samples.

Test results shall be expressed in mg pollutant per kg biosolids on a 100% dry weight basis.

Biosolids to be land applied shall be tested for Organic-N, ammonium-N, and nitrate-N at the frequencies required above.

12.2. Prior to land application, the Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in Part 503.32. Prior to disposal in a surface disposal site, the Discharger shall demonstrate that the biosolids meet Class B levels or shall ensure that the site is covered at the end of each operating day.

12.3. For biosolids that are land applied or placed in a surface disposal site, the Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction requirements in Part 503.33 (b).

12.4. Class 1 facilities (facilities with pretreatment programs or others designated as Class 1 by the Regional Administrator) and Federal facilities with > 5 mgd influent flow shall sample biosolids for pollutants listed under Section 307(a) of the Clean Water Act (as required in the pretreatment section of the permit for POTWs with pretreatment programs.) Class 1 facilities and Federal facilities with > 5 mgd influent flow shall test dioxins/dibenzofurans using a detection limit of < 1 pg/g during their next sampling period if they have not done so within the past 5 years and once per 5 years thereafter.

12.5. The biosolids shall be tested annually, or more frequently if necessary, to determine hazardousness in accordance with California Law.

12.6. If biosolids are placed in a surface disposal site (dedicated land disposal site or monofill), a qualified groundwater scientist shall develop a groundwater monitoring program for the site, or shall certify that the placement of biosolids on the site will not contaminate an aquifer.

12.7. Biosolids placed in a municipal landfill shall be tested semi-annually by the Paint Filter Test (SW-846, Method 9095) to demonstrate that there are no free liquids.

13. The Discharger either directly or through contractual arrangements with their biosolids management contractors shall comply with the following 40 C.F.R. 503 notification requirements:

13.1. A reuse/disposal plan shall be submitted to EPA Region IX Coordinator and, in the absence of other state or regional reporting requirements, to the state permitting agency, prior to the use or disposal of any biosolids from this facility to a new or previously unreported site. The plan shall be submitted by the land applier of the biosolids and shall include a description and a topographic map of the proposed site(s) for reuse or disposal, names and addresses of the applier(s) and site owner(s), and a list of any state or local permits which must be obtained. For land application sites, the plan shall include a description of the crops or vegetation to be grown, proposed nitrogen loadings to be used for the crops, and a groundwater monitoring plan if one exists.

13.2. If the Discharger biosolids do not meet Part 503.13 Table 3 metals concentration limitations, the Discharger must require their land applier to contact the state permitting authority to determine whether bulk biosolids subject to the cumulative pollutant loading rates in Part 503.12(b)(2) have been applied to the site since July 20, 1993, and, if so, the cumulative amount of pollutants applied to date, and background concentration, if known. The Discharger shall then notify EPA Region IX Coordinator of this information.

13.3. For biosolids that are land applied, the Discharger shall notify the applier in writing of the nitrogen content of the biosolids, and the applier's requirements under Part 503, including the requirements that the applier certify that the requirement to obtain information in Subpart A, and that the management practices, site restrictions, and any applicable vector attraction reduction requirements Subpart D have been met. The Discharger shall require the applier to certify at the end of 38 months following application of Class B biosolids that those harvesting restrictions in effect for up to 38 months have been met.

13.4. If bulk biosolids are shipped to another State or to Indian Lands, the Discharger must send written notice prior to the initial application of bulk biosolids to the permitting

authorities in the receiving State or Indian Land (the EPA Regional Office for the area and the State/Indian authorities).

13.5. Notification of Part 503 non-compliance: The Discharger shall require applicators of their biosolids to notify EPA Region IX and their state permitting agency of any noncompliance within 24 hours if the non-compliance may seriously endanger health or the environment. For other instances of non-compliance, the Discharger shall require applicators of their biosolids to notify EPA Region IX and their state permitting agency of the non-compliance in writing within 10 working days of becoming aware of the non-compliance.

14. The Discharger shall submit an annual biosolids report to EPA Region IX Biosolids Coordinator by February 19 of each year for the period covering the previous calendar year. The report shall include:

14.1. The amount of biosolids generated that year, in dry metric tons, and the amount accumulated from previous years.

14.2. Results of all pollutant monitoring required in the Monitoring Section above.

14.3. Descriptions of pathogen reduction methods, and vector attraction reduction methods, as required in Parts 503.17 and 503.27.

14.4. Results of any groundwater monitoring or certification by groundwater scientist that the placement of biosolids in a surface disposal site will not contaminate an aquifer.

14.5. Names and addresses of land applicators and surface disposal site operators, and volumes applied (dry metric tons).

14.6. Names and addresses of persons who received biosolids for storage, further treatment, disposal in a municipal waste landfill, or for other reuse/disposal methods not covered in 14.c, above, and volumes delivered to each.

15. The Discharger shall require all parties contracted to manage their biosolids to submit an annual biosolids report to EPA Region IX Biosolids Coordinator by February 19 of each year for the period covering the previous calendar year. The report shall include:

15.1. Names and addresses of land applicators and surface disposal site operators, name, location (latitude/longitude), and size (hectares) of site(s), volumes applied/disposed (dry metric tons) and for land application, biosolids loading rates (metric tons per hectare), nitrogen loading rates (kg/ha), dates of applications, crops grown, dates of seeding and harvesting and certifications that the requirement to obtain information in Part 503.12(e)(2), management practices in Part 503.14 and site restrictions in Part 503.32(b)(5) have been met.

ATTACHMENT J. SUMMARY OF REASONABLE POTENTIAL ANALYSIS

Water Quality Criteria Comparison

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
1	Antimony	µg/L	0.6	0.67	6	NONE	NONE	4300	6
2	Arsenic	µg/L	0.3	1.25	10	340	150	NONE	10
3	Beryllium	µg/L	0.6	<0.03	4	NONE	NONE	Narrative	4
4	Cadmium	µg/L	0.6	<i>0.065</i>	5	12.3	4.9	Narrative	4.9
5a	Chromium III*	µg/L	0.6	0.71		3593	428	Narrative	428
5b	Chromium VI	µg/L	0.7	0.04	50	16.3	11.4	Narrative	11.4
6	Copper	µg/L	0.3	3.84		32	20	NONE	20
7	Lead	µg/L	0.4	<i>0.11</i>		253	10	Narrative	10
8	Mercury	ng/L	0.8	0.00586	0.012 note a	Reserved	Reserved	0.051	0.012
9	Nickel*	µg/L	0.2	3.49	100	994	110	4600	100
10	Selenium	µg/L	0.3	0.67	50	RESERVED	5	Narrative	5
11	Silver*	µg/L	0.6	0.01		18.6	none	NONE	18.6
12	Thallium	µg/L	0.6	<0.25	2	NONE	NONE	6.3	2
13	Zinc	µg/L	0.2	32.8		254.2	254	NONE	254
14	Cyanide	µg/L	0.5	6.4	200	22	5.2	220,000	5.2
15	Asbestos	Fibers/L			7x10^6	NONE	NONE	NONE	7x10^6
16	2,3,7,8-TCDD (Dioxin)	pg/L	0.6	<i>2.1</i>	30	NONE	NONE	0.014	0.014
17	Acrolein	µg/L	0.6	<1.3		NONE	NONE	780	780
18	Acrylonitrile	µg/L	0.6	<0.2		NONE	NONE	0.66	0.66

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
19	Benzene	µg/L	0.6	<0.1	1	NONE	NONE	71	1
20	Bromoform	µg/L	0.7	4	--	NONE	NONE	360	360
21	Carbon Tetrachloride	µg/L	0.6	<0.11	0.5	NONE	NONE	4.4	0.5
22	Chlorobenzene	µg/L	0.6	<0.13	--	NONE	NONE	21,000	21,000
23	Dibromochloromethane	µg/L	0.6	15.6	--	NONE	NONE	34	34
24	Chloroethane	µg/L	0.6	ND	--	NONE	NONE	NONE	NONE
25	2-chloroethyl vinyl ether	µg/L	0.6	ND	--	NONE	NONE	NONE	NONE
26	Chloroform	µg/L	0.3	23.2	--	NONE	NONE	Reserved	Reserved
27	Dichlorobromomethane	µg/L	0.5	25.8	--	NONE	NONE	46	46
28	1,1-Dichloroethane	µg/L	0.6	ND	5	NONE	NONE	NONE	5
29	1,2-dichloroethane	µg/L	0.6	ND	0.5	NONE	NONE	99	0.5
30	1,1-Dichloroethylene	µg/L	0.6	ND	6	NONE	NONE	3.2	3.2
31	1,2-dichloropropane	µg/L	0.6	<0.5	5	NONE	NONE	39	5
32	1,3-dichloropropylene	µg/L	0.6	<0.5	0.5	NONE	NONE	1,700	0.5
33	Ethylbenzene	µg/L	0.6	<0.5	300	NONE	NONE	29,000	300
34	Methyl bromide	µg/L	0.6	<0.5	--	NONE	NONE	4,000	4,000
35	Methyl chloride	µg/L	0.6	ND	--	NONE	NONE	Narrative	Narrative
36	Methylene chloride	µg/L	0.6	ND	--	NONE	NONE	1,600	1,600
37	1,1,2,2-tetrachloroethane	µg/L	0.6	<0.5	1	NONE	NONE	11	1
38	Tetrachloroethylene	µg/L	0.6	<0.5	5	NONE	NONE	8.85	5
39	Toluene	µg/L	0.6	0.34	150	NONE	NONE	200,000	150
40	Trans 1,2-Dichloroethylene	µg/L	0.6	<0.6	10	NONE	NONE	140,000	10

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
41	1,1,1-Trichloroethane	µg/L	0.6	<0.16	200	NONE	NONE	Narrative	200
42	1,1,2-trichloroethane	µg/L	0.6	<0.09	5	NONE	NONE	42	5
43	Trichloroethylene	µg/L	0.6	<0.15	5	NONE	NONE	81	5
44	Vinyl chloride	µg/L	0.6	<0.25	0.5	NONE	NONE	525	0.5
45	2-chlorophenol	µg/L	0.6	<0.18	--	NONE	NONE	400	400
46	2,4-dichlorophenol	µg/L	0.6	<0.6	--	NONE	NONE	790	790
47	2,4-dimethylphenol	µg/L	0.6	<0.3	--	NONE	NONE	2,300	2,300
48	4,6-dinitro-o-resol (aka 2-methyl-4,6-Dinitrophenol)	µg/L	0.6	<0.13	--	NONE	NONE	765	765
49	2,4-dinitrophenol	µg/L	0.6	<1.5	--	NONE	NONE	14,000	14,000
50	2-nitrophenol	µg/L	0.6	<0.1	--	NONE	NONE	NONE	None
51	4-nitrophenol	µg/L	0.6	<1.2	--	NONE	NONE	NONE	None
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-cresol)	µg/L	0.6	<0.23	--	NONE	NONE	NONE	None
53	Pentachlorophenol	µg/L	0.6	<0.4	1	pH dependent	pH dependent	8.2	1
54	Phenol	µg/L	0.5	0.26	--	NONE	NONE	4,600,000	4.6x10^6
55	2,4,6-trichlorophenol	µg/L	0.6	0.24	--	NONE	NONE	6.5	6.5
56	Acenaphthene	µg/L	0.6	<0.22	--	NONE	NONE	2,700	2,700
57	Acenaphthylene	µg/L	0.6	<0.17	--	NONE	NONE	NONE	NONE
58	Anthracene	µg/L	0.6	<0.12	--	NONE	NONE	110,000	110,000
59	Benzidine	µg/L	0.6	<0.77	--	NONE	NONE	0.00054	0.00054
60	Benzo(a)Anthracene	µg/L	0.6	<0.14	--	NONE	NONE	0.049	0.049

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
61	Benzo(a)Pyrene	µg/L	0.6	<0.007	--	NONE	NONE	0.049	0.049
62	Benzo(b)Fluoranthene	µg/L	0.6	<0.004	--	NONE	NONE	0.049	0.049
63	Benzo(ghi)Perylene	µg/L	0.6	<0.12	--	NONE	NONE	NONE	NONE
64	Benzo(k)Fluoranthene	µg/L	0.6	<0.005	--	NONE	NONE	0.049	0.049
65	Bis(2-Chloroethoxy) methane	µg/L	0.6	<0.11	--	NONE	NONE	NONE	NONE
66	Bis(2-Chloroethyl)Ether	µg/L	0.6	<0.2	--	NONE	NONE	1.4	1.4
67	Bis(2-Chloroisopropyl) Ether	µg/L	0.6	<0.2	--	NONE	NONE	170,000	170,000
68	Bis(2-Ethylhexyl) Phthalate	µg/L	0.8	2	4	NONE	NONE	5.9	4
69	4-Bromophenyl Phenyl Ether	µg/L	0.6	<0.27	--	NONE	NONE	NONE	NONE
70	Butylbenzyl Phthalate	µg/L	0.6	0.52	--	NONE	NONE	5,200	5,200
71	2-Chloronaphthalene	µg/L	0.6	<0.13	--	NONE	NONE	4,300	4,300
72	4-Chlorophenyl Phenyl Ether	µg/L	0.6	<0.32	--	NONE	NONE	NONE	NONE
73	Chrysene	µg/L	0.6	<0.005	--	NONE	NONE	0.049	0.049
74	Dibenzo(a,h)Anthracene	µg/L	0.6	<0.004	--	NONE	NONE	0.049	0.049
75	1,2-Dichlorobenzene	µg/L	0.6	<0.07	600	NONE	NONE	17,000	600
76	1,3-Dichlorobenzene	µg/L	0.6	<0.08	--	NONE	NONE	2,600	2,600
77	1,4-Dichlorobenzene	µg/L	0.6	<0.16	5	NONE	NONE	2,600	5
78	3,3'-Dichlorobenzidine	µg/L	0.6	<0.54	--	NONE	NONE	0.077	0.077
79	Diethyl Phthalate	µg/L	0.6	1.3	--	NONE	NONE	120,000	120,000
80	Dimethyl Phthalate	µg/L	0.6	<0.28	--	NONE	NONE	2,900,000	2.9x10^6

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
81	Di-n-Butyl Phthalate	µg/L	0.6	0.43	--	NONE	NONE	12,000	12,000
82	2,4-Dinitrotoluene	µg/L	0.6	<0.18	--	NONE	NONE	9.1	9.1
83	2,6-Dinitrotoluene	µg/L	0.6	<0.27	--	NONE	NONE	NONE	NONE
84	Di-n-Octyl Phthalate	µg/L	0.6	<0.11	--	NONE	NONE	NONE	NONE
85	1,2-Diphenylhydrazine	µg/L	0.6	<0.2	--	NONE	NONE	0.54	0.54
86	Fluoranthene	µg/L	0.6	<0.08	--	NONE	NONE	370	370
87	Fluorene	µg/L	0.6	<0.35	--	NONE	NONE	14,000	14,000
88	Hexachlorobenzene	µg/L	0.6	<0.17	--	NONE	NONE	0.00077	0.00077
89	Hexachlorobutadiene	µg/L	0.6	<0.33	--	NONE	NONE	50	50
90	Hexachlorocyclopentadiene	µg/L	0.6	<0.53	--	NONE	NONE	17,000	17,000
91	Hexachloroethane	µg/L	0.6	<0.13	--	NONE	NONE	8.9	8.9
92	Indeno(1,2,3-cd)Pyrene	µg/L	0.6	<0.13	--	NONE	NONE	0.049	0.049
93	Isophorone	µg/L	0.6	<0.11	--	NONE	NONE	600	600
94	Napthalene	µg/L	0.6	<0.13	--	NONE	NONE	NONE	NONE
95	Nitrobenzene	µg/L	0.6	<0.17	--	NONE	NONE	1,900	1,900
96	N-Nitrosodimethylamine	µg/L	0.8	2.3	--	NONE	NONE	8.1	8.1
97	N-Nitrosodi-n-Propylamine	µg/L	0.6	<0.0006	--	NONE	NONE	1.4	1.4
98	N-Nitrosodiphenylamine	µg/L	0.6	<0.0013	--	NONE	NONE	16	16
99	Phenanthrene	µg/L	0.6	<0.31	--	NONE	NONE	NONE	NONE
100	Pyrene	µg/L	0.6	<0.25	--	NONE	NONE	11,000	11,000
101	1,2,4-Trichlorobenzene	µg/L	0.6	<0.19	--	NONE	NONE	NONE	NONE
102	Aldrin	µg/L	0.6	<0.002	--	3	NONE	0.00014	0.00014

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
103	alpha-BHC	µg/L	0.6	<0.0005	--	NONE	NONE	0.013	0.013
104	beta-BHC	µg/L	0.6	<0.002	--	NONE	NONE	0.046	0.046
105	gamma-BHC (aka Lindane)	µg/L	0.4	0.007	0.2	0.95	NONE	0.063	0.063
106	delta-BHC	µg/L	0.6	<0.001	--	NONE	NONE	NONE	NONE
107	Chlordane	µg/L	0.6	<0.01	--	2.4	0.0043	0.00059	0.00059
108	4,4'-DDT	µg/L	0.6	<0.001	--	1.1	0.001	0.00059	0.00059
109	4,4'-DDE	µg/L	0.6	<0.001	--	NONE	NONE	0.00059	0.00059
110	4,4'-DDD	µg/L	0.6	<0.001	--	NONE	NONE	0.00084	0.00083
111	Dieldrin	µg/L	0.6	<0.001	--	0.24	0.056	0.00014	0.00014
112	alpha-Endosulfan	µg/L	0.6	<0.001	--	0.22	0.056	240	0.056
113	beta-Endosulfan	µg/L	0.6	<0.001	--	0.22	0.056	240	0.056
114	Endosulfan Sulfate	µg/L	0.6	<0.002	--	NONE	NONE	240	240
115	Endrin	µg/L	0.6	<0.001	--	0.086	0.036	0.81	0.036
116	Endrin Aldehyde	µg/L	0.6	<0.001	--	NONE	NONE	0.81	0.81
117	Heptachlor	µg/L	0.6	<0.0008	--	0.52	0.0038	0.00021	0.00021
118	Heptachlor Epoxide	µg/L	0.6	<0.001	--	0.52	0.0038	0.00011	0.00011
119	PCB Aroclor 1016	µg/L	0.6	<0.02	--	NONE	0.014	0.00017	0.00017
120	PCB Aroclor 1221	µg/L	0.6	<0.08	--	NONE	0.014	0.00017	0.00017
121	PCB Aroclor 1232	µg/L	0.6	<0.08	--	NONE	0.014	0.00017	0.00017
122	PCB Aroclor 1242	µg/L	0.6	<0.04	--	NONE	0.014	0.00017	0.00017
123	PCB Aroclor 1248	µg/L	0.6	<0.02	--	NONE	0.014	0.00017	0.00017
124	PCB Aroclor 1254	µg/L	0.6	<0.01	--	NONE	0.014	0.00017	0.00017

CTR #	Parameters	Units	CV	MEC	Title 22 MCLs	Freshwater C acute = CMC tot	Freshwater C chronic = CCC tot	Human health for consumption of Organisms only	Lowest C
125	PCB Aroclor 1260	µg/L	0.6	<0.01	--	NONE	0.014	0.00017	0.00017
126	Toxaphene	µg/L	0.6	<0.04	3	0.73	0.0002	0.00075	0.00075
	Iron	µg/L	0.2	140	300	--	--	--	300
	Trihalomethanes	µg/L	0.4	61.5	80	--	--	--	80

Notes:

MEC = Maximum Effluent Concentration

C = Criteria

B = Background Receiving Water Concentration

a = State Water Resources Control Board Mercury Provisions

* = Hardness-dependent Criteria with Hardness = 176 mg/L

Reasonable Potential Tiers

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
1	Antimony	µg/L	NO	0.34	NO	NO	NO	MEC<C
2	Arsenic	µg/L	NO	6.07	NO	NO	NO	MEC<C
3	Beryllium	µg/L	NO	<0.02	NO	NO	NO	MEC<C
4	Cadmium	µg/L	NO	0.14	NO	NO	NO	MEC<C
5a	Chromium III*	µg/L	NO	0.61	NO	NO	NO	MEC<C
5b	Chromium VI	µg/L	NO	0.13	NO	NO	NO	MEC<C
6	Copper	µg/L	NO	9.43	NO	NO	NO	MEC<C

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
7	Lead	µg/L	NO	0.49	NO	NO	NO	MEC<C
8	Mercury	ng/L	NO	0.37	NO	NO	NO	MEC<C
9	Nickel*	µg/L	NO	5.21	NO	NO	NO	MEC<C
10	Selenium	µg/L	NO	22	YES	NO	Yes	Limit required, B>C & pollutant detected in effluent
11	Silver*	µg/L	NO	0.01	NO	NO	NO	MEC<C
12	Thallium	µg/L	NO	<0.01	NO	NO	NO	MEC<C
13	Zinc	µg/L	NO	6.09	NO	NO	NO	MEC<C
14	Cyanide	µg/L	YES	3.8	--	NO	Yes	Limit required, MEC>C
15	Asbestos	Fibers/L	NO	NA	NO	NO	NO	MEC<C
16	2,3,7,8-TCDD (Dioxin)	pg/L	Yes	0.98		NO	Yes	Limit required, MEC>C
17	Acrolein	µg/L	NO	<0.7	NO	NO	NO	MEC<C
18	Acrylonitrile	µg/L	NO	<0.2	NO	NO	NO	MEC<C
19	Benzene	µg/L	NO	<0.09	NO	NO	NO	MEC<C
20	Bromoform	µg/L	NO	0.15	NO	NO	NO	MEC<C
21	Carbon Tetrachloride	µg/L	NO	<0.11	NO	NO	NO	MEC<C
22	Chlorobenzene	µg/L	NO	<0.13	NO	NO	NO	MEC<C
23	Dibromochloromethane	µg/L	NO	ND	NO	NO	NO	MEC<C
24	Chloroethane	µg/L	Uc	ND	NO	NO	NO	No criteria
25	2-chloroethyl vinyl ether	µg/L	Uc	ND	NO	NO	NO	No criteria
26	Chloroform	µg/L	Uc	0.17	NO	NO	NO	No criteria
27	Dichlorobromomethane	µg/L	NO	ND	NO	NO	NO	MEC<C

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
28	1,1-Dichloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
29	1,2-dichloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
30	1,1-Dichloroethylene	µg/L	NO	ND	NO	NO	NO	MEC<C
31	1,2-dichloropropane	µg/L	NO	ND	NO	NO	NO	MEC<C
32	1,3-dichloropropylene	µg/L	NO	ND	NO	NO	NO	MEC<C
33	Ethylbenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
34	Methyl bromide	µg/L	NO	ND	NO	NO	NO	MEC<C
35	Methyl chloride	µg/L	Uc	ND	NO	NO	NO	No criteria
36	Methylene chloride	µg/L	NO	ND	NO	NO	NO	MEC<C
37	1,1,2,2-tetrachloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
38	Tetrachloroethylene	µg/L	NO	ND	NO	NO	NO	MEC<C
39	Toluene	µg/L	NO	ND	NO	NO	NO	MEC<C
40	Trans 1,2-Dichloroethylene	µg/L	NO	ND	NO	NO	NO	MEC<C
41	1,1,1-Trichloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
42	1,1,2-trichloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
43	Trichloroethylene	µg/L	NO	ND	NO	NO	NO	MEC<C
44	Vinyl chloride	µg/L	NO	ND	NO	NO	NO	MEC<C
45	2-chlorophenol	µg/L	NO	ND	NO	NO	NO	MEC<C
46	2,4-dichlorophenol	µg/L	NO	ND	NO	NO	NO	MEC<C
47	2,4-dimethylphenol	µg/L	NO	ND	NO	NO	NO	MEC<C
48	4,6-dinitro-o-resol (aka2-methyl-4,6-Dinitrophenol)	µg/L	NO	ND	NO	NO	NO	MEC<C
49	2,4-dinitrophenol	µg/L	NO	ND	NO	NO	NO	MEC<C
50	2-nitrophenol	µg/L	Uc	ND	NO	NO	NO	No criteria
51	4-nitrophenol	µg/L	Uc	ND	NO	NO	NO	No criteria

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-cresol)	µg/L	Uc	ND	NO	NO	NO	No criteria
53	Pentachlorophenol	µg/L	NO	ND	NO	NO	NO	MEC<C
54	Phenol	µg/L	NO	ND	NO	NO	NO	MEC<C
55	2,4,6-trichlorophenol	µg/L	NO	ND	NO	NO	NO	MEC<C
56	Acenaphthene	µg/L	NO	ND	NO	NO	NO	MEC<C
57	Acenaphthylene	µg/L	Uc	ND	NO	NO	NO	No criteria
58	Anthracene	µg/L	NO	ND	NO	NO	NO	MEC<C
59	Benzidine	µg/L	NO	ND	NO	NO	NO	MEC<C
60	Benzo(a)Anthracene	µg/L	NO	ND	NO	NO	NO	MEC<C
61	Benzo(a)Pyrene	µg/L	NO	ND	NO	NO	NO	MEC<C
62	Benzo(b)Fluoranthene	µg/L	NO	ND	NO	NO	NO	MEC<C
63	Benzo(ghi)Perylene	µg/L	Uc	ND	NO	NO	NO	No criteria
64	Benzo(k)Fluoranthene	µg/L	NO	0.008	NO	NO	NO	MEC<C
65	Bis(2-Chloroethoxy) methane	µg/L	Uc	ND	NO	NO	NO	No criteria
66	Bis(2-Chloroethyl)Ether	µg/L	NO	ND	NO	NO	NO	MEC<C
67	Bis(2-Chloroisopropyl) Ether	µg/L	NO	ND	NO	NO	NO	MEC<C
68	Bis(2-Ethylhexyl) Phthalate	µg/L	NO	0.17	NO	NO	NO	MEC<C
69	4-Bromophenyl Phenyl Ether	µg/L	Uc	ND	NO	NO	NO	No criteria
70	Butylbenzyl Phthalate	µg/L	NO	ND	NO	NO	NO	MEC<C
71	2-Chloronaphthalene	µg/L	NO	ND	NO	NO	NO	MEC<C
72	4-Chlorophenyl Phenyl Ether	µg/L	Uc	ND	NO	NO	NO	No criteria
73	Chrysene	µg/L	NO	0.006	NO	NO	NO	MEC<C
74	Dibenzo(a,h)Anthracene	µg/L	NO	0.005	NO	NO	NO	MEC<C

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
75	1,2-Dichlorobenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
76	1,3-Dichlorobenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
77	1,4-Dichlorobenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
78	3,3'-Dichlorobenzidine	µg/L	NO	ND	NO	NO	NO	MEC<C
79	Diethyl Phthalate	µg/L	NO	ND	NO	NO	NO	MEC<C
80	Dimethyl Phthalate	µg/L	NO	ND	NO	NO	NO	MEC<C
81	Di-n-Butyl Phthalate	µg/L	NO	ND	NO	NO	NO	MEC<C
82	2,4-Dinitrotoluene	µg/L	NO	ND	NO	NO	NO	MEC<C
83	2,6-Dinitrotoluene	µg/L	Uc	ND	NO	NO	NO	No criteria
84	Di-n-Octyl Phthalate	µg/L	Uc	ND	NO	NO	NO	No criteria
85	1,2-Diphenylhydrazine	µg/L	NO	ND	NO	NO	NO	MEC<C
86	Fluoranthene	µg/L	NO	ND	NO	NO	NO	MEC<C
87	Fluorene	µg/L	NO	ND	NO	NO	NO	MEC<C
88	Hexachlorobenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
89	Hexachlorobutadiene	µg/L	NO	ND	NO	NO	NO	MEC<C
90	Hexachlorocyclopentadiene	µg/L	NO	ND	NO	NO	NO	MEC<C
91	Hexachloroethane	µg/L	NO	ND	NO	NO	NO	MEC<C
92	Indeno(1,2,3-cd)Pyrene	µg/L	NO	ND	NO	NO	NO	MEC<C
93	Isophorone	µg/L	NO	ND	NO	NO	NO	MEC<C
94	Napthalene	µg/L	Uc	ND	NO	NO	NO	No criteria
95	Nitrobenzene	µg/L	NO	ND	NO	NO	NO	MEC<C
96	N-Nitrosodimethylamine	µg/L	NO	ND	NO	NO	NO	MEC<C
97	N-Nitrosodi-n-Propylamine	µg/L	NO	ND	NO	NO	NO	MEC<C
98	N-Nitrosodiphenylamine	µg/L	NO	ND	NO	NO	NO	MEC<C
99	Phenanthrene	µg/L	Uc	ND	NO	NO	NO	No criteria

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
100	Pyrene	µg/L	NO	ND	NO	NO	NO	MEC<C
101	1,2,4-Trichlorobenzene	µg/L	Uc	ND	NO	NO	NO	No criteria
102	Aldrin	µg/L	NO	0.007	NO	NO	NO	MEC<C
103	alpha-BHC	µg/L	NO	ND	NO	NO	NO	MEC<C
104	beta-BHC	µg/L	NO	ND	NO	NO	NO	MEC<C
105	gamma-BHC (aka Lindane)	µg/L	NO	ND	NO	NO	NO	MEC<C
106	delta-BHC	µg/L	Uc	ND	NO	NO	NO	No criteria
107	Chlordane	µg/L	NO	ND	NO	NO	NO	MEC<C
108	4,4'-DDT	µg/L	NO	ND	NO	NO	NO	MEC<C
109	4,4'-DDE	µg/L	NO	ND	NO	NO	NO	MEC<C
110	4,4'-DDD	µg/L	NO	ND	NO	NO	NO	MEC<C
111	Dieldrin	µg/L	NO	ND	NO	NO	NO	MEC<C
112	alpha-Endosulfan	µg/L	NO	ND	NO	NO	NO	MEC<C
113	beta-Endosulfan	µg/L	NO	ND	NO	NO	NO	MEC<C
114	Endosulfan Sulfate	µg/L	NO	ND	NO	NO	NO	MEC<C
115	Endrin	µg/L	NO	ND	NO	NO	NO	MEC<C
116	Endrin Aldehyde	µg/L	NO	ND	NO	NO	NO	MEC<C
117	Heptachlor	µg/L	NO	ND	NO	NO	NO	MEC<C
118	Heptachlor Epoxide	µg/L	NO	ND	NO	NO	NO	MEC<C
119	PCB Aroclor 1016	µg/L	NO	ND	NO	NO	NO	MEC<C
120	PCB Aroclor 1221	µg/L	NO	ND	NO	NO	NO	MEC<C
121	PCB Aroclor 1232	µg/L	NO	ND	NO	NO	NO	MEC<C
122	PCB Aroclor 1242	µg/L	NO	ND	NO	NO	NO	MEC<C
123	PCB Aroclor 1248	µg/L	NO	ND	NO	NO	NO	MEC<C
124	PCB Aroclor 1254	µg/L	NO	ND	NO	NO	NO	MEC<C

CTR #	Parameters	Units	Tier 1 MEC > = Lowest C	B	Tier 2 B>C & present in Effluent	Tier 3 other info	RPA result	Reason
125	PCB Aroclor 1260	µg/L	NO	ND	NO	NO	NO	MEC<C
126	Toxaphene	µg/L	NO	ND	NO	NO	NO	MEC<C
	Iron	µg/L	No	7120	YES	NO	Yes	Limit required, B>C & pollutant detected in effluent
	Trihalomethanes	µg/L	Yes	0.17	--	NO	Yes	Limit required, TSD RPA

Notes:

MEC = Maximum Effluent Concentration

C = Criteria

B = Background Receiving Water Concentration

Ud = Undetermined due to lack of data

Uc = Undetermined due to lack of criteria

RPA = Reasonable Potential Analysis

TSD = USEPA Technical Support Document

Aquatic Life Limit Calculations

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
1	Antimony	--	--	--	--	--	--	--	--	--
2	Arsenic	--	--	--	--	--	--	--	--	--
3	Beryllium	--	--	--	--	--	--	--	--	--
4	Cadmium	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
5a	Chromium III*	--	--	--	--	--	--	--	--	--
5b	Chromium VI	--	--	--	--	--	--	--	--	--
6	Copper	--	--	--	--	--	--	--	--	--
7	Lead	--	--	--	--	--	--	--	--	--
8	Mercury	--	--	--	--	--	--	--	--	--
9	Nickel*	--	--	--	--	--	--	--	--	--
10	Selenium	0.527	NA	0.715	3.575	3.575	1.264	4.5188	1.896	6.7782
11	Silver*	--	--	--	--	--	--	--	--	--
12	Thallium	--	--	--	--	--	--	--	--	--
13	Zinc	--	--	--	--	--	--	--	--	--
14	Cyanide	0.373	8.206	0.581	3.0212	3.0212	1.455	4.39585	2.684	8.1089
15	Asbestos	--	--	--	--	--	--	--	--	--
16	2,3,7,8-TCDD (Dioxin)	--	--	--	--	--	--	--	--	--
17	Acrolein	--	--	--	--	--	--	--	--	--
18	Acrylonitrile	--	--	--	--	--	--	--	--	--
19	Benzene	--	--	--	--	--	--	--	--	--
20	Bromoform	--	--	--	--	--	--	--	--	--
21	Carbon Tetrachloride	--	--	--	--	--	--	--	--	--
22	Chlorobenzene	--	--	--	--	--	--	--	--	--
23	Dibromochloromethane	--	--	--	--	--	--	--	--	--
24	Chloroethane	--	--	--	--	--	--	--	--	--
25	2-chloroethyl vinyl ether	--	--	--	--	--	--	--	--	--
26	Chloroform	--	--	--	--	--	--	--	--	--
27	Dichlorobromomethane	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
28	1,1-Dichloroethane	--	--	--	--	--	--	--	--	--
29	1,2-dichloroethane	--	--	--	--	--	--	--	--	--
30	1,1-Dichloroethylene	--	--	--	--	--	--	--	--	--
31	1,2-dichloropropane	--	--	--	--	--	--	--	--	--
32	1,3-dichloropropylene	--	--	--	--	--	--	--	--	--
33	Ethylbenzene	--	--	--	--	--	--	--	--	--
34	Methyl bromide	--	--	--	--	--	--	--	--	--
35	Methyl chloride	--	--	--	--	--	--	--	--	--
36	Methylene chloride	--	--	--	--	--	--	--	--	--
37	1,1,2,2-tetrachloroethane	--	--	--	--	--	--	--	--	--
38	Tetrachloroethylene	--	--	--	--	--	--	--	--	--
39	Toluene	--	--	--	--	--	--	--	--	--
40	Trans 1,2-Dichloroethylene	--	--	--	--	--	--	--	--	--
41	1,1,1-Trichloroethane	--	--	--	--	--	--	--	--	--
42	1,1,2-trichloroethane	--	--	--	--	--	--	--	--	--
43	Trichloroethylene	--	--	--	--	--	--	--	--	--
44	Vinyl chloride	--	--	--	--	--	--	--	--	--
45	2-chlorophenol	--	--	--	--	--	--	--	--	--
46	2,4-dichlorophenol	--	--	--	--	--	--	--	--	--
47	2,4-dimethylphenol	--	--	--	--	--	--	--	--	--
48	4,6-dinitro-o-resol (aka2-methyl-4,6-Dinitrophenol)	--	--	--	--	--	--	--	--	--
49	2,4-dinitrophenol	--	--	--	--	--	--	--	--	--
50	2-nitrophenol	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
51	4-nitrophenol	--	--	--	--	--	--	--	--	--
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-cresol)	--	--	--	--	--	--	--	--	--
53	Pentachlorophenol	--	--	--	--	--	--	--	--	--
54	Phenol	--	--	--	--	--	--	--	--	--
55	2,4,6-trichlorophenol	--	--	--	--	--	--	--	--	--
56	Acenaphthene	--	--	--	--	--	--	--	--	--
57	Acenaphthylene	--	--	--	--	--	--	--	--	--
58	Anthracene	--	--	--	--	--	--	--	--	--
59	Benzidine	--	--	--	--	--	--	--	--	--
60	Benzo(a)Anthracene	--	--	--	--	--	--	--	--	--
61	Benzo(a)Pyrene	--	--	--	--	--	--	--	--	--
62	Benzo(b)Fluoranthene	--	--	--	--	--	--	--	--	--
63	Benzo(ghi)Perylene	--	--	--	--	--	--	--	--	--
64	Benzo(k)Fluoranthene	--	--	--	--	--	--	--	--	--
65	Bis(2-Chloroethoxy) methane	--	--	--	--	--	--	--	--	--
66	Bis(2-Chloroethyl)Ether	--	--	--	--	--	--	--	--	--
67	Bis(2-Chloroisopropyl) Ether	--	--	--	--	--	--	--	--	--
68	Bis(2-Ethylhexyl) Phthalate	--	--	--	--	--	--	--	--	--
69	4-Bromophenyl Phenyl Ether	--	--	--	--	--	--	--	--	--
70	Butylbenzyl Phthalate	--	--	--	--	--	--	--	--	--
71	2-Chloronaphthalene	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
72	4-Chlorophenyl Phenyl Ether	--	--	--	--	--	--	--	--	--
73	Chrysene	--	--	--	--	--	--	--	--	--
74	Dibenzo(a,h)Anthracene	--	--	--	--	--	--	--	--	--
75	1,2-Dichlorobenzene	--	--	--	--	--	--	--	--	--
76	1,3-Dichlorobenzene	--	--	--	--	--	--	--	--	--
77	1,4-Dichlorobenzene	--	--	--	--	--	--	--	--	--
78	3,3'-Dichlorobenzidine	--	--	--	--	--	--	--	--	--
79	Diethyl Phthalate	--	--	--	--	--	--	--	--	--
80	Dimethyl Phthalate	--	--	--	--	--	--	--	--	--
81	Di-n-Butyl Phthalate	--	--	--	--	--	--	--	--	--
82	2,4-Dinitrotoluene	--	--	--	--	--	--	--	--	--
83	2,6-Dinitrotoluene	--	--	--	--	--	--	--	--	--
84	Di-n-Octyl Phthalate	--	--	--	--	--	--	--	--	--
85	1,2-Diphenylhydrazine	--	--	--	--	--	--	--	--	--
86	Fluoranthene	--	--	--	--	--	--	--	--	--
87	Fluorene	--	--	--	--	--	--	--	--	--
88	Hexachlorobenzene	--	--	--	--	--	--	--	--	--
89	Hexachlorobutadiene	--	--	--	--	--	--	--	--	--
90	Hexachlorocyclopentadiene	--	--	--	--	--	--	--	--	--
91	Hexachloroethane	--	--	--	--	--	--	--	--	--
92	Indeno(1,2,3-cd)Pyrene	--	--	--	--	--	--	--	--	--
93	Isophorone	--	--	--	--	--	--	--	--	--
94	Napthalene	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
95	Nitrobenzene	--	--	--	--	--	--	--	--	--
96	N-Nitrosodimethylamine	--	--	--	--	--	--	--	--	--
97	N-Nitrosodi-n-Propylamine	--	--	--	--	--	--	--	--	--
98	N-Nitrosodiphenylamine	--	--	--	--	--	--	--	--	--
99	Phenanthrene	--	--	--	--	--	--	--	--	--
100	Pyrene	--	--	--	--	--	--	--	--	--
101	1,2,4-Trichlorobenzene	--	--	--	--	--	--	--	--	--
102	Aldrin	--	--	--	--	--	--	--	--	--
103	alpha-BHC	--	--	--	--	--	--	--	--	--
104	beta-BHC	--	--	--	--	--	--	--	--	--
105	gamma-BHC (aka Lindane)	--	--	--	--	--	--	--	--	--
106	delta-BHC	--	--	--	--	--	--	--	--	--
107	Chlordane	--	--	--	--	--	--	--	--	--
108	4,4'-DDT	--	--	--	--	--	--	--	--	--
109	4,4'-DDE	--	--	--	--	--	--	--	--	--
110	4,4'-DDD	--	--	--	--	--	--	--	--	--
111	Dieldrin	--	--	--	--	--	--	--	--	--
112	alpha-Endosulfan	--	--	--	--	--	--	--	--	--
113	beta-Endosulfan	--	--	--	--	--	--	--	--	--
114	Endosulfan Sulfate	--	--	--	--	--	--	--	--	--
115	Endrin	--	--	--	--	--	--	--	--	--
116	Endrin Aldehyde	--	--	--	--	--	--	--	--	--
117	Heptachlor	--	--	--	--	--	--	--	--	--
118	Heptachlor Epoxide	--	--	--	--	--	--	--	--	--

CTR #	Parameters	ECA acute multiplier (p.7)	LTA acute	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier (n=4)	AMEL aq.life	MDEL multiplier (n=4)	MDEL aqlife
119	PCB Aroclor 1016	--	--	--	--	--	--	--	--	--
120	PCB Aroclor 1221	--	--	--	--	--	--	--	--	--
121	PCB Aroclor 1232	--	--	--	--	--	--	--	--	--
122	PCB Aroclor 1242	--	--	--	--	--	--	--	--	--
123	PCB Aroclor 1248	--	--	--	--	--	--	--	--	--
124	PCB Aroclor 1254	--	--	--	--	--	--	--	--	--
125	PCB Aroclor 1260	--	--	--	--	--	--	--	--	--
126	Toxaphene	--	--	--	--	--	--	--	--	--
	Iron	--	--	--	--	--	--	--	--	--
	Trihalomethanes	--	--	--	--	--	--	--	--	--

Notes:

ECA =

LAT = Long Term Average

AMEL = Average Monthly Effluent Limitation

MDEL = maximum Daily Effluent Limitation

Human Health Limit Calculations and Final Limitation Determination

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
1	Antimony	--	--	--	--	--	No reasonable potential
2	Arsenic	--	--	--	--	--	No reasonable potential
3	Beryllium	--	--	--	--	--	No reasonable potential
4	Cadmium	--	--	--	--	--	No reasonable potential
5a	Chromium III*	--	--	--	--	--	No reasonable potential
5b	Chromium VI	--	--	--	--	--	No reasonable potential
6	Copper	--	--	--	--	--	No reasonable potential
7	Lead	--	--	--	--	--	No reasonable potential
8	Mercury	--	--	--	--	--	No reasonable potential
9	Nickel*	--	--	--	--	--	No reasonable potential
10	Selenium	--	--	--	4.5	6.8	Reasonable potential
11	Silver*	--	--	--	--	--	No reasonable potential
12	Thallium	--	--	--	--	--	No reasonable potential
13	Zinc	--	--	--	--	--	No reasonable potential
14	Cyanide	220000	1.845	405900	4.4	8.1	Reasonable potential
15	Asbestos	--	--	--	--	--	No reasonable potential
16	2,3,7,8-TCDD (Dioxin)	0.014	2.01	0.028	0.014	0.028	Reasonable potential
17	Acrolein	--	--	--	--	--	No reasonable potential
18	Acrylonitrile	--	--	--	--	--	No reasonable potential
19	Benzene	--	--	--	--	--	No reasonable potential
20	Bromoform	--	--	--	--	--	No reasonable potential
21	Carbon Tetrachloride	--	--	--	--	--	No reasonable potential
22	Chlorobenzene	--	--	--	--	--	No reasonable potential

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
23	Dibromochloromethane	--	--	--	--	--	No reasonable potential
24	Chloroethane	--	--	--	--	--	No reasonable potential
25	2-chloroethyl vinyl ether	--	--	--	--	--	No reasonable potential
26	Chloroform	--	--	--	--	--	No reasonable potential
27	Dichlorobromomethane	--	--	--	--	--	No reasonable potential
28	1,1-Dichloroethane	--	--	--	--	--	No reasonable potential
29	1,2-dichloroethane	--	--	--	--	--	No reasonable potential
30	1,1-Dichloroethylene	--	--	--	--	--	No reasonable potential
31	1,2-dichloropropane	--	--	--	--	--	No reasonable potential
32	1,3-dichloropropylene	--	--	--	--	--	No reasonable potential
33	Ethylbenzene	--	--	--	--	--	No reasonable potential
34	Methyl bromide	--	--	--	--	--	No reasonable potential
35	Methyl chloride	--	--	--	--	--	No reasonable potential
36	Methylene chloride	--	--	--	--	--	No reasonable potential
37	1,1,2,2-tetrachloroethane	--	--	--	--	--	No reasonable potential
38	Tetrachloroethylene	--	--	--	--	--	No reasonable potential
39	Toluene	--	--	--	--	--	No reasonable potential
40	Trans 1,2-Dichloroethylene	--	--	--	--	--	No reasonable potential
41	1,1,1-Trichloroethane	--	--	--	--	--	No reasonable potential
42	1,1,2-trichloroethane	--	--	--	--	--	No reasonable potential
43	Trichloroethylene	--	--	--	--	--	No reasonable potential
44	Vinyl chloride	--	--	--	--	--	No reasonable potential
45	2-chlorophenol	--	--	--	--	--	No reasonable potential

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
46	2,4-dichlorophenol	--	--	--	--	--	No reasonable potential
47	2,4-dimethylphenol	--	--	--	--	--	No reasonable potential
48	4,6-dinitro-o-resol (aka 2-methyl-4,6-Dinitrophenol)	--	--	--	--	--	No reasonable potential
49	2,4-dinitrophenol	--	--	--	--	--	No reasonable potential
50	2-nitrophenol	--	--	--	--	--	No reasonable potential
51	4-nitrophenol	--	--	--	--	--	No reasonable potential
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-cresol)	--	--	--	--	--	No reasonable potential
53	Pentachlorophenol	--	--	--	--	--	No reasonable potential
54	Phenol	--	--	--	--	--	No reasonable potential
55	2,4,6-trichlorophenol	--	--	--	--	--	No reasonable potential
56	Acenaphthene	--	--	--	--	--	No reasonable potential
57	Acenaphthylene	--	--	--	--	--	No reasonable potential
58	Anthracene	--	--	--	--	--	No reasonable potential
59	Benzidine	--	--	--	--	--	No reasonable potential
60	Benzo(a)Anthracene	--	--	--	--	--	No reasonable potential
61	Benzo(a)Pyrene	--	--	--	--	--	No reasonable potential
62	Benzo(b)Fluoranthene	--	--	--	--	--	No reasonable potential
63	Benzo(ghi)Perylene	--	--	--	--	--	No reasonable potential
64	Benzo(k)Fluoranthene	--	--	--	--	--	No reasonable potential
65	Bis(2-Chloroethoxy) methane	--	--	--	--	--	No reasonable potential
66	Bis(2-Chloroethyl)Ether	--	--	--	--	--	No reasonable potential

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
67	Bis(2-Chloroisopropyl) Ether	--	--	--	--	--	No reasonable potential
68	Bis(2-Ethylhexyl) Phthalate	--	--	--	--	--	No reasonable potential
69	4-Bromophenyl Phenyl Ether	--	--	--	--	--	No reasonable potential
70	Butylbenzyl Phthalate	--	--	--	--	--	No reasonable potential
71	2-Chloronaphthalene	--	--	--	--	--	No reasonable potential
72	4-Chlorophenyl Phenyl Ether	--	--	--	--	--	No reasonable potential
73	Chrysene	--	--	--	--	--	No reasonable potential
74	Dibenzo(a,h)Anthracene	--	--	--	--	--	No reasonable potential
75	1,2-Dichlorobenzene	--	--	--	--	--	No reasonable potential
76	1,3-Dichlorobenzene	--	--	--	--	--	No reasonable potential
77	1,4-Dichlorobenzene	--	--	--	--	--	No reasonable potential
78	3,3'-Dichlorobenzidine	--	--	--	--	--	No reasonable potential
79	Diethyl Phthalate	--	--	--	--	--	No reasonable potential
80	Dimethyl Phthalate	--	--	--	--	--	No reasonable potential
81	Di-n-Butyl Phthalate	--	--	--	--	--	No reasonable potential
82	2,4-Dinitrotoluene	--	--	--	--	--	No reasonable potential
83	2,6-Dinitrotoluene	--	--	--	--	--	No reasonable potential
84	Di-n-Octyl Phthalate	--	--	--	--	--	No reasonable potential
85	1,2-Diphenylhydrazine	--	--	--	--	--	No reasonable potential
86	Fluoranthene	--	--	--	--	--	No reasonable potential
87	Fluorene	--	--	--	--	--	No reasonable potential
88	Hexachlorobenzene	--	--	--	--	--	No reasonable potential
89	Hexachlorobutadiene	--	--	--	--	--	No reasonable potential

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
90	Hexachlorocyclopentadiene	--	--	--	--	--	No reasonable potential
91	Hexachloroethane	--	--	--	--	--	No reasonable potential
92	Indeno(1,2,3-cd)Pyrene	--	--	--	--	--	No reasonable potential
93	Isophorone	--	--	--	--	--	No reasonable potential
94	Napthalene	--	--	--	--	--	No reasonable potential
95	Nitrobenzene	--	--	--	--	--	No reasonable potential
96	N-Nitrosodimethylamine	--	--	--	--	--	No reasonable potential
97	N-Nitrosodi-n-Propylamine	--	--	--	--	--	No reasonable potential
98	N-Nitrosodiphenylamine	--	--	--	--	--	No reasonable potential
99	Phenanthrene	--	--	--	--	--	No reasonable potential
100	Pyrene	--	--	--	--	--	No reasonable potential
101	1,2,4-Trichlorobenzene	--	--	--	--	--	No reasonable potential
102	Aldrin	--	--	--	--	--	No reasonable potential
103	alpha-BHC	--	--	--	--	--	No reasonable potential
104	beta-BHC	--	--	--	--	--	No reasonable potential
105	gamma-BHC (aka Lindane)	--	--	--	--	--	No reasonable potential
106	delta-BHC	--	--	--	--	--	No reasonable potential
107	Chlordane	--	--	--	--	--	No reasonable potential
108	4,4'-DDT	--	--	--	--	--	No reasonable potential
109	4,4'-DDE	--	--	--	--	--	No reasonable potential
110	4,4'-DDD	--	--	--	--	--	No reasonable potential
111	Dieldrin	--	--	--	--	--	No reasonable potential
112	alpha-Endosulfan	--	--	--	--	--	No reasonable potential

CTR #	Parameters	Human health AMEL _{hh} = ECA = C _{hh} O	MDEL/ AMEL multiplier	Human health MDEL _{hh}	Lowest AMEL	Lowest MDEL	Recommendation
113	beta-Endosulfan	--	--	--	--	--	No reasonable potential
114	Endosulfan Sulfate	--	--	--	--	--	No reasonable potential
115	Endrin	--	--	--	--	--	No reasonable potential
116	Endrin Aldehyde	--	--	--	--	--	No reasonable potential
117	Heptachlor	--	--	--	--	--	No reasonable potential
118	Heptachlor Epoxide	--	--	--	--	--	No reasonable potential
119	PCB Aroclor 1016	--	--	--	--	--	No reasonable potential
120	PCB Aroclor 1221	--	--	--	--	--	No reasonable potential
121	PCB Aroclor 1232	--	--	--	--	--	No reasonable potential
122	PCB Aroclor 1242	--	--	--	--	--	No reasonable potential
123	PCB Aroclor 1248	--	--	--	--	--	No reasonable potential
124	PCB Aroclor 1254	--	--	--	--	--	No reasonable potential
125	PCB Aroclor 1260	--	--	--	--	--	No reasonable potential
126	Toxaphene	--	--	--	--	--	No reasonable potential
	Iron	--	--	--	300	--	Reasonable potential
	Trihalomethanes	--	--	--	80	--	Reasonable potential